SILENT SCALE SHIFT: WHEN SHARED UPDATES BREAK EXISTING READOUTS

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ABSTRACT

Shared encoders can keep learning while breaking the readouts that depend on them. We study this failure under auxiliary-task updates and identify a scale-sensitive component, *silent scale shift*: task information remains decodable, but an existing readout loses accuracy. In matched HAR continuations across ten seeds, frozen-readout accuracy falls from 0.946 to 0.453, while the current head reaches 0.948; unlabeled scale correction restores 0.938. With a real reconstruction auxiliary, error falls to 0.765 of detachment while frozen accuracy drops to 0.826; paired linear alignment restores 0.947. Exact affine boundaries explain the scale mechanism, and a repair hierarchy separates it from richer drift. A signed update audit shows why task-gradient alignment and unsigned variance probes can miss these changes. Finally, an independently confirmed SNR experiment retains a 0.89-percentage-point auxiliary gain and the old logits under a reference-anchored constraint, at $2.37\times$ CPU inference cost. Folding the constraint into a frozen-trunk encoder removes the reference cost but loses the gain. Together, these results make readout compatibility a distinct requirement for shared representation learning.

1 INTRODUCTION

An encoder and its downstream readouts need not update together. Auxiliary training changes the features; a deployed classifier may retain the weights and input statistics fitted to an earlier version. The current task head can adapt to this change and remain accurate. What happens to the readout already in use?

We isolate a large compatibility failure by cloning a healthy HAR model and its Adam state, retaining the fitted readout, and continuing matched branches on identical minibatches. Across ten seeds at variance-auxiliary dose 3, frozen-readout accuracy falls from 0.946 to 0.453, although the jointly trained head reaches 0.948. A single global input-scale correction, fitted without new labels, restores 0.938 (Figure 1b). We call this scale-sensitive component *silent scale shift*. The information needed for the task remains available; the existing readout has become incompatible with its input.

This failure also occurs with a real auxiliary objective. HAR reconstruction lowers held-out MSE to 0.765 of the dose-matched detached decoder, while frozen-readout accuracy falls to 0.826. A linear map fitted to paired old and new training features restores 0.947, leaving the encoder and reconstruction head unchanged (Figure 1c). None of the five seeds meets the scatter-collapse criterion. Auxiliary progress and readout failure can therefore coexist before that criterion signals a problem.

We study the failure through three complementary tests. First, exact affine-readout boundaries and a fixed repair hierarchy identify the contribution of scale. Second, a signed chain-rule audit explains what task-gradient relations and an unsigned radial probe leave unresolved. Methods such as PCGrad (Yu et al., 2020) and CAGrad (Liu et al., 2021a) act on task-gradient relations; variance change also depends on the update's alignment with the variance gradient. For Adam, we additionally measure the applied parameter increment. Third, we test whether useful auxiliary learning can coexist with an unchanged readout.

For the third test, a reference-anchored readout-nullspace control (Shen et al., 2020; Farajtabar et al., 2020) preserves healthy logits while leaving 53 of 64 feature directions free. After validation fixes the SNR recipe, ten new seeds confirm a 0.89-percentage-point auxiliary gain at $2.37\times$ CPU in-

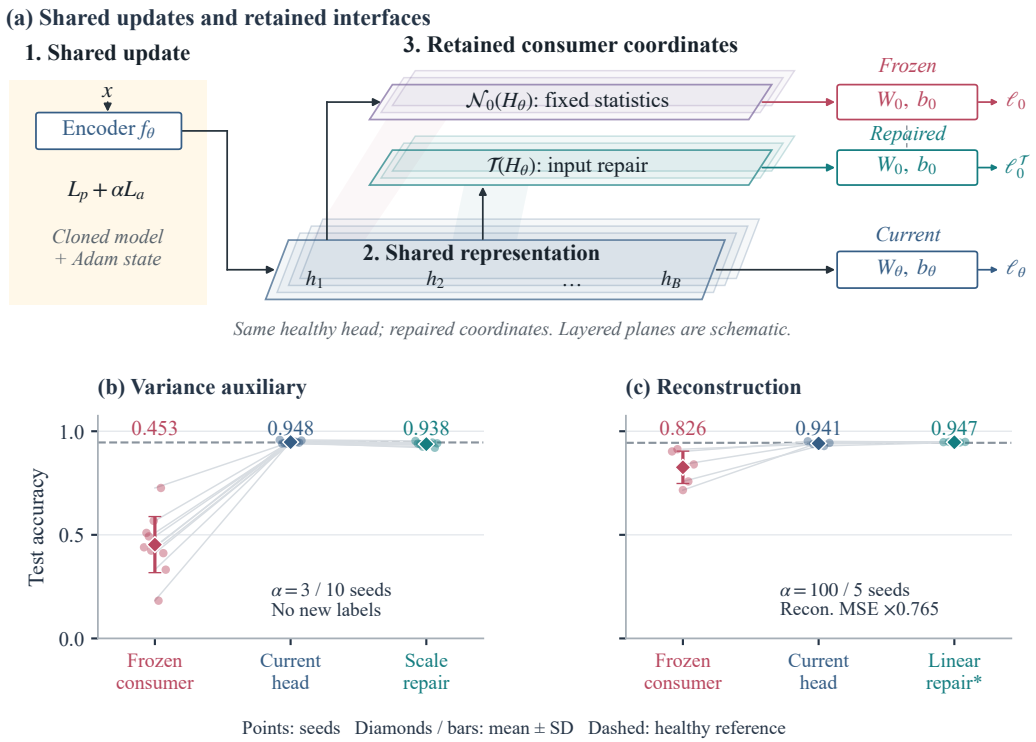


Figure 1: An updated representation can support the task while breaking an existing readout. (a) Three paths through shared features: the current head, the frozen readout, and the same readout after input repair. $\mathcal{N}_0$ is the healthy normalizer; $\mathcal{T}$ repairs normalization or alignment while retaining (W_0, b_0). (b) HAR variance continuation: global scale correction restores much of the lost frozen accuracy. (c) HAR reconstruction: auxiliary MSE falls to 0.765 of detachment; paired linear alignment restores readout accuracy. Repairs use training features without new labels and leave both learned models fixed. *Linear repair requires paired old/new anchors. Accuracies use the full test set; bars show population seed SD.

ference cost. A single-encoder follow-up preserves the logits but loses the gain when its nonlinear trunk is frozen. The resulting picture is concrete: task decodability, existing-readout compatibility, and the cost of preserving both require separate measurements.

2 When Existing Readouts Fail

Let $h_\theta(x) \in \mathbb{R}^m$ be the shared features. An existing readout applies the fixed map $W_0 D_0^{-1}(h_\theta(x) - \mu_0) + b_0$, with healthy means μ_0, positive diagonal scales D_0, weights W_0, and bias b_0. We use *consumer* for this complete input-to-output map. During continuation, the current task head learns with the encoder, while the old consumer stays fixed. A refitted probe tests what the new features can support; the old consumer tests whether its interface still works.

Pure contraction already separates these two questions. For $H_c = cH$, an affine readout has logits $cWh + b$, which approach b as $c \to 0$. A unique bias winner then makes the prediction constant, although every $c > 0$ preserves linear decodability. The following boundaries locate the relevant class changes.

Proposition 1 (Two scale boundaries of a frozen affine readout)**.** *Write the logits after fixed input statistics as* $z_i(c) = z_0 + c\, s_i$. *Assume unique winners* $t = \arg\max_j z_0[j]$ *and* $p_i = \arg\max_j z_i(1)[j]$. *The bias-only class first loses as* c *increases from zero at*

$$c_i^{\text{bias}} = \min_{j:\, s_i[j] > s_i[t]} \frac{z_0[t] - z_0[j]}{s_i[j] - s_i[t]}, \tag{1}$$

with the minimum of an empty set taken as ∞. The first class change while shrinking from $c = 1$ instead occurs at

$$c_i^{\text{first}} = \max_{\substack{j:\, s_i[p_i] > s_i[j] \\ 0 < \xi_{ij} < 1}} \xi_{ij}, \qquad \xi_{ij} = \frac{z_0[j] - z_0[p_i]}{s_i[p_i] - s_i[j]}, \tag{2}$$

with zero denoting no positive-scale flip. These boundaries can differ in multiclass problems. With nonzero feature variances and exact statistics, refitting the input statistics makes the readout invariant to every $c > 0$.

Proof. Each pairwise margin is affine in c. Increasing c from zero reaches the smallest positive crossing of the bias winner; decreasing c from one reaches the largest crossing of the healthy winner in $(0, 1)$. Standardising cH cancels c from numerator and denominator. For example, logits $(1, 10c, -1 + 12c)$ give $c^{\text{bias}} = 0.1$ but $c^{\text{first}} = 0.5$. □

The imposed-scale sweep in Figure 10 tests the scale mechanism (Appendix H.6). It isolates scaling from the additional drift learned during training.

The matched HAR trajectory in Figure 11a starts from one healthy checkpoint and excludes calibration examples from encoder and readout fitting. At $\alpha = 3$, the current head recovers while the frozen consumer remains inaccurate. Repairs fitted to training features without new labels distinguish mean and scale changes: mean correction reaches 0.813, global scale correction 0.938, and their combination 0.944, close to per-feature recalibration (0.947). At $\alpha = 100$, per-feature repair reaches 0.829 versus 0.935 after refitting. Scale explains much of the moderate-dose failure; stronger or projected contraction requires richer alignment (Appendix H.8).

Separately trained encoders show the same decodability–compatibility gap at $m = 32, 128$ and with a Transformer (Appendix H.6). Those historical studies include calibration examples in encoder fitting. We report negative log-likelihood (NLL) and expected calibration error (ECE), using 15 equal-width bins for ECE. Positive temperature scaling preserves predicted classes; input normalization can repair their scale dependence.

3 Auditing Shared Updates

3.1 Signed variance change

Let $H(\theta) \in \mathbb{R}^{B \times m}$ be the outputs of a shared encoder on a fixed diagnostic batch; a linear primary head and an auxiliary head consume H. Training minimizes $L_p + \alpha L_a$, where L_p is the primary loss, L_a the auxiliary loss, and $\alpha \geq 0$ the dose; multiple auxiliaries can be absorbed into their weighted sum. Write $G_p = \nabla_\theta L_p$ and $G_a = \nabla_\theta L_a$. Define

$$V(\theta) = \frac{1}{2B} \|C_B H(\theta)\|_F^2, \qquad C_B = I - \mathbf{1}\mathbf{1}^\top / B, \qquad v = \nabla_\theta V. \tag{3}$$

Here $2V$ is the covariance trace with divisor B: a rotation-invariant measure of variation across examples. Under gradient flow,

$$\dot{V} = r - \alpha c, \qquad r = -v^\top G_p, \qquad c = v^\top G_a. \tag{4}$$

We call $c > 0$ *auxiliary contraction*, and $r > 0$ *primary restoration*. When both are positive, $\alpha c / r > 1$ defines the instantaneous contraction-dominated regime. Reporting (r, c) also covers a primary update that itself reduces variance. This identity retains the encoder Jacobian; it defines a signed diagnostic without a reduced dynamical model.

Proposition 2 (Conflict information does not identify contraction)**.** *The task-gradient Gram matrix does not determine the sign of $\dot{V}$. In particular, there are two auxiliary updates with identical task-gradient norms and conflict, but opposite effects on variance under the same primary update.*

Proof. Take $G_p = e_1$, $G_a^\pm = \pm e_2$, and $v = -e_1 + \beta e_2$ for $\beta > 0$. Both pairs have identity Gram matrix and zero conflict, whereas $r = 1$ and $c^\pm = \pm\beta$. For $\alpha\beta > 1$, $\dot{V}^+ < 0 < \dot{V}^-$. These gradients are locally realizable by a smooth encoder and losses (Appendix A). □

Task-gradient relations therefore leave variance change unidentified. A combiner may preserve variance by reducing the auxiliary dose, but preserving a particular auxiliary update requires checking its projection onto v. The fixed-dose comparisons below test this distinction.

The local balance also differs from eventual collapse. In the symmetric linear model $\dot{z} = (R - \alpha A)z$, $R \succ 0$, $A \succeq 0$, some mode contracts when $\alpha\lambda_{\max}(R^{-1/2}AR^{-1/2}) > 1$; all modes contract when $\alpha\lambda_{\min}(R^{-1/2}AR^{-1/2}) > 1$ (Theorem 3). A head can exploit the surviving modes. In HAR, even a $4{,}600\times$ trace contraction leaves standardized-probe accuracy unchanged (Appendix H.4). We therefore measure variance, effective rank, task decodability, and old-readout utility separately.

3.2 What an unsigned probe leaves out

Let h and u stack the rows of H and $C_B H$, respectively, and let $g_p = \nabla_h L_p$, $g_a = \nabla_h L_a$. A cheap feature-space probe uses the centered radial direction:

Definition 1 (Unsigned radial toxicity proxy)**.** *For $u \neq 0$ and $\|g_p\| > 0$, define*

$$\rho_b = \frac{\|\Pi_u g_a\|}{\|g_p\|}, \qquad \Pi_u g = \frac{u^\top g}{\|u\|^2} u. \tag{5}$$

The experimental estimator $\hat{\rho}$ instead averages regularized per-example ratios. The frozen decision uses a recorded value $q = \hat{\rho}_{t_0}$ and the nominal dose $\alpha_{\text{probe}} = 1/q$.

The proxy discards the sign of the auxiliary direction and uses the entire primary-gradient norm as its reference. Both choices matter: expansion and contraction can give the same ρ_b, and a large primary gradient need not restore centered variation. There is also a measurement mismatch. The controlled driver probes the isotropic surrogate $\|C_B H\|_F^2/(Bm)$ but trains with fixed projected-variance auxiliaries; the real-head driver probes the actual auxiliary. The signed audit below accounts for these choices.

3.3 A local validity boundary

The discrepancy can be exposed without assuming radial training dynamics. Let $J = \partial h/\partial\theta$, $K = JJ^\top$, $e = u/\|u\|$, and $n_p = \|g_p\| > 0$. At an audited checkpoint choose any $k > 0$ and write $K = k(I + E)$. Define

$$a = \frac{e^\top g_a}{n_p}, \quad d = -\frac{e^\top g_p}{n_p}, \quad \epsilon_a = \frac{e^\top E g_a}{n_p}, \quad \epsilon_p = -\frac{e^\top E g_p}{n_p}. \tag{6}$$

Here a is signed radial loading relative to the primary norm, d measures restoration alignment, and the ϵ terms measure the effect of encoder geometry. For the actual auxiliary, $|a| = \rho_b$. Products with K can be evaluated by Jacobian–vector and vector–Jacobian products, without constructing the full matrix.

Theorem 1 (Local validity boundary of a frozen probe)**.** *For any frozen probe value $q \geq 0$, the variance dynamics satisfy*

$$\frac{\dot{V}}{N} = 1 - \alpha q + \mathcal{E}, \qquad N = \frac{k\|u\| n_p}{B} > 0, \qquad \mathcal{E} = (d - 1) + \epsilon_p - \alpha(a - q + \epsilon_a). \tag{7}$$

Consequently, with

$$\Delta = |d - 1| + |\epsilon_p| + \alpha\big(|a - q| + |\epsilon_a|\big), \qquad |1 - \alpha q| > \Delta, \tag{8}$$

the probe and the exact variance derivative agree in sign. If $1 - \alpha q > \Delta$, variance increases; if $1 - \alpha q < -\Delta$, it decreases. The remaining region requires the signed audit.

Proof. The chain rule gives $\dot{V} = -u^\top K(g_p + \alpha g_a)/B$. Substitute $K = k(I + E)$, divide by N, and add and subtract $1 - \alpha q$. The triangle inequality bounds $|\mathcal{E}|$ by Δ. □

The margin is sufficient and checkpoint-local. Computing it requires the exact flux, so it serves as an offline explanation of the cheap probe. On 405 HAR checkpoints, the frozen certificate fires on 2; the

signed balance $\psi = d - \alpha a$ with a combined residual certifies 10 at the same scale. Per-checkpoint scale optimization raises the counts to 15 and 154 (Table 2). Every certified sign is correct. The corrected balance alone agrees in sign on 58% of checkpoints, leaving substantial encoder-geometry mismatch.

In the identity metric ($E = 0$), $\dot{V}/N = d - \alpha a$. If $a = q$ and $0 < d < \alpha q < 1$, the nominal gate stays closed while variance contracts: the primary norm overstates restoration. If $a = -q$, the same unsigned gate can open while the auxiliary expands variance. Drift after calibration contributes through $a - q$. These examples identify concrete sources of a wrong sign.

The mismatch term also separates into actionable components. If a_0 is signed loading at calibration for the actual auxiliary, then

$$|a_t - q| \le |a_t - a_0| + |a_0 - |a_0|| + \big||a_0| - q\big|. \tag{9}$$

The terms separate temporal drift, discarded sign, and surrogate or aggregation mismatch. A radial approximation is informative when the auxiliary contracts, the primary gradient restores centered variation, the encoder metric is compatible with the approximation, and calibration transfers to the audited checkpoint.

A scalar reduction, $\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma)$, yields a saddle-node threshold $\alpha_c = \lambda_\star/\kappa$ under the explicit shape and endpoint assumptions of Theorem 2. Equating $1/q$ with this threshold additionally requires calibration to $\kappa/\lambda_\star$. Empirically, we compare $1/q$ with the half-height crossing $\alpha_{1/2}$ and report the operational collapse threshold separately.

4 Evidence Across Tasks and Protocols

4.1 Protocols and observables

The controlled benchmark spans HAR (Anguita et al., 2012), RadioML (O'Shea et al., 2016), battery health (Saha & Goebel, 2007), and NYUv2. A three-layer MLP with a fixed projected-variance auxiliary isolates contraction; Transformer, CNN, real-head, CelebA, and financial studies test its scope. Appendices F–H give the protocols and optimizer grids. Error bars show population seed SD unless stated otherwise.

The operational label is relative scatter $s = \mathcal{S}_h/\mathcal{S}_{\text{single}} < 0.3$, using the same-seed single-task reference and

$$\mathcal{S}_h = \mathbb{E}_b\Big[\mathrm{std}_j\Big(h_j^{(b)} - \mathbb{E}_{b'} h_j^{(b')}\Big)\Big]. \tag{10}$$

This coordinate-dependent proxy misses variation along the all-ones feature direction. We use "collapse" for the declared 0.3 criterion and measure information separately. The log-dose interpolated crossing at $s = 0.5$, $\alpha_{1/2}$, summarizes the dose curve.

4.2 Calibration is task- and width-dependent

Across sixteen domain–width cells, the ratio of predicted dose to observed half-height crossing is 0.95–3.01 for battery and 30–116 for RadioML; HAR reverses bias with width (Figure 7a). All cells train projected auxiliaries and probe the isotropic surrogate. The calibration error varies substantially across domains even in this controlled setting.

4.3 Calibration and drift change the decision

On HAR at $m = 64$, detached warm-up moves $1/\bar{\rho}$ from 1.34 at $W = 1$ to 0.00907 at $W = 4$, against a half-height crossing of 0.00141. At dose 0.03, $W = 1, 2$ leave all gates closed and all ten seeds collapse; $W \ge 4$ opens every gate and prevents collapse. At dose 100, all tested warm-ups protect scatter (Figure 7b,d). An extreme dose can therefore hide a calibration error.

Recomputing the probe during training can produce false alarms. HAR ratios grow even at zero auxiliary dose, and running gates activate in every seed of ten configurations with final scatter 0.50–1.04. Battery has one activation among 35 seeds ending above half-height (Appendix H.2). A shrinking primary-gradient denominator can inflate the ratio. On 405 HAR checkpoints, the unsigned predictor agrees with the exact variance derivative on 51%, with median restoration alignment 0.065.

Table 1: Independent confirmation of the validation-selected SNR auxiliary: 20% of fitting primary labels, dose 1, ten new seeds. D/J denote detachment/joint training; G denotes the reference-anchored readout-nullspace constraint. Own accuracy uses the current task head. The same healthy consumer is retained for frozen accuracy. Means with population seed SD; bounds use a separate fixed 40-consumer bank and a five-point relative-error budget. The guarded representation requires both encoder evaluations at inference.

Arm	own accuracy	frozen accuracy	frozen NLL	ECE	certified
D	0.275 ± 0.010	0.259 ± 0.016	2.176	0.153	0/10
J	0.285 ± 0.009	0.258 ± 0.019	2.135	0.155	0/10
GD	0.279 ± 0.010	0.259 ± 0.013	2.031	0.048	10/10
GJ	0.288 ± 0.009	0.259 ± 0.013	2.031	0.048	10/10

4.4 Real auxiliaries and stronger encoders

Four real auxiliary heads further separate loading from contraction (Figure 5a). Mean per-example task-gradient cosines have magnitude at most 0.06, while SNR radial loading is 88%. Yet SNR's endpoint signed radial cosine is negative, and $c_t = v^\top G_a < 0$ at $74/75$ historical epoch-end checkpoints. Its relative collapse cannot be attributed to persistent positive sampled flow contraction. Flow and actual Adam signs also disagree on $33/75$ sampled paired updates. We record both the applied increment and a primary-only Adam counterfactual (Appendix H.7).

Protocol changes alter the observed failure. Historical SNR scatter onset lies between doses 3 and 10, whereas I/Q reconstruction stays above the criterion through 10000. Under healthy-checkpoint continuation, I/Q at 3000 gives frozen accuracy 0.301 versus 0.304 detached, with trace 1.061; SNR at 30 gives 0.243 versus 0.304, with scatter 0.870. Resetting auxiliary weights or their Adam state also fails to reproduce the historical severe collapse (Appendix H.11). Warm-up, split, architecture, and horizon define the operating regime.

A waveform CNN on the same I/Q split reaches 0.424 healthy-readout accuracy. At reconstruction dose 3000, frozen accuracy is 0.415 versus 0.418 detached, while the current head reaches 0.437. This stronger encoder shows little consumer damage on the fixed real-auxiliary grid (Appendix H.10).

5 Repairing and Preserving Compatibility

Real reconstruction tests repair while the auxiliary makes progress (Table 34, Appendix H.9). At HAR dose 100, frozen accuracy falls from 0.941 under detachment to 0.826, while the current head reaches 0.941. Test reconstruction MSE falls to 0.765 of detachment, with $0/5$ scatter collapses. Per-feature recalibration reaches 0.863; paired linear alignment reaches 0.947 (Figure 1c). Repair changes only the consumer input, retaining the auxiliary improvement.

Let $A = W_0 \operatorname{diag}(1/s_0)$ and let $P_\perp$ project onto $\ker A$. The reference-anchored control $H_0 + (H_\theta - H_0)P_\perp$ preserves all healthy logits while emitting one shared 64-vector (Proposition 10, Figure 2). Validation selects SNR dose 1 from six fixed candidates, using 20% of fitting primary labels. Ten new seeds confirm ordinary auxiliary gain J–D of 0.97 percentage points, paired 95% interval $[0.63, 1.31]$, and guarded gain GJ–GD of 0.89, $[0.60, 1.18]$ (Table 1). The guarded model meets the declared 0.5-point noninferiority margin against J. All guarded predictions, NLL, and ECE match the reference; maximum logit error is 4.6×10^{-14}. The ten guarded joint endpoints also pass a separate five-point disagreement budget. Ordinary frozen accuracy barely changes, but ECE rises from 0.048 to 0.155. Both encoders remain at inference, at a measured $2.37\times$ CPU cost (Appendix H.12).

With a frozen nonlinear trunk, the last-affine constraint folds into one encoder. Ten new seeds per task on SNR and HAR retain logits but confirm no compiled auxiliary benefit (Appendix H.13). On SNR, compiled joint training (CJ) falls below full joint training (J) by -3.53 points, $[-4.23, -2.83]$: freezing accounts for -3.53, and the added constraint changes accuracy by -0.003. A separately fixed nonlinear-side development grid also fails its benefit screen (Appendix H.14). These controls locate the practical difficulty in retaining useful adaptation at lower deployment cost.

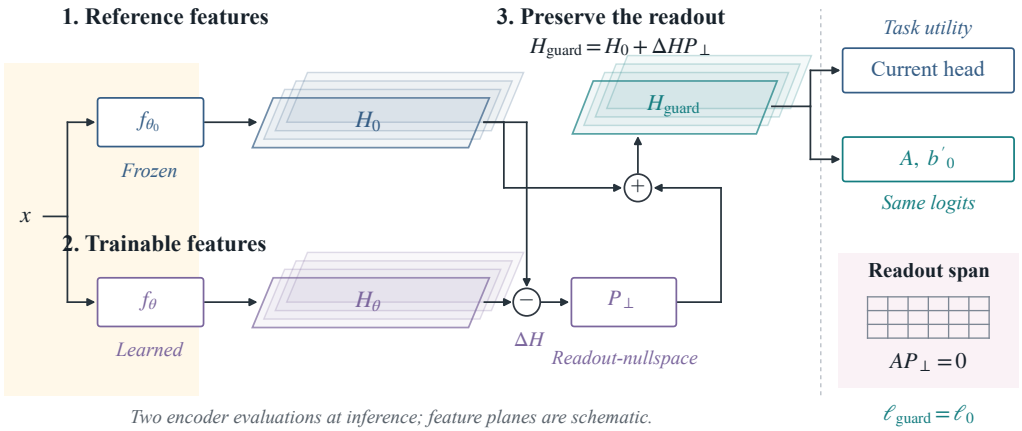


Figure 2: A fixed readout leaves a nullspace for learning. Projecting $\Delta H = H_\theta - H_0$ into $\ker A$ and adding H_0 gives one shared 64-vector with unchanged reference logits in real arithmetic. The normalized head is $A = W_0 D_0^{-1}$ and its effective bias is $b'_0 = b_0 - A\mu_0$. The inset shows the annihilation condition; planes are schematic. Both encoders are evaluated at inference.

Detachment blocks auxiliary gradients to the encoder and provides the reference when the auxiliary brings no primary-task value. The spectral toxicity filter (STF) instead removes a chosen component of the auxiliary channel. Its idealized backward form is

$$F_w[g_a] = g_a - w\Pi_u g_a, \qquad w \in [0, 1], \tag{11}$$

The implemented selective arms transform the auxiliary input, $\widetilde{H} = H - wC_B HP$, using a frozen projector P or a centered radial direction. The transformation changes the forward input as well as its gradient, while the auxiliary head continues training (Appendix E).

The calibrated arm freezes the warm-up probe and opens the gate when $\alpha\hat{\rho} \geq 1$. The rank variant then selects

$$\text{rank}_{\text{sel}} = \min\{\text{rank} \leq \text{rank}_{\max} : \alpha\hat{\rho}_{\text{res}}[\text{rank}] < 1\}, \tag{12}$$

from nested candidate subspaces. If none passes, it uses the cap and records the unresolved residual. The radial arm is a separate intervention. Theorem 5 gives the minimum deletion rank for a specified PSD operator; using an empirical PCA basis adds a basis-identification question. Gate calibration and subspace choice therefore need separate evaluation.

The low-label SNR dose scan shows both the value and limits of filtering (Figure 8a,b). With 20% primary labels, joint training is healthy at low dose but collapses in every seed from $\alpha = 30$. Calibrated STF avoids collapse at every sampled dose and exceeds detachment by 0.5–1.0 percentage points over $\alpha \in [15, 100]$; the advantage disappears at extreme doses. This five-seed scan uses a different label budget from the fully labeled study in Appendix H.17.

The fully labeled SNR comparison shows why clearing the scatter threshold is insufficient (Figure 8c). Detachment reaches 0.307 accuracy with $0/10$ collapses. The radial arm reaches 0.199 with $1/10$, and the rank-3 arm 0.211 with $0/10$. Basis choice also changes how much auxiliary learning is removed: in HAR reconstruction, an auxiliary-gradient basis uses 30.75 directions versus PCA's 62.0 among the same four gate-open seeds, at similar accuracy (Table 7).

At the controlled high doses, calibrated filtering reduces collapse from $10/10$ to $0/10$ on HAR, battery, and RadioML, and from $5/5$ to $0/5$ on CelebA Smiling (Liu et al., 2015). With a Transformer at RadioML dose 3000, joint training collapses in $10/10$ seeds at accuracy 0.156; filtering gives $0/10$ collapses and relative scatter 0.970. Reliable gate activation at these extreme doses coexists with the calibration errors in Figure 7.

Algorithm 1 Auditing representation survival and probe validity

Require: Diagnostic batch, healthy reference, checkpoint θ, operating losses and dose α, recorded calibration value q.

1: Report task metrics, covariance trace $2V$, effective rank, standardized linear probe, and the declared operational scatter criterion.
2: Audit healthy-readout accuracy/NLL/ECE, mean/global/per-feature repairs, and any paired alignment; compare with a refitted probe.
3: With unlabeled monitor inputs and healthy reference predictions, bound consumer disagreement; select a certified candidate or abstain.
4: Report gradient-flow fluxes (r, c) from (4). For Adam, audit $v^\top s_\theta$ and the actual post-step V change; specify any auxiliary counterfactual.
5: Record $1-\alpha q$, the signed balance $\psi = d-\alpha a$, and the corresponding residuals. Form $q^\star = a/d$ only when $d \neq 0$.
6: **if** $|1 - \alpha q| > \Delta$ **then**
7: Mark the gradient-flow sign locally certified for this batch.
8: **else**
9: Use the signed audit; report restoration misalignment, geometry, and probe mismatch separately.
10: **end if**
11: Compare joint training, detachment, and any selective intervention using both survival and task utility at matched doses and seeds.

6 Evaluating a Shared Representation

The evidence suggests three questions for evaluation: how did the features change, what task information remains decodable, and does the existing consumer still work? Algorithm 1 combines these checks with the signed update audit.

When Δ exceeds the probe margin, the unsigned decision remains unresolved. The signed audit attributes its current error to restoration alignment, encoder geometry, discarded sign, or calibration mismatch. Repeating the same unsigned ratio cannot separate these effects.

The intervention comparison also needs both compatibility and task utility. A rank-3 SNR filter can pass the scatter criterion while underperforming detachment; the low-label radial filter can lose its advantage at extreme doses. An open gate alone says little about the quality of the selected subspace.

For a present checkpoint, unlabeled inputs can support a conservative consumer decision. On a fixed corpus, the increase in classification error is bounded by disagreement with healthy predictions. We sample $1{,}024$ monitor inputs, select the first candidate in a fixed repair hierarchy whose disagreement bound meets the budget, and otherwise abstain. Across 105 joint endpoints, a 10-percentage-point budget certifies 49, with 0 observed violations and 56 abstentions; a healthy-reference scatter heuristic makes 11 false-safe declarations (Appendix H.15).

The checks have different scopes. The signed flux diagnoses the current gradient-flow direction; Adam needs its applied increment. The STF gate is an empirical calibration rule. The disagreement bound covers the fixed checkpoint family and unmonitored corpus remainder, with access to healthy predictions; it leaves changed populations and future updates open. The CNN and single-encoder controls likewise limit the evidence for broadly useful, inexpensive protection.

7 Related Work

Multi-task learning (Caruana, 1997) and task grouping (Standley et al., 2020) ask when shared training helps. PCGrad and CAGrad address gradient conflict (Yu et al., 2020; Liu et al., 2021a); MGDA, Nash-MTL, and FAMO optimize Pareto, bargaining, or progress-balancing criteria (Sener & Koltun, 2018; Navon et al., 2022; Liu et al., 2023). Other methods adapt task-gradient geometry or loss weights (Senushkin et al., 2023; Liu et al., 2021b; Chen et al., 2018; Kendall et al., 2018; Wang et al., 2021; Limarenko et al., 2026; Zhou et al., 2025). We add the variance gradient to this picture: task-gradient relations alone leave the sign of variance change unidentified.

Backward-compatible representation learning explicitly studies model-version compatibility (Shen et al., 2020), including jointly trained transformations (Hu et al., 2022b). We examine the scale-sensitive component of compatibility loss during auxiliary updates. CORAL-style covariance alignment (Sun et al., 2016), paired linear alignment, calibration under shift (Guo et al., 2017; Ovadia et al., 2019), normalization (Ioffe & Szegedy, 2015), and linear probes (Alain & Bengio, 2016) provide established diagnostic controls. Negative transfer (Wang et al., 2019) concerns task performance; here the old readout can fail while a current or refitted head succeeds. Our fixed-corpus consumer check complements work on unlabeled performance estimation (Garg et al., 2022) and update certification (Usama & Qadir, 2026) through reference disagreement and abstention.

Self-supervised learning studies collapse through stop-gradient mechanisms (Grill et al., 2020; Chen & He, 2021; Caron et al., 2021), variance and covariance regularization (Zbontar et al., 2021; Bardes et al., 2022; Ermolov et al., 2021), and spectral dynamics (Jing et al., 2022; Dong et al., 2021; Kim et al., 2025). RankMe evaluates representations by effective rank (Garrido et al., 2023). Our observable is variation across examples, distinct from terminal classifier geometry in neural collapse (Papyan et al., 2020; Wang et al., 2026). Input repair shows how reduced variation can coexist with retained decodability and lost readout compatibility.

Auxiliary-gradient projection (Dery et al., 2021; Zhu et al., 2025), auxiliary selection (Navon et al., 2021; Jiang et al., 2023; Shin et al., 2024), weight-space nullspace adaptation (Pham et al., 2026), and mergeable frozen-backbone updates (Hu et al., 2022a) provide the context for our intervention controls. Activity recognition (Anguita et al., 2012), modulation recognition (O'Shea et al., 2016), and battery health (Saha & Goebel, 2007; Severson et al., 2019; Attia et al., 2020; Pradhan & Chakraborty, 2022) provide the test domains. We use these controls to measure compatibility, useful adaptation, and cost.

8 Conclusion

An accurate current head can conceal a broken existing readout. Matched continuation and fixed input repairs expose this gap, with exact affine boundaries explaining its scale-sensitive component. Signed audits identify what gradient alignment and unsigned probes miss. The SNR confirmation shows that auxiliary benefit and unchanged old logits can coexist, while the single-encoder controls expose the cost of preserving both. Shared-representation evaluation should therefore measure what new features can predict and whether existing readouts can still use them.

AI USE STATEMENT

Generative AI tools assisted with drafting, language editing, code, literature retrieval, proof development, and figure design. The new diagnostic identities were checked with executable numerical examples; assumptions are stated explicitly with each claim and in Appendix D. Figures are drawn from recorded experimental results. The authors remain responsible for verifying the final text, claims, code, and artifacts.

ETHICS STATEMENT

We use existing datasets: UCI-HAR, RadioML2016.10A, the NASA PCoE and MATR battery archives, public A-share market data, NYUv2, and CelebA. We collected no new human-subject data. The HAR identity auxiliary is evaluated as a training-head probe, not a held-out privacy attack. Battery prediction is safety-relevant; Section 4.3 describes the observed limitations, and deployment requires task-specific validation.

REPRODUCIBILITY STATEMENT

The artifact contains theoretical assumptions, code, per-seed records, the fixed development/confirmation plan, checkpoint and consumer replays, scale-boundary checks, actual-step audits, and the corrected battery protocol. Historical battery checkpoint selection and readout calibration are exploratory. The anonymized repository supplies released code and earlier records: `https://github.com/code-2026l/RepFix`.

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A Proofs and Audit Details for the Diagnostic Boundary

Let $\theta \in \mathbb{R}^2$, $v_0 = -e_1 + \beta e_2$, and $z(\theta) = v_0 + \theta$. Choose a two-example encoder $H(\theta) = (z^\top, -z^\top)^\top$. Then $V = \|z\|^2/2$ and $\nabla_\theta V = v_0$ at $\theta = 0$. The losses $L_p = \|z - (v_0 - e_1)\|^2/2$ and $L_a^\pm = \|z - (v_0 \mp e_2)\|^2/2$ are nonnegative smooth functions of the representation. Their gradients at zero are e_1 and $\pm e_2$. Thus Proposition 2 concerns realizable encoder/loss configurations, not arbitrary incompatible gradients. Both task pairs have the same Gram matrix, while the signed variance derivatives differ.

Vectorization of the centered batch gives $u = (C_B \otimes I_m)h$ and $\nabla_\theta V = J^\top u/B$. Hence $\dot{V} = -u^\top J J^\top (g_p + \alpha g_a)/B$. For any fixed positive scale k, write $JJ^\top = k(I + E)$. This is a decomposition, not an assumption of isotropy. With the definitions in (6), the normalized derivative is $d + \epsilon_p - \alpha(a + \epsilon_a)$, which proves (7). Taking absolute values term by term proves (8). The exact discrepancy $\mathcal{E}$ can be smaller than Δ because signed terms cancel. Failure of the sufficient margin condition therefore leaves the probe undecided; it does not prove the probe wrong. Choosing k must be reported, since it changes the decomposition and the tightness of this conservative envelope, although it cannot change $\dot{V}$.

Record the batch, checkpoint, losses, auxiliary dose, probe value q, and scale k. Compute g_p, g_a for the actual operating losses and u from the unfiltered representation. Evaluate Kg by first computing $J^\top g$, then a Jacobian–vector product with J; a dense K is unnecessary. This yields $a, d, \epsilon_a, \epsilon_p$, the exact signed flux, and the validity margin. When $u = 0$ or $\|g_p\| = 0$, the normalized decomposition is undefined and the unnormalized identity (4) remains the appropriate diagnostic. This audit costs more than a frozen feature-space probe: at $B = 256$ and $m = 64$ on CPU the exact flux costs $0.30\,\mathrm{s}$ while the exact $k = \mathrm{tr}(K)/(Bm)$ certificate costs $20.7\,\mathrm{s}$, a factor of 69.6 (`results/CERTCOST/`). The trace enters only through the canonical scale: an envelope at any other k, including the per-checkpoint minimizer of Table 2, needs the exact flux and the head-level gradients but no trace of K. The certificate is therefore an offline audit, not a cheap substitute for measuring the flux; the cheap object in the deployed rule is the probe q. Its purpose is to identify why the cheap decision can or cannot be transported to an operating checkpoint. The archive contains cheap proxies and intervention outcomes for 405 HAR checkpoints. The frozen predictor agrees in sign with the exact variance derivative on 51%. Its sufficient certificate fires on 2 checkpoints at the trace-normalized scale $k = \mathrm{tr}(K)/(Bm)$ and 15 under per-checkpoint scale optimization, with no incorrect firing.

The signed balance $\psi = d - \alpha a$ replaces the unsigned loading and primary-norm reference with directional quantities. When $d \neq 0$, it equals $d(1 - \alpha q^\star)$ with $q^\star = a/d$. Sign agreement rises to 58% overall and 61% at positive dose. Computing the balance needs the same two head-level backward passes as the original probe; certifying it additionally needs the exact flux.

Table 2 varies the predictor, residual envelope, and scale separately. At canonical k, replacing the frozen predictor with ψ raises coverage from 2 to 3 with separate absolute geometry terms and to 10 when their signed cancellation is retained. Per-checkpoint scale optimization gives 15, 121, and 154. All rows have zero false firings, as required by the sufficient condition. For comparison, the frozen predictor with directly measured $|\mathcal{E}|$ certifies 128 at canonical k. Every arm uses the common normalization N, so these counts compare the full retrospective audits on the same prediction scale.

The optimized combined certificate covers $57/226$ contracting checkpoints (25%) and $97/179$ expanding checkpoints (54%). Restricting to $|d| \geq 0.1$ leaves 168 checkpoints, of which $47/97$ contracting and $52/71$ expanding points certify. The archive also contains 51 points with $|d| \leq 0.01$. Using ψ directly avoids division by weak restoration and remains defined at $d = 0$. These subsets describe the same audit archive.

Figure 3 ranks observed sign agreement using margins that already consume the exact derivative. AUC is 0.60 for the canonical corrected combined arm, 0.80 after scale optimization, and 0.63 for the canonical frozen arm. Scale optimization can itself improve these retrospective scores: whenever ψ and $r - \alpha c$ have matching nonzero signs, choosing $k = (r - \alpha c)/(\|u\| n_p \psi/B) > 0$ makes the combined residual zero. Unrestricted optimization would therefore certify all 236 matching-sign points; the recorded 154 uses a finite 241-point grid. Script `certificate_matched_audit.py` regenerates the arms from stored primitives.

Table 2: Matched retrospective certificate ablations on 405 HAR checkpoints. "Contracting" counts certified points among the 226 negative exact derivatives. All rows have zero false firings. Canonical $k = \mathrm{tr}(K)/(Bm)$ is fixed; optimized k is selected per checkpoint on the same 241-point grid. Triangle and combined geometry residuals are $|\epsilon_p| + \alpha|\epsilon_a|$ and $|\epsilon_p - \alpha\epsilon_a|$. Every envelope uses the exact flux.

Predictor	scale	residual	fires	contracting
frozen $1 - \alpha q$	canonical	original triangle	2	2
frozen $1 - \alpha q$	optimized	original triangle	15	10
corrected ψ	canonical	geometry triangle	3	3
corrected ψ	optimized	geometry triangle	121	41
corrected ψ	canonical	geometry combined	10	10
corrected ψ	optimized	geometry combined	154	57
frozen $1 - \alpha q$	canonical	exact $\lvert\mathcal{E}\rvert$	128	115

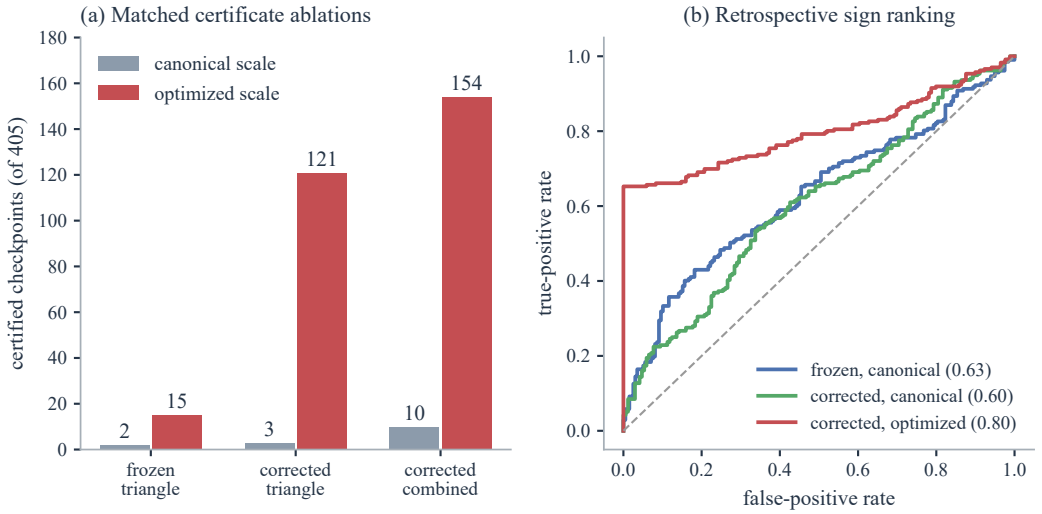


Figure 3: Certificate coverage depends on the predictor, residual, and scale. (a) Frozen and signed predictors under canonical and optimized scales, with separate and combined geometry residuals. (b) ROC curves for observed sign agreement. Every margin consumes the exact flux, so the curves describe retrospective audit discrimination.

The operational gate and the rank rule rely on different assumptions. For the gate, Theorem 1 requires the actual operating losses and encoder Jacobian, a recorded calibration value q, and discrepancy smaller than the sign margin at the audited checkpoint. For the minimum-rank claim, Theorem 5 requires a specified PSD operator; an empirical PCA basis needs separate identification. Deployed STFCAL evaluates neither certificate online. Table 3 records the scope of each arm.

Table 3: Assumptions behind the gate, rank rule, and deployed filters. The sign certificate and minimum-rank theorem concern different mathematical objects; the empirical arms retain their own calibration and basis assumptions.

Arm	Claim	Required assumption
gate $\alpha\hat{\rho} \geq 1$	local sign of $\dot{V}$ (Theorem 1)	frozen q; actual operating losses and encoder Jacobian; drift inside Δ
rank rule $\mathrm{rank}_{\mathrm{sel}}$	minimum deletion rank (Theorem 5)	specified PSD operator; nested candidate subspaces
PCA basis	empirical surrogate	identified separately; not assumed to be the operator's eigenbasis
deployed STFCAL	local heuristic	no online certificate; calibration audit only

Corollary 1 (Transport to a later audited checkpoint)**.** *Let q_0 be the probe frozen at calibration, and let q_t and Δ_t be the live probe and the envelope of Theorem 1 at a later checkpoint. If*

$$|1 - \alpha q_0| > \Delta_t + \alpha|q_t - q_0|,$$

then the frozen decision has the same sign as the exact variance derivative at that checkpoint.

Proof. $|(1 - \alpha q_0) - (1 - \alpha q_t)| = \alpha|q_t - q_0|$. If the displayed inequality holds, then $|1 - \alpha q_t| \geq |1 - \alpha q_0| - \alpha|q_t - q_0| > \Delta_t$, so Theorem 1 applies at t with the live terms. □

Corollary 2 (Signed directional balance)**.** *Let $\psi = d - \alpha a$. With the decomposition of Theorem 1,*

$$\frac{\dot{V}}{N} = \psi + \epsilon_p - \alpha\epsilon_a.$$

If $|\psi| > |\epsilon_p - \alpha\epsilon_a|$ at a specified $k > 0$, then $\dot{V}$ has the sign of ψ, including when $d = 0$. When $d \neq 0$, define $q^\star = a/d$ and $\Delta^\star = |\epsilon_p - \alpha\epsilon_a|/|d|$; the same condition is $|1 - \alpha q^\star| > \Delta^\star$, with predicted sign $\operatorname{sign}[d(1 - \alpha q^\star)]$*. The $|d - 1|$ discrepancy is absent.*

Proof. Regroup (7) as $d - \alpha a + \epsilon_p - \alpha\epsilon_a$. A perturbation smaller in magnitude than ψ cannot change its sign. Substitution of $q^\star$ gives the ratio form when $d \neq 0$. □

The quantities $a \propto \sum_i (h_i - \bar{h})^\top g_{a,i}$ and $d \propto -\sum_i (h_i - \bar{h})^\top g_{p,i}$ are the centered auxiliary and primary radial derivatives. Their signed ratio $q^\star$ allows cancellation across examples. The archived probe instead averages per-example magnitudes divided by per-example primary norms. Replacing that primary reference by $d\, n_p$ removes the $|d - 1|$ discrepancy.

Corollary 1 requires the frozen margin to exceed the current envelope plus probe drift. It certifies $2/405$ checkpoints at the canonical trace-normalized scale and $14/405$ under scale optimization, with no incorrect firing. The corresponding frozen certificate without the added drift term fires on $15/405$. Outside the certified checkpoints, calibration remains an empirical decision rule.

Figure 4 shows how closely the margin approaches the transport envelope. The 14 certified checkpoints in (a) lie just above it; the margin/envelope ratio in (b) has median 0.86 and upper quartile 0.996. Across the 27 width–dose cells in (c), sign agreement is 0.56 above median restoration alignment and 0.47 below it. Signed mismatch $|a - q|$ varies largely independently of d (log-scale correlation -0.05). Panel (d) compares $|\mathcal{E}|$ and the minimum envelope at the same per-checkpoint scale, with median ratio 0.885. Matching the normalization is essential to interpreting envelope tightness. These archive associations describe where the offline audit is informative; they leave prospective performance to a separate evaluation.

For $K = I$, the local condition is $d - \alpha a < 0$. Take $d = 0.1$, $a = q = 0.2$, and $\alpha = 1$: the nominal gate is closed while the exact derivative is negative. This is a false negative caused by treating the primary norm as radial restoration. Conversely, with $d = 1$, $a = -q = -2$, and $\alpha = 1$, the gate is open but the exact derivative is positive. For a fixed q, the theorem applies at each time with time-indexed components. If $\sup_{t \in I} \Delta_t < |1 - \alpha q|$, its sign conclusion holds throughout the interval I. Obtaining this uniform bound is an additional task; a single frozen observation does not supply it. Even a negative variance derivative on a finite interval does not by itself imply convergence to zero.

Let s_θ be the actual proposed parameter increment, including any optimizer preconditioning, momentum, and weight decay. If ∇V is L-Lipschitz on the segment from θ to $\theta + s_\theta$, then

$$\left|V(\theta + s_\theta) - V(\theta) - v^\top s_\theta\right| \leq \tfrac{L}{2}\|s_\theta\|^2. \tag{13}$$

This follows by integrating the gradient along the segment. A first-order margin exceeding this remainder certifies the sign of the finite-step change. The gradient-flow decomposition applies directly to plain gradient updates; the actual-step audit is the corresponding observable for Adam. Separating primary and auxiliary contributions through a nonlinear optimizer requires specifying the counterfactual update and its state. A feature-gradient norm ratio is not an Adam-step certificate.

The controlled driver computes the probe in `rho_probe` within `cross_domain_mtl.py`:

$$\hat{\rho} = \frac{1}{B}\sum_{i=1}^{B} \frac{\|\Pi_{u_i} \widetilde{g}_{a,i}\|}{\|g_{p,i}\| + 10^{-8}}, \qquad \widetilde{g}_a = \nabla_H \frac{\|C_B H\|_F^2}{Bm}. \tag{14}$$

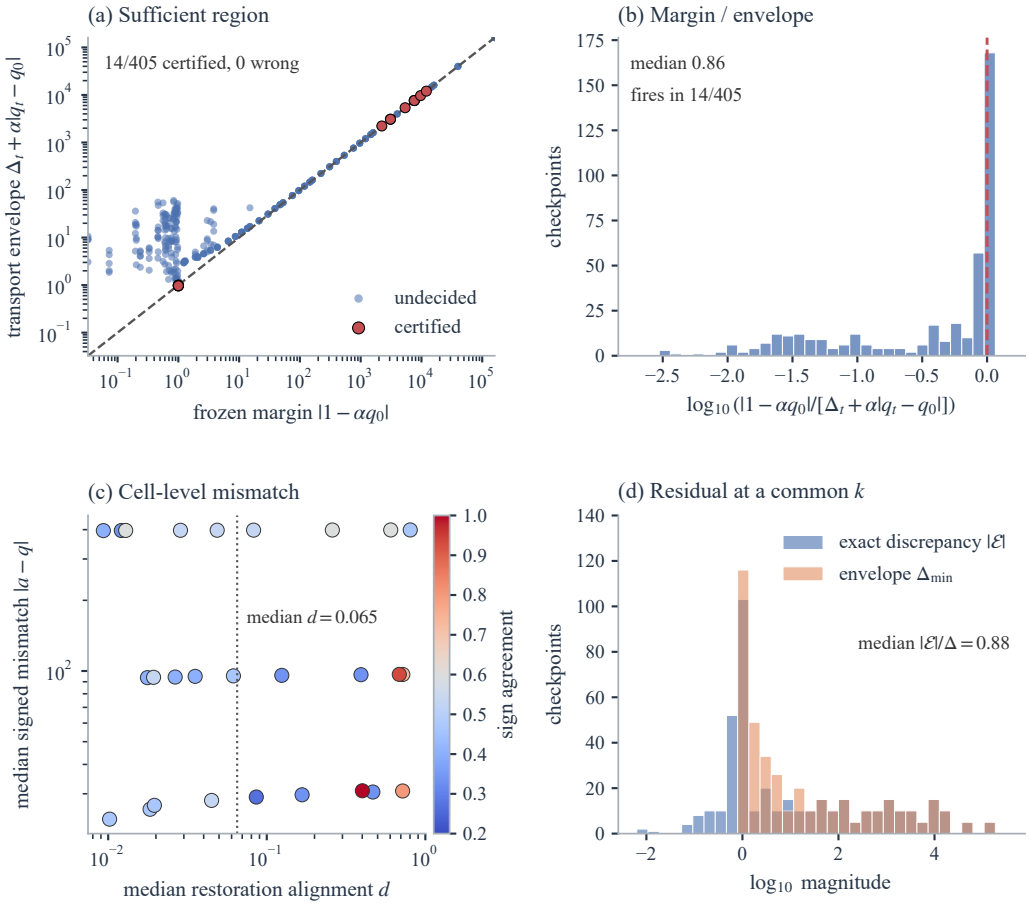


Figure 4: Most audited checkpoints fall outside the sufficient transport margin. Audit of 405 HAR checkpoints. (a) Frozen margin versus the optimized transport envelope; 14 checkpoints certify, all correctly. (b) Margin/envelope ratio: median 0.86, upper quartile 0.996. (c) Restoration alignment and signed mismatch across 27 width–dose cells. (d) Exact discrepancy and envelope at the same optimized scale, with median $|\mathcal{E}|/\Delta_{\min} = 0.885$.

Training instead uses the model's fixed random projections. Thus a contracting surrogate, per-example averaging, and regularization all enter q. The real-head driver differentiates the specified reconstruction, identity, or SNR loss. The A1 alignment records are probes *after* joint training; their mean cosine is an average of examplewise cosines, not the cosine of the aggregate parameter gradients. Their unsigned loading cannot establish the sign of $v^\top G_a$. The complete main-figure audit is generated by `paper/revision/build_figures.py`, with source hashes and reconstructed statistics. Duplicate low-label dose logs are reconciled by $(\alpha, \text{mode}, \text{seed})$ and checked for equality before removal.

`paper/revision/check_boundary.py` checks the exact discrepancy, the sufficient sign certificate, the realizable conflict example, and the finite-step remainder on seeded linear encoders. These are algebraic checks of the new theory, separate from the archived neural-network experiments.

A.1 Conditional Dynamical Models

Definition 2 (Degenerate minimizer)**.** *A loss has a degenerate minimizer if a constant representation attains its minimum. For a differentiable interior minimum, stationarity is necessary but insufficient. Variance shrinkage has such minima; reconstruction with varying targets generally does not, be-*

cause a constant code need not be stationary for a fixed decoder. Accordingly, we evaluate trained reconstruction heads empirically.

Theorem 2 (Conditional scalar collapse transition)**.** *Assume an independently justified scalar reduction*

$$\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma), \qquad \kappa > 0, \tag{15}$$

where $b(0) = 0$. Let $f(\sigma) = b(\sigma)/\sigma$ on $(0, \infty)$ tend to zero at both ends, increase strictly to a unique nondegenerate maximum $\lambda_\star = f(\sigma_\star) > 0$, and decrease strictly afterwards. Then $\alpha_c = \lambda_\star/\kappa$. For $0 < \alpha < \alpha_c$ there are two positive equilibria, the smaller unstable and the larger stable; initial values below the smaller one still collapse. At α_c the equilibria coalesce in a saddle-node; for $\alpha > \alpha_c$, every positive trajectory tends to zero.

Proof. The vector field is $\sigma(f(\sigma)-\alpha\kappa)$. The crossings and their stability follow from monotonicity; the nonzero second derivative gives the nondegenerate fold. Above the maximum the field is strictly negative for every $\sigma > 0$, whose only possible limiting equilibrium is zero. □

Lemma 1 (Departure from a primary optimum)**.** *At $G_p = 0$, any nonzero G_a makes the joint gradient nonzero when $\alpha > 0$. This implies departure from the primary optimum, but neither collapse nor irreversibility.*

Corollary 3 (Fixed-dose attenuation)**.** *In Theorem 2, a constant auxiliary multiplier $q \in (0, 1]$ replaces the effective dose by $q\alpha$ and the threshold by α_c/q. Positive attenuation can therefore preserve the positive equilibrium at a fixed dose. Exact deletion removes the contraction term but is not necessary for fixed-dose protection.*

Theorem 3 (First modal contraction versus full linear collapse)**.** *For the linear symmetric model $\dot{z} = (R - \alpha A)z$, assume $R \succ 0$ and $A \succeq 0$ in the same coordinates. Along an eigenvector of $R - \alpha A$ with eigenvalue ν_j, the modal variance obeys*

$$\dot{\varsigma}_j^2 = 2\nu_j\varsigma_j^2. \tag{16}$$

Let

$$T = R^{-1/2}AR^{-1/2}, \qquad \rho_\star = \lambda_{\max}(T). \tag{17}$$

Some mode contracts iff $\alpha\rho_\star > 1$. Every mode contracts exponentially iff $\alpha\lambda_{\min}(T) > 1$. Equality leaves a neutral mode.

Proof. $R - \alpha A = R^{1/2}(I - \alpha T)R^{1/2}$ has the same inertia as $I - \alpha T$ by congruence. Since the generator is symmetric, its eigenvalue signs determine modal growth or decay. □

The scalar model and the symmetric linear model answer different questions. Neither follows from measuring the proxy $\hat{\rho}$ alone. For example, with $R = I$, $A = \mathrm{diag}(2, 1/2)$, and $\alpha = 1$, one mode contracts and another expands. This permits task performance on a surviving direction without preserving the full representation. The NYUv2 result is compatible with this mechanism but does not identify its dynamical eigenvectors.

B Detailed Intervention Comparisons

This section collects the per-method intervention comparisons (Table 4) and the cross-domain diagnostics (Figure 6).

The battery application uses a temporal encoder with a mixer and gated state-space block, feeding SOH, RUL, reconstruction, and distributional heads. The corrected protocol keeps train, validation, and test cells disjoint, selects checkpoints on validation SOH, and separates observed-lifetime MAE from censored shortfall. Across ten initializations, detachment minus joint training is -3.47 SOH percentage points (95% t interval $[-4.20, -2.74]$) and -34.8 observed-RUL cycles ($[-57.8, -11.8]$). These intervals describe initialization variability on the fixed split.

On this MATR protocol (Severson et al., 2019), detachment improves SOH RMSE from 5.57 to 2.10 percentage points and observed-RUL MAE from 81.5 to 46.7 cycles (Table 5); selective filtering gives no further improvement. Joint training has relative covariance trace 3.55, so the gain is a

Table 4: Controlled optimizer comparisons, ten seeds per method. Cells give collapse rate and mean task metric. The optimizer grid uses $(m, \alpha) = (128, 0.1)$ for HAR, $(32, 3000)$ for battery, and $(64, 10000)$ for RadioML. Battery uses the controlled variance auxiliary. Detachment's collapse counts come from separate $m = 64$ controls at doses $100, 100, 3000$; its task metrics belong to those protocols (Tables 5–6). Detachment removes the encoder's dose dependence, but these widths differ from the optimizer grid.

Combiner	HAR acc. ↑	Battery log-SOH RMSE ↓	RadioML acc. ↑
JOINT (plain summation)	1.00 (0.945)	1.00 (0.611)	1.00 (0.256)
conflict resolution			
PCGrad	1.00 (0.585)	1.00 (0.601)	1.00 (0.091)
CAGrad (c=0.5)	1.00 (0.952)	1.00 (0.607)	1.00 (0.091)
Pareto			
MGDA	1.00 (0.200)	1.00 (0.385)	1.00 (0.091)
gradient geometry			
Aligned-MTL	1.00 (0.865)	1.00 (0.609)	1.00 (0.091)
IMTL-G	1.00 (0.419)	1.00 (0.601)	1.00 (0.091)
Nash-MTL	1.00 (0.280)	1.00 (0.552)	1.00 (0.091)
loss weighting			
FAMO	1.00 (0.182)	1.00 (0.613)	1.00 (0.091)
UW (uncert.)	1.00 (0.940)	1.00 (0.611)	1.00 (0.091)
GradNorm	1.00 (0.958)	1.00 (0.611)	1.00 (0.170)
IMTL-L	1.00 (0.947)	1.00 (0.611)	1.00 (0.091)
Detachment (STF-0)	0.0	0.0	0.0

Table 5: Corrected MATR test results: ten initialization seeds on a fixed cell-disjoint split, with validation-based checkpoint selection. $S = \mathrm{tr}\,\mathrm{Cov}(H)$; relative S is the mean same-seed ratio to the SOH-only reference; PR is participation rank. Censored shortfall measures lower-bound violations in cycles. Table 12 reports uncertainty.

Mode	SOH RMSE% ↓	obs. RUL MAE ↓	shortfall ↓	S	relative S	PR
SOH-only	0.61	—	—	1.01	1.00	1.35
JOINT	5.57	81.5	17.60	2.86	3.55	2.49
STF-0	**2.10**	**46.7**	**2.06**	4.77	6.24	1.86
STFCAL	2.71	63.6	5.73	1.40	1.89	2.03
STFCALRANK	4.98	107.6	17.57	617.90	651.77	1.21

negative-transfer result without representation contraction. The rank arm's large trace and low participation rank also illustrate why amplitude and diversity need separate measurement. Checkpoint selection and censoring-aware evaluation changed together relative to the historical protocol; their individual contributions to the changed ranking remain unresolved.

Figure 6 collects the dose sweeps, width crossings, NYUv2 gate boundary, intervention counts, and financial outcomes. Width changes both empirical crossing locations and probe calibration (Yang & Hu, 2021); Appendix D examines their scaling separately.

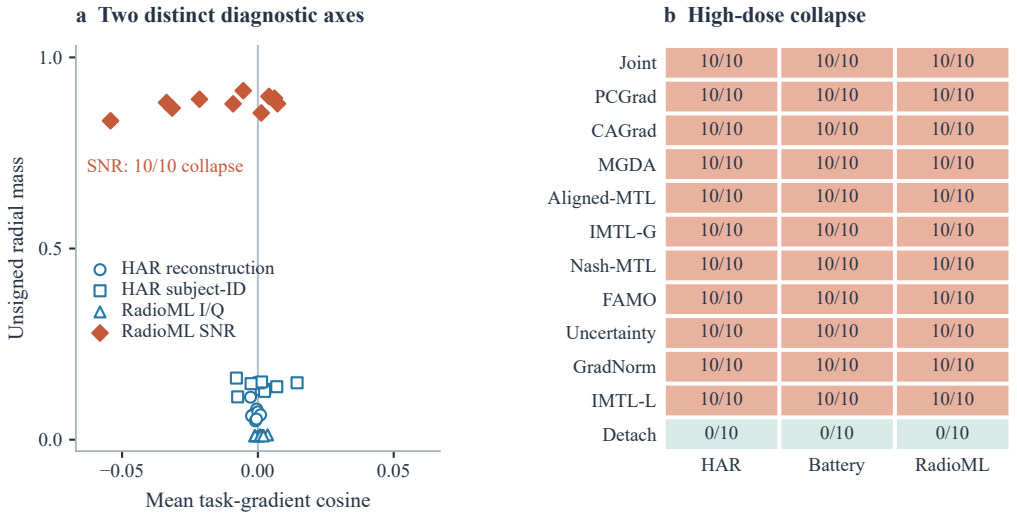


Figure 5: Task-gradient alignment leaves contraction unresolved. (a) Terminal probes of four real auxiliary heads, ten seeds each: mean per-example task-gradient cosine versus unsigned radial loading. SNR has high loading and $10/10$ collapsed endpoints; its signed flux is audited separately. (b) All ten optimizer baselines meet the scatter-collapse criterion in $10/10$ seeds at controlled doses 0.1 (HAR), 3000 (battery), and 10000 (RadioML). Detachment uses separate ten-seed controls. Table 4 gives widths and task metrics.

Table 6: Real auxiliary heads at width 64, ten seeds: mean primary accuracy and collapse rate in parentheses. The listed radial selective arms do not match detachment on high-dose SNR regression. The separate rank-3 arm also avoids collapse in all ten SNR seeds, but reaches only 0.211 accuracy (Figure 8c).

Auxiliary head	α	JOINT	STF-0	STF-HARD	STF-SOFT	STFCAL
HAR input-reconstruction	0.1	.942 (0.0)	.943 (0.0)	.942 (0.0)	.942 (0.0)	.942 (0.0)
HAR subject-ID probe	0.1	.941 (0.0)	.943 (0.0)	.944 (0.0)	.943 (0.0)	.942 (0.0)
HAR subject-ID probe	10	.830 (0.0)	.940 (0.0)	.940 (0.3)	.939 (0.3)	.940 (0.0)
RadioML I/Q-reconstruction	3000	.343 (0.0)	.307 (0.0)	.329 (0.0)	.327 (0.0)	.315 (0.0)
RadioML SNR regression	3000	.205 (1.0)	**.307 (0.0)**	.208 (1.0)	.200 (0.9)	.199 (0.1)

The remaining appendices give the financial application (Appendix C), proofs and scaling analysis (Appendix D), operator and sampling results (Appendix E), protocols (Appendix F), NYUv2 (Appendix G), supplementary experiments (Appendix H), and regularization baselines (Appendix I).

C Financial Ranking Application

The motivating observation came from multi-task equity selection. A production model shared an encoder of A-share factors across pairwise ranking, distributional (MDN (Bishop, 1994)), reconstruction (AE), and view-contrastive (SCARF-style (Bahri et al., 2022)) heads. After deployment, predicted rankings became nearly constant under the unchanged training objective: cross-sectional prediction spreads fell by approximately 93% relative to validation while individual training losses stayed low. The observation prompted us to measure feature and prediction variation alongside loss; identifying its cause required the controlled studies.

The separate financial experiment uses walk-forward training with real distributional, reconstruction, and contrastive heads. Its R5fin records set `fixed_aux=0`, so Table 15 evaluates real auxiliaries.

Table 7: Real HAR reconstruction ($m = R = 64$, ten seeds). Candidate ranks include closed-gate seeds; active ranks average only gate-open seeds. Accuracy and scatter average all ten seeds. PCA and auxiliary-gradient bases activate on identical seeds; these results establish neither superiority over joint training nor globally optimal deletion.

α	basis	gate open	candidate rank	active rank	rel. scatter	accuracy
12	PCA	4/10	57.0	62.00	0.842	0.9492
12	Aux. grad.	4/10	22.6	30.75	0.811	0.9497
30	PCA	8/10	62.8	63.62	0.754	0.9512
30	Aux. grad.	8/10	44.1	47.88	0.746	0.9531
100	PCA	10/10	64.0	64.00	0.630	0.9524
100	Aux. grad.	10/10	59.4	59.40	0.654	0.9520

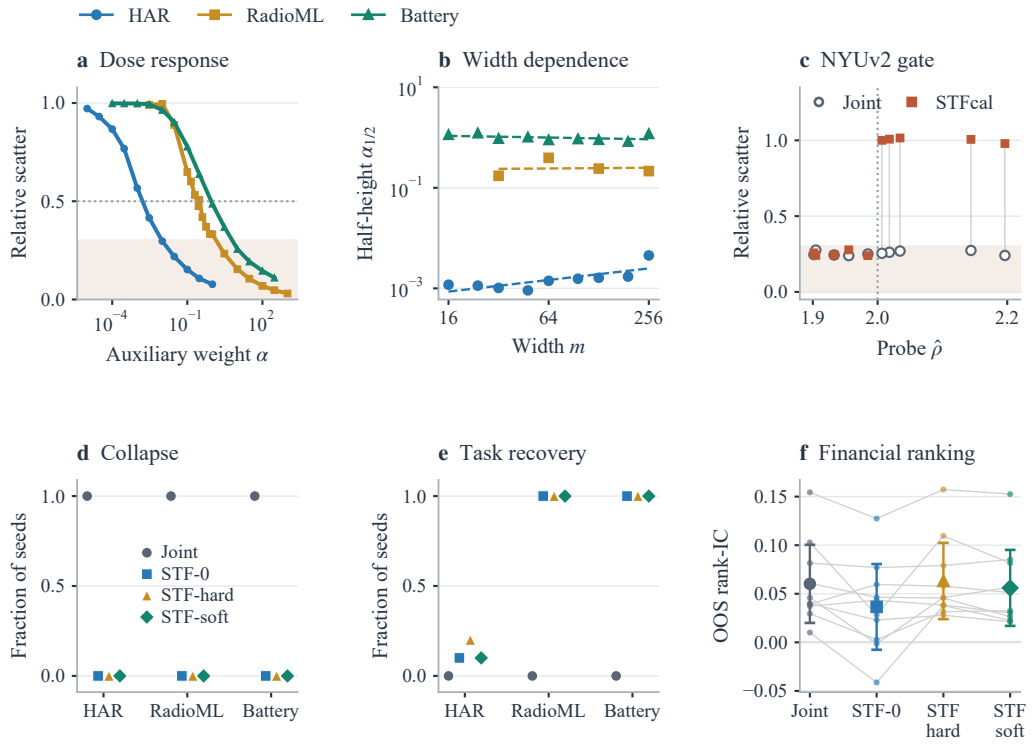


Figure 6: Additional diagnostics across domains. (a) Dose–scatter curves and empirical half-height crossings. (b) Width dependence. (c) Paired NYUv2 gate-boundary outcomes at $\alpha = 0.5$. (d,e) Collapse and task-recovery rates for the controlled interventions. (f) Financial rank-IC, paired by temporal fold. The corresponding tables specify protocols, seed counts, and the separate 0.3 collapse criterion.

Across ten folds and ten initialization seeds, mean out-of-sample (OOS) rank-IC is 0.060 for joint training, 0.036 for detachment, 0.063 for the hard gate, and 0.056 for the soft gate. Averaging seeds within each fold gives paired differences from joint of -0.0237, $+0.0029$, and -0.0042. Two-sided Wilcoxon p values are 0.037, 0.432, and 0.695; none survives Holm correction across the three comparisons. Dependence between temporal folds makes these sensitivity analyses. Detachment is costly in this sample, while a benefit from retaining a particular subspace remains unresolved. Script `finance_fold_audit.py` reproduces the comparisons from all 40 source files.

In the separate synthetic testbed, initialization probes give $\rho = 0.74 \pm 0.19$ over 12 seeds and predicted critical weight $\alpha_c = \langle 1/\rho \rangle = 1.45 \pm 0.37$, consistent with the observed onset. Other domains use the projected-gradient diagnostic of Definition 1, but its numerical value depends on the task, encoder, and calibration time.

D Proofs and Additional Analysis

D.1 Statements Deferred from the Main Text

We formalize several statements discussed in the main text. Theorem numbering continues from the main text.

Proposition 3 (Sufficient conditions for gradient vanishing)**.** *For a differentiable prediction map with Jacobian J and output loss gradient ℓ, the parameter gradient vanishes iff $J^\top \ell = 0$. Sufficient conditions are $\ell = 0$, or identical Jacobian rows together with $\mathbf{1}^\top \ell = 0$. Equal prediction values alone imply neither condition.*

Proof. Apply the chain rule. For a counterexample to the stronger assertion, $\hat{y}_i = \theta x_i$ is constant at $\theta = 0$ but has Jacobian rows x_i. The pairwise logistic loss $\log(1 + e^{-(\hat{y}_1 - \hat{y}_2)})$ has output gradient $(-1/2, 1/2)$ at a tie, so its parameter gradient is $(x_2 - x_1)/2$, generally nonzero. □

Proposition 4 (Scope of positive attenuation)**.** *Under Theorem 2, a fixed positive multiplier q leaves a finite threshold α_c/q but can prevent the transition at a specified dose. A dose-dependent positive multiplier can keep the effective dose subcritical for every α: for $0 < \epsilon < 1$, take $q(\alpha) = (1 - \epsilon)/(1 + \alpha/\alpha_c)$.*

Proof. Substitute $q\alpha$ for α. The proposed choice gives $q\alpha/\alpha_c = (1-\epsilon)\alpha/(\alpha+\alpha_c) < 1$. A positive stable equilibrium remains, although its lower basin boundary persists. Preserving gradient direction therefore does not imply that positive task weighting must fail. □

Proposition 5 (Rank-dilution scaling)**.** *Assume the effective ratio scales as $\rho_\star \asymp (m_\Pi/m)^{\gamma_0}$, $\gamma_0 \geq 1$, and $m_\Pi \propto m^\beta$, $0 \leq \beta < 1$. Then $\alpha_c = 1/\rho_\star$ implies*

$$\gamma \;=\; \gamma_0\,(1-\beta), \qquad \gamma_0 \geq 1, \tag{18}$$

The separable scaling is assumed; rank alone does not imply it. More generally, restoration $\asymp m^a$ and contraction $\asymp m^b m_\Pi^c$ yield $\gamma = a - b - c\beta$, with the displayed form requiring $c = a - b = \gamma_0$. If the toxic norm scales as m_Π^p, then $c = 2p$ only when contraction depends quadratically on that norm and has no additional rank dependence. The equality $\gamma_0 = 2p$ further requires $a - b = 2p$. Neither exponent is identified by the toxic eigenvalue spectrum alone. A fixed effective rank gives $\beta = 0$; a fixed per-head output dimension need not.

Corollary 4 (Composition of rank-one PSD operators)**.** *Let $T = \sum_k \rho_k u_k u_k^\top$, with unit u_k and $\rho_k \geq 0$, in common restoration-normalized coordinates. If the directions are mutually orthogonal, $\lambda_{\max}(T) = \max_k \rho_k$; if aligned, it is $\sum_k \rho_k$. An additional orthogonal direction leaves the maximum unchanged only if its coefficient does not exceed the previous maximum.*

Proof. For orthogonal directions, the u_k are eigenvectors. For aligned directions, the sum is a single rank-one operator. Neither case establishes the geometry of the production model's heads. □

For an explicit scalar example, take $b(\sigma) = \lambda\sigma^2/(1+\sigma^2)$. Then

$$\sigma_\pm = \frac{\lambda \pm \sqrt{\lambda^2 - 4(\alpha\kappa)^2}}{2\alpha\kappa}, \qquad \alpha_c = \lambda/(2\kappa). \tag{19}$$

This verifies the fold and its basin dependence. Radial auxiliary behavior,

$$g_a = \kappa\delta h + o(\|\delta h\|), \tag{20}$$

requires isotropic curvature; stationarity at a constant representation is insufficient. For a general encoder, the Jacobian still determines its effect on variance through (4). In discrete time, linearizing a scalar stable equilibrium (Strogatz, 1994) with derivative $-\ell$ gives

$$e_{t+1} = (1 - \eta\ell)e_t, \qquad 0 < \eta\ell < 2, \tag{21}$$

where e_t is a deviation from that equilibrium, not total representation scatter.

Power-law spectra do not determine the capacity exponent. For eigenvalues $\lambda_i = Ci^{-(1+s)}$ with C independent of rank, $\|A_r\|_F^2 = C^2 \sum_{i=1}^{r} i^{-2(1+s)}$. Hence the Frobenius norm is bounded for $s > -1/2$, grows as $\sqrt{\log r}$ for $s = -1/2$, and as $r^{-s-1/2}$ for $s < -1/2$. A decaying spectrum ($s > -1$) therefore has power exponent at most $1/2$; for the commonly intended $s > 0$ it has exponent zero. The trace instead involves $\sum_i i^{-(1+s)}$, and the operator norm is C for a decreasing spectrum. Rank-dependent normalization $C(r, m)$ changes these conclusions. Thus neither a universal $\gamma_0(s)$ nor $\gamma_0 = 2p$ follows without specifying the norm, normalization, restoration and coupling to dynamics. Concretely, if $\kappa \asymp m^b \|A_r\|^q$, $\|A_r\| \asymp r^p$, and $r \asymp m^\beta$, then $\gamma = a - b - qp\beta$. The restricted form $\gamma_0(1 - \beta)$ requires $a - b = qp = \gamma_0$. Hence $\gamma_0 = 2p$ holds only under the additional quadratic coupling $q = 2$ and matching restoration exponent $a - b = 2p$; these are not implied by the measured width slope.

The original two-head generator has $W_k \in \mathbb{R}^{m \times r}$ with independent entries of variance $1/r$. Its combined span has rank $\min(m, 2r)$ almost surely. For the averaged quadratic loss the sample-normalized Hessian is $A = 2 \sum_k W_k W_k^\top / (2r)$, with $\mathbb{E}A = 2I/r$. At $r = 32$, widths $32, 64, 128, 256$ therefore have ranks $32, 64, 64, 64$. Fixed per-head rank is not fixed combined rank over this range, and changing r also changes curvature. The supplied orthonormal control instead constructs $A = Q \operatorname{diag}(\lambda) Q^\top$ at exactly specified rank and normalization.

The RS1 experiment fixes per-head rank $r \in \{4, 8, 16\}$ across widths $32, 48, 64, 96$, giving combined ranks $8, 16, 32$. At anchor 3×10^{-4}, descriptive half-height slopes are -0.31, 0.45, and 3.02; minimum relative scatter ranges are 0.040–0.068, 0.013–0.018, and 0.0056–0.0132. Using the unpenalized anchor changes the slopes to 0.65, 5.55, and 3.88. The fitted exponent is therefore sensitive to both rank and anchor. The analyzer retains all twelve curves and reports crossings and scatter floors separately.

The orthonormal control constructs the sample-normalized auxiliary Hessian $A = Q \operatorname{diag}(\lambda) Q^\top$ at exact rank r and verifies its amplitude. Three normalizations hold different quantities fixed: $\lambda = 1$ fixes curvature per direction, so the Frobenius norm grows as $\sqrt{r}$; $\lambda = 1/r$ fixes the trace; and $\lambda = 1/\sqrt{r}$ fixes the Frobenius norm. The training grid tests the first two at $m \in \{32, 64, 128\}$, $r \in \{4, 16\}$, seven doses, and three seeds, holding the loss, optimizer, and data generator otherwise fixed. Three observations follow. First, at fixed width and per-direction curvature, the $r = 16$ crossing is lower than the $r = 4$ crossing by factors of 4.8 at $m = 64$ and 228 at $m = 32$. Second, the fitted width slope for $r = 4$ changes from -0.29 under $\lambda = 1$ to $+0.91$ under $\lambda = 1/r$. Third, minimum scatter is 3.5–11% of the healthy reference for $r = 4$, versus 0.4–1.1% for $r = 16$. Thus normalization affects the estimated exponent even when rank is controlled exactly. Each three-width fit has only one residual degree of freedom and cannot identify a precise exponent. The commands `controlled_rank_scan.py --self-test` and `run_controlled_rank_grid.py` verify the construction and reproduce the twelve configurations. The historical squared pairwise hinge has zero margin, so tied scores can attain zero primary loss. These controls concern that objective; they do not establish the same transition for informative ranking objectives. The separate `--primary-loss logistic` option removes this degeneracy, but no unfinished logistic sweep is counted as evidence.

The synthetic testbed uses random toxic projections of rank $\lfloor m/2 \rfloor$ and a low-rank primary task with 2–8 active directions across seeds (mean 5.3). It sweeps 17 logarithmically spaced doses over $[3 \times 10^{-4}, 10]$ at six widths $m \in \{8, \ldots, 256\}$ under a pairwise-ranking loss. The toxic-rank schedule is proportional to width. The relative half-height rule gives $\gamma = 0.07$ across six widths and $\gamma = 0.01$ across the four widths used below. This nearly flat trend is consistent with the restricted law at $\beta = 1$, but does not independently establish that law.

Using a fixed absolute scatter threshold instead gives $\gamma = 1.53$ ($R^2 = 0.91$) on the same curves. At $\sigma_h = 1.5 \times 10^{-3}$, the crossing depends on the width-dependent healthy amplitude: the fixed-rank family has $\sigma_h(0) \propto m^{0.95}$. The absolute rule also gives nearly constant slopes $1.94, 2.02, 2.04, 1.96, 1.89$ across rank schedules $\beta = 0, \ldots, 1$, varying by only 0.15 and largely masking schedule dependence. Using relative crossings, the fixed-per-head-rank family instead gives $\gamma_0 = 2.43$ ($R^2 = 0.90$, four widths). Its combined span can still grow as $\min(m, 2r)$, so this is not an exact fixed-effective-rank estimate.

Both synthetic families extend the dose grid to $\alpha = 0$ (21 points), using four widths $m \in \{32, 64, 128, 256\}$, five seeds, and 800 epochs. The fixed-rank family is the $\beta = 0$ case with constant per-head rank r. A corresponding ten-seed run yields crossings differing by $+3.1\%, +1.8\%, -5.8\%, +19.7\%$ across these widths. This comparison quantifies sensitivity to seed count, including variation in the healthy anchor at larger widths.

Two checks qualify the crossing estimator. First, a half-height crossing can occur without near-total collapse. For fixed per-head rank $r \leq 4$, scatter plateaus at 2–23% of its healthy value; only $r \geq 8$ reaches a floor of at most 0.9%. These floors must be distinguished from the operational 0.3 collapse threshold. Second, the synthetic fits use the first penalized dose, $\alpha = 3 \times 10^{-4}$, as the anchor. The unpenalized value is set by the ranking loss, is shared across toxic ranks, and lies 30–40% above the penalized trend. Varying the anchor over $\alpha \in [0, 3 \times 10^{-3}]$ preserves the decreasing ordering of $\gamma(\beta)$: $\gamma_0 \in [1.55, 3.23]$ at $\beta = 0$, versus $\gamma(1) \in [-0.06, +0.74]$. The ordering is more stable than the fitted coefficients.

The rank-schedule fits test a conditional scaling hypothesis. Given $\alpha_c \asymp (m/m_\Pi)^{\gamma_0}$ and an independently measured $m_\Pi \asymp m^\beta$, substitution gives $\gamma = \gamma_0(1 - \beta)$. Testing the premise requires measuring effective rank separately from the observed slope. The generator audit above specifies how per-head and combined ranks differ.

Crossing rules affect the fitted magnitudes. For HAR, an absolute threshold $T \in [0.20, 0.50]$ gives $\gamma \in [0.66, 0.82]$, compared with 0.66 under the four-width relative rule. Battery requires a grid that reaches a healthy branch: earlier sweeps started at $\alpha = 300$, where relative scatter was already 0.07–0.11, making thresholds left-censored. Extending to $\alpha \in [10^{-4}, 300]$ with five seeds gives a healthy branch at $\alpha \leq 10^{-3}$ (relative scatter 1.000 ± 0.005) and crossings $0.95, 0.89, 0.90, 1.18$ at $m = 32, \ldots, 256$. The fitted slope is $\gamma_{\text{bat}} = 0.10$ ($R^2 = 0.43$), varying over $[-0.10, +0.12]$ across crossing rules. These four widths are consistent with weak width dependence but do not resolve a precise exponent.

Treating each width's crossing as one observation gives four-width OLS 95% intervals $[-0.07, +4.93]$ for the $\beta = 0$ schedule, $[-0.16, +1.49]$ for HAR, $[-0.24, +0.43]$ for battery, and $[-1.16, +1.20]$ for RadioML. None excludes zero. Interior schedules yield positive slopes: 2.57 (CI $[0.87, 4.27]$) at $\beta = 0.25$ and 1.86 ($[0.81, 2.91]$) at $\beta = 0.50$; $\beta = 1$ remains consistent with zero. At the reported anchor, all four remaining schedule intervals contain $\gamma_0(1 - \beta)$, whereas the $\beta = 0.75$ interval excludes it at anchor 10^{-4}. These results support schedule dependence without identifying an anchor-independent coefficient.

The denser SYN8 family supplies 26 completed (m, β) cells, ten seeds each. Since $r(64) = 32$ for every schedule, we fit $\log \alpha_c = c + \gamma_0(1 - \beta) \log(m/64)$. At anchor 3×10^{-4}, the constrained fit gives $\gamma_0 = 2.50$ (nominal OLS 95% CI $[1.94, 3.06]$). Across anchors $\{10^{-4}, 3 \times 10^{-4}\}$ and inclusion/exclusion of $m = 128$, all four conditional intervals exclude zero (lowest bound 1.04). In the unconstrained model $c + ax + b\beta x$, $x = \log(m/64)$, every interval for $a + b$ contains zero and $-b/a = 0.78$–1.03: the linear form is not rejected, not established. These are cell-level fit intervals, not independent-seed uncertainty: the schedules share seeds and the $m = 64$ curves are identical. Joint/anchor-row point estimates range from 1.35 to 2.87. At $m \geq 96$ the sampled curve never resolves the collapsed floor; half-height crossings interpolate the observed decline, not a verified collapse threshold. Thus SYN8 sharpens a conditional trend without supplying a precise physical exponent. `synth_beta_joint.py` reproduces the fits.

Nine-width HAR and battery sweeps use $m \in \{16, 24, 32, 48, 64, 96, 128, 192, 256\}$ with the same driver, seeds, epochs, and dose grids. Shared configurations agree to floating-point precision (maximum difference 2×10^{-16}). HAR gives $\gamma_{\text{HAR}} = 0.38$ (95% CI $[+0.10, +0.67]$, $R^2 = 0.60$, $n = 9$); excluding the two narrowest widths gives 0.60. Battery gives $\gamma_{\text{bat}} = -0.06$ (CI $[-0.17, +0.06]$), consistent with weak width dependence. The denser grid narrows conditional fit intervals but does not eliminate crossing uncertainty. For example, HAR at $m = 32$, $\alpha = 10^{-3}$ has relative scatter 0.5026 ± 0.0524, placing the half-height crossing close to a sampled grid point and making interpolation sensitive to seed variation. We report both four- and nine-width estimates rather than treating either as an identified dynamical exponent.

We also examine probe scaling across encoders. Repeating calibration with a Transformer at $m \in \{32, 64, 128\}$ and ten seeds gives probe exponents -0.93 for HAR ($R^2 = 0.99$), $+0.39$ for battery ($R^2 = 0.87$), and -0.17 for RadioML ($R^2 = 0.79$). The corresponding MLP values are

-1.63 ($R^2 = 0.96$), $+0.67$ ($R^2 = 0.86$), and $+0.28$ ($R^2 = 0.40$). Signs agree for HAR and battery; RadioML is consistent with zero for both encoders, and the 95% intervals overlap in all three domains. This is evidence of broadly similar probe trends in the tested encoders, not architecture invariance. The script `probe_exponent.py` reproduces the analysis.

Probe exponents and scatter-crossing exponents estimate different quantities. Their MLP point estimates agree in sign for RadioML ($+0.28$ versus $+0.02$), but disagree for HAR (-1.63 versus $+0.38$) and battery ($+0.67$ versus -0.06). A half-height crossing locates a dose at which trained scatter reaches half its reference value; the probe estimates $1/\hat{\rho}$ at the warm-up anchor. Agreement within a factor of a few at one width (a factor of 7 for NYUv2 at $m = 64$) does not imply agreement in width scaling. Negative probe slopes do not satisfy the restricted nonnegative exponent assumed in (18), but that assumption is not a general constraint on empirical slopes. The width analysis here therefore reports crossing exponents separately, while Appendix G reports probe values as calibration statistics.

Several sweeps show bimodal per-seed scatter near the empirical crossing. Such outcomes are compatible with a saddle-node model, but can also arise from initialization, finite training time, and optimization. We report the heterogeneity without assigning it a unique dynamical cause.

The original GPU experiments use fixed integer seeds, bf16 autocast with TF32 matrix multiplication, and Adam (lr $= 10^{-3}$, batch 256) on an NVIDIA A100 80 GB. Exceptions are specified with their protocols. Operational collapse means scatter below 0.3 times the same-seed single-task reference; the released recovery rate compares each seed with the mean joint-training metric. Code, seeds, hyperparameters, and per-fold results are included in the repository.

E Generalizations and Finite-Size Theory

Proposition 6 (Fixed attenuation in a specified model)**.** *For $T \succeq 0$ in restoration-normalized coordinates, uniform attenuation $T \mapsto (1-\lambda)T$, $0 \le \lambda < 1$, multiplies the first-modal-contraction threshold by $1/(1-\lambda)$. At fixed dose the smallest λ meeting $\alpha(1-\lambda)\|T\| \le 1$ is $\max(0, 1 - 1/(\alpha\|T\|))$ when $\alpha\|T\| > 0$.*

Proof. All eigenvalues scale by $1 - \lambda$; solve the displayed inequality. This statement concerns a fixed PSD model, not arbitrary signed or nonsymmetric filtered dynamics. □

Theorem 4 (Annihilation by orthogonal compression)**.** *Let $T \succeq 0$, $U = \mathrm{range}(T)$, and $Q = I - P_S$ be an orthogonal projector. Then $QTQ = 0$ iff $U \subseteq S$. Among such deletions the unique smallest deleted subspace is U, and $I - P_U$ preserves its orthogonal complement. This is an algebraic statement about two-sided compression.*

Proof. $QTQ = (T^{1/2}Q)^\top (T^{1/2}Q) = 0$ iff $\mathrm{range}(Q) \subseteq \ker T$, equivalently $U \subseteq S$. A general one-sided backward operator QT need not describe symmetric neural-network dynamics; a signed filter can even reverse contraction, so its operator norm cannot supply the missing implication. □

Corollary 5 (Subspace estimation bound)**.** *If $\|P_U - P_{\hat{U}}\| \le \epsilon$ and $Q = I - P_{\hat{U}}$, then*

$$\|QT\| \le \epsilon\|T\|, \qquad \|QTQ\| \le \epsilon^2\|T\|. \tag{22}$$

Proof. Use $T = P_U T P_U$ and $\|QP_U\| \le \epsilon$ on one or both sides. Whitening must precede this argument if T is restoration-normalized; an angle bound in other coordinates is insufficient. These are upper bounds on residual magnitude, not evidence of a nonzero residual or inevitable failure at a finite dose. □

Theorem 5 (Minimum rank for a PSD compression)**.** *Fix a PSD operator T in specified Euclidean coordinates, with eigenvalues $\kappa_1 \ge \cdots \ge \kappa_d \ge 0$, and a threshold $\tau \ge 0$. Among orthogonal projectors $Q = I - P_S$ satisfying $\|QTQ\| \le \tau$, the minimum deletion rank is*

$$r^\star = \#\{j : \kappa_j > \tau\}.$$

Deleting $S_c = \mathrm{span}\{u_j : \kappa_j > \tau\}$ attains it. Within the class of spectral deletions, S_c also minimizes any inclusion-monotone deleted energy. Uniqueness of the energy optimum requires positive energy in every additionally deleted eigendirection.

Proof. The spectral choice leaves only eigenvalues at most τ. For any $\dim S < r^\star$, dimension counting gives a unit vector $v \in S^\perp \cap S_c$. Its Rayleigh quotient satisfies $v^\top QTQv = v^\top Tv > \tau$, violating feasibility. For spectral deletions feasibility requires containing S_c, giving the energy statement by inclusion. Rotated subspaces can also attain the same minimum rank; no global minimum-energy claim over rotations follows. □

Corollary 6 (Saturation and its limits)**.** *For $\tau = 1/\alpha$, the optimal rank changes in steps as α crosses inverse positive eigenvalues and eventually equals $\dim U$. The limiting projector is $I - P_U$, which equals full detachment only on U, or globally when U is the entire gradient space. Exhausting a finite probe depth does not prove saturation of U.*

The theorem concerns a specified operator, not a mean of gradient-norm ratios. For a linear reconstruction decoder Wh with squared loss, $g_a = W^\top \delta$ (up to loss normalization) and the Hessian is $W^\top W$. The uncentered second moment relevant to normalized residual energy is

$$C_g = \mathbb{E}\left[\frac{g_a g_a^\top}{(\|g_p\| + \varepsilon)^2}\right] = W^\top \mathbb{E}\left[\frac{\delta\delta^\top}{(\|g_p\| + \varepsilon)^2}\right] W.$$

For any rank-r projector P, the squared residual is $\mathrm{tr}((I - P)C_g)$, minimized by the top r eigenvectors of C_g (the Rayleigh–Ritz principle). Its square root upper-bounds the corresponding mean norm by Jensen's inequality. It does not generally minimize that mean itself. Hessian eigenvectors coincide with this basis under isotropic weighted residual covariance; representation PCA requires its leading subspace to coincide with the leading eigenspace of C_g. Commutation gives a common eigenbasis but does not ensure the same ordering. A nonlinear decoder additionally contributes residual-weighted second derivatives to its Hessian. Thus a nearly flat PCA residual curve is consistent with gradient energy outside leading representation components, but does not prove that the reconstruction residual is orthogonal to those components.

Table 9 gives the variance-shrinkage rank sweep. For real HAR reconstruction at $\alpha = 12$, auxiliary-gradient and PCA bases give mean candidate ranks 22.6 and 57.0. Both gates open in four of ten seeds, with active ranks 30.75 and 62.0 at similar aggregate accuracy (Table 7). This comparison isolates basis sensitivity. The artifact also includes the three-seed RMS-basis pilot and its negative accuracy result.

In the ten-seed battery cap32 experiment, `stfcalrank` reaches its cap in every seed and gives SOH RMSE 5.26, versus 5.24 for joint training. The larger cap therefore does not establish minimum-rank protection. A single-seed full-depth audit gives representation covariance traces 3.131, 4.065, 0.906, and 0.821 for joint training, detachment, the radial calibrated arm, and the rank arm, respectively, versus 1.147 for the SOH-only reference. Corresponding normalized SOH prediction variances are 0.00826, 0.01380, 0.01253, and 0.01019, versus 0.01379. These distinct quantities establish neither representation recovery nor constant predictions. Historical checkpoints were selected on the evaluation split and remain exploratory. The corrected cell-disjoint protocol fits preprocessing on training cells and evaluates test data after validation-based checkpoint selection.

Forward and backward-only filters differ even when their Jacobians agree. For fixed U, let $\widetilde{H}_f = H - C_B HUU^\top$ and $\widetilde{H}_b = \mathrm{sg}(H) + D - C_B DUU^\top$, where $D = H - \mathrm{sg}(H)$ and sg stops gradients. Both induce the backward map $G \mapsto G - C_B GUU^\top$, but only $\widetilde{H}_b$ preserves the auxiliary input. Consequently, the loss gradients are evaluated at different forward values and need not produce the same update. Even at full rank, both retain a batch-mean gradient channel, unlike full detachment.

On HAR reconstruction at $\alpha = 100$, $m = R = 64$, paired forward and backward-only filters share warm-up, seed, basis, and selected rank. All ten gates open. Forward filtering gives primary accuracy 0.9507 ± 0.0033, reconstruction MSE 0.05112 ± 0.01083, and relative scatter 0.681 ± 0.039; backward-only filtering gives 0.9478 ± 0.0058, 0.02053 ± 0.00115, and 0.597 ± 0.061. Auxiliary errors use unfiltered features. The paired accuracy difference (backward minus forward) is -0.00285 (95% t interval $[-0.00815, 0.00245]$); reconstruction improves in every seed while scatter decreases. Holm-adjusted Wilcoxon p values across these outcomes are 0.242, 0.00586, and 0.0195. Thus the forward choice changes the auxiliary-fit/scatter tradeoff, with no resolved primary-accuracy advantage. Joint and detached accuracy in the same harness are 0.9461 and 0.9431. A three-seed $\alpha = 12$ pilot, including two closed-gate pairs, is also retained. Scripts `filter_semantics_audit.py` and `summarize_filter_semantics.py` provide the checks and paired records. Probe algebra and eigendecomposition run in fp32, with invariance to enclosing bf16 autocast checked separately.

A complementary experimental intervention uses $v = \nabla_\theta V$, proposed update $u_0 = G_p + \alpha G_a$, and slack $h = V - V_{\min}$. For $k > 0$ and $v \neq 0$, the closest update satisfying $v^\top u \leq kh$ is

$$u^\star = u_0 - \frac{(v^\top u_0 - kh)_+}{\|v\|^2} v.$$

The KKT conditions prove this Euclidean halfspace projection. Under a well-defined continuous flow, $\dot{h} \geq -kh$ preserves feasibility on this *fixed batch*. At $v = 0$ feasibility requires $kh \geq 0$. Discrete updates require checking the actual post-step variance: the supplied SGD implementation backtracks and restores parameters on failure. The argument covers neither unseen data nor Adam without modeling its applied update. This applies established constrained-gradient ideas (Allibhoy & Cortés, 2024), without claiming halfspace projection as novel; empirical benefits remain to be established.

Table 8 compares deletion with positive attenuation on UCI-HAR ($m = 64$, sixty epochs). The calibrated multiplier uses the unprojected mean gradient ratio, so its $c = 1$ is a calibration setting rather than an identified dynamical threshold. The $c = 0.1$ setting is competitive with the tested fixed multipliers. This exploratory grid has three seeds per configuration and includes non-finite runs. At $\alpha = 300$, seed 0 diverges for every arm, including detachment, whose auxiliary heads still train during detached warm-up. We record numerical divergence separately from representation collapse.

Table 8: Matched attenuation on UCI-HAR ($m = 64$, sixty epochs). Accuracy and calibrated multiplier q are averaged over completed seeds; `div` indicates non-finite evaluation. There are three seeds per configuration, except at $\alpha = 300$, where seed 0 diverges for every arm and is excluded. Bold identifies the best fixed setting and the best calibrated setting when it exceeds the best fixed accuracy.

	accuracy			multiplier q		
arm	30	100	300	30	100	300
plain summation	.9293	`div`	`div`	1	1	1
detachment (STF-0)	.9276	.9276	`div`	0	0	0
fixed 10^{-4}	.9275	.9276	.9272	—	—	—
fixed $3{\cdot}10^{-3}$	.9278	**.9304**	.9286	—	—	—
fixed 10^{-2}	**.9301**	.9299	**.9293**	—	—	—
fixed $3{\cdot}10^{-2}$	**.9301**	.9294	.9265	—	—	—
fixed 10^{-1}	.9295	.9291	.9260	—	—	—
fixed $3{\cdot}10^{-1}$	.9242	.8842	`div`	—	—	—
calibrated $c = 0.1$	.9299	**.9306**	.9289	.076	.016	.004
calibrated $c = 0.3$	.9261	.9285	.9281	.228	.049	.012
calibrated $c = 1.0$	.9257	.9266	.9276	.759	.163	.039

Proposition 7 (Sampling fluctuation of a frozen statistic)**.** *For independent identically distributed scalar diagnostics X_i with finite variance σ_X^2, $\hat{\rho} = B^{-1}\sum_i X_i$ has variance σ_X^2/B and obeys a central limit theorem. Under a normal approximation the probability that the* gate $\alpha\hat{\rho} > 1$ *opens is* $\Phi(\sqrt{B}(\rho - 1/\alpha)/\sigma_X)$. *This is not the probability of training collapse. Batch-dependent centering or fitted bases require additional analysis or an independent frozen calibration sample.*

Proof. Use variance additivity and the central limit theorem. For a ratio A/P, the delta-method variance includes $-2\mu_A \operatorname{Cov}(A, P)/\mu_P^3$ as well as the two marginal variance terms. Mean per-example ratios and ratios of means are distinct estimands. □

Proposition 8 (Conditional calibration bound)**.** *Assume independent $X_i \in [0, M]$ with common mean ρ and population variance at most v_0. Bernstein's inequality gives*

$$\mathbb{P}(|\hat{\rho} - \rho| \geq \varepsilon) \leq 2\exp\left[-\frac{B\varepsilon^2}{2v_0 + 2M\varepsilon/3}\right]. \tag{23}$$

Thus an error probability at most δ follows if

$$B \geq (2v_0 + 2M\varepsilon/3)\varepsilon^{-2}\log(2/\delta). \tag{24}$$

Table 9: Selected rank versus auxiliary dose on UCI-HAR ($m = 64$, five seeds, probe depth $R = 64$). Candidate ranks are given for representation-PCA and auxiliary-gradient bases. A closed gate removes no directions; full detachment blocks all 64. Relative scatter is $\mathcal{S}_h/\mathcal{S}_{single}$ for `stfcalrank`. The PSD optimum in Theorem 5 additionally requires identifying the operator and basis.

α	gate	$r^\star$ (principal)	$r^\star$ (aux. basis)	rel. scatter (rank arm)	rel. scatter (JOINT)
0.003	0	–	–	0.404	0.402
0.01	0	–	–	0.287	0.280
0.03	1	1.6	1.6	1.007	0.200
0.1	1	3.8	3.8	1.026	0.138
0.3	1	4.2	4.2	1.018	0.101
1.0	1	11.6	11.6	0.978	0.070
3.0	1	25.8	25.8	0.987	0.051
10	1	39.8	40.0	0.986	0.035
30	1	47.4	47.8	0.968	0.025
100	1	61.6	61.0	0.950	0.016

Table 10: Historical battery results with evaluation-selected checkpoints, retained as exploratory comparisons. The legacy RUL score combines errors against observed lifetimes and censored lower bounds. Reference rows use four seeds; other rows use ten. SOH and legacy RUL are separate outcomes; single-task rows provide references.

	MATR		NASA	
Mode	SOH RMSE (%)	legacy RUL score	SOH RMSE (%)	legacy RUL score
single-task reference	0.51	694.6	9.17	18.5
JOINT (aux heads)	5.24 ± 0.54	209.0 ± 59.8	11.42 ± 0.85	17.5 ± 1.1
JOINT (toxifier)	1.23 ± 0.44	306.6 ± 48.2	9.66 ± 0.44	17.6 ± 2.4
STF-0	1.68 ± 0.41	574.7 ± 80.7	9.35 ± 0.55	18.0 ± 2.0
STF-0W (tri-weight)	1.54 ± 0.44	541.2 ± 91.3	9.40 ± 0.62	18.6 ± 1.4
STFCAL ($\hat{\rho}$=17.4/0.40)	1.77 ± 0.47	419.8 ± 33.6	11.40 ± 0.78	18.3 ± 1.7

All MATR rows: censoring-aware RUL supervision, all 185 cells retained; bf16 autocast training, fp32 evaluation.

A nonzero primary-gradient denominator alone does not provide a uniform bound M. Population bounds cannot be replaced by observed sample moments without a separate valid empirical-concentration result.

Proof. Apply the bounded-variable Bernstein inequality to the centered independent sum and solve for B. □

Corollary 7 (Certifying a frozen gate or nested rank rule)**.** *Under Proposition 8, a gate with population margin $d = |\rho - 1/\alpha| > 0$ has error probability at most $2\exp[-Bd^2/(2v_0 + 2Md/3)]$. For $R + 1$ fixed residual statistics with uniformly positive margins, a union bound controls all their threshold decisions and hence the selected nested rank. This does not identify the statistics with Hessian eigenvalues, certify an adaptively fitted basis without conditioning, or certify a subsequent training trajectory.*

Proof. A wrong threshold decision requires an estimation error at least its margin; apply the bound and then the union bound. Strict margins exclude equality cases. □

Proposition 9 (Local linear noise floor)**.** *For deviations $e_{t+1} = ae_t + \eta\xi_t$ from a fixed equilibrium, assume $|a| < 1$ and independent mean-zero innovations of variance D. The stationary variance is $\eta^2 D/(1 - a^2)$.*

Proof. The second-moment recursion is $\mathbb{E}e_{t+1}^2 = a^2\mathbb{E}e_t^2 + \eta^2 D$; sum the convergent geometric series. This is variance of deviations, not total representation scatter. Divergence of a local linear

Table 11: Frozen-anchor sampling experiment on HAR ($m = 64$, four detached warm-up epochs, five seeds). Each seed uses 2000 batches of size B, sampled with replacement. Standard deviation and CV describe variation across batches at a fixed anchor. "Meas." is the observed frequency of $|\hat{\rho} - \rho| > \eta\rho$; "bound" is the Bernstein plug-in envelope from (23), using estimated probe moments ($\kappa^2 = 2\rho^2 c_0^2 + \frac{2}{3} R_{\max}\varepsilon$). Values above one are uninformative probability bounds. The envelopes exceed observed frequencies but are not population confidence guarantees. The fitted CV slope is -0.500 ($R^2 = 0.96$), and the mean slope is $+0.006$. The norm of $\sum_i g_a^{(i)}$ is at most 4×10^{-8} of $\sum_i \|g_a^{(i)}\|$. Raising nominal confidence from 90% to 99.9% multiplies the sample-size expression by $\ln(2/10^{-3})/\ln(2/0.1) = 2.54$.

				$\eta = 0.2$		$\eta = 0.1$	
B	$\hat{\rho}$	sd($\hat{\rho}$)	CV($\hat{\rho}$)	meas.	bound	meas.	bound
16	74.30	69.52	0.947	0.820	1.94	0.911	1.98
32	76.69	50.40	0.664	0.740	1.90	0.866	1.97
64	77.03	36.00	0.471	0.657	1.83	0.836	1.94
128	77.00	25.34	0.332	0.526	1.71	0.756	1.90
256	77.74	18.24	0.237	0.378	1.49	0.657	1.83
512	77.71	12.92	0.168	0.221	1.15	0.538	1.69
1024	77.97	9.20	0.119	0.094	0.70	0.384	1.45
2048	78.00	6.34	0.082	0.022	0.29	0.218	1.07
4096	77.90	4.54	0.059	0.003	0.06	0.095	0.61

approximation as $|a| \to 1$ does not prove a universal ordering of half-height and binary-collapse thresholds in a nonlinear experiment. □

At a frozen HAR anchor, 2000 resampled batches per size over five seeds give slopes -0.500 for probe CV, -0.494 for standard deviation, and $+0.006$ for the mean. Table 11 compares observed fluctuations with plug-in envelopes based on estimated variance and range. Shared batch centering and estimated moments limit these envelopes to diagnostic use. The records also distinguish mean per-example ratios from ratios of batch-mean gradients: the latter can vanish for centered variance losses.

E.1 Prediction Variance Normalization and Target z-Scoring

For $\hat{y} = w^\top h + b$, shrinking centered features by $1/c$ and expanding w by c preserves predictions. Shrinking both changes the regression loss, and constant predictions generally cannot fit varying targets. This scale ambiguity motivates reporting prediction variance and across-example feature covariance alongside task error, and testing prediction normalization and target scaling as controls.

The historical regression recipe standardizes targets and divides predictions by a detached batch standard deviation. This changes optimization and can reduce sensitivity to scale, but detached normalization does not implement the derivative of a truly scale-invariant loss. Softmax also does not normalize logit scale: changing that scale changes probabilities. These operations therefore require empirical ablations, with scaling statistics fit only on training data.

F Additional Tables and Protocol

The corrected battery protocol uses cell-disjoint train/validation/test partitions: 130/27/28 MATR cells and 24/4/5 NASA cells. MATR test data contain 1069 observed and 1797 censored cycles; all 344 NASA test cycles are observed. Variation is over initializations on these fixed splits. Checkpoints minimize validation SOH, prioritizing that task. Participation rank is $(\operatorname{tr}\Sigma)^2/\operatorname{tr}(\Sigma^2)$; prediction variance uses squared SOH percentage points. Trace measures amplitude and participation rank measures spectral concentration; both depend on the encoder–head parameterization. Script `current_evidence_audit.py` checks split disjointness, seed pairing, and source hashes and regenerates the table. The remaining tables give datasets, historical censoring and head comparisons, auxiliary-weight sweeps, and the training protocol.

Table 12: Corrected battery test results. Means ± population standard deviations over ten initialization seeds on one fixed cell split. RUL is observed-only MAE; S is across-example covariance trace, PR is participation rank, and V_{pred} is SOH prediction variance in squared percentage points. The SOH-only head does not train RUL.

Data	Mode	SOH RMSE (%)	RUL MAE	S	PR	V_{pred}
MATR	SOH-only	0.61 ± 0.19	—	1.01 ± 0.50	1.35 ± 0.14	141.67 ± 5.92
MATR	JOINT	5.57 ± 0.48	81.54 ± 26.84	2.86 ± 0.57	2.49 ± 0.65	66.10 ± 8.59
MATR	STF-0	2.10 ± 0.62	46.73 ± 11.32	4.77 ± 1.33	1.86 ± 0.22	109.50 ± 13.01
MATR	STFCAL	2.71 ± 0.54	63.59 ± 19.43	1.40 ± 0.44	2.03 ± 0.20	99.26 ± 9.69
MATR	STFCALRANK	4.98 ± 1.09	107.58 ± 31.10	617.90 ± 489.76	1.21 ± 0.25	70.11 ± 20.06
NASA	SOH-only	12.66 ± 0.99	—	3.39 ± 1.57	2.59 ± 0.74	183.51 ± 44.46
NASA	JOINT	14.43 ± 0.64	20.05 ± 1.59	5.41 ± 0.85	3.25 ± 0.49	127.49 ± 24.32
NASA	STF-0	12.74 ± 1.20	19.43 ± 2.66	4.38 ± 1.28	3.16 ± 0.58	135.70 ± 37.07
NASA	STFCAL	14.47 ± 1.22	19.20 ± 2.40	4.55 ± 1.42	2.75 ± 0.62	109.72 ± 24.59
NASA	STFCALRANK	14.47 ± 1.22	19.20 ± 2.40	4.55 ± 1.42	2.75 ± 0.62	109.72 ± 24.59

Algorithm 2 STF training protocol (per domain, per configuration)

1: **Input:** domain data $\mathcal{D}$, encoder width m, toxicity schedule $\{\alpha\}$, seeds
2: **Phase 1 – health anchor:** train the single-task encoder ($\alpha = 0$), record healthy scatter $\mathcal{S}_{single} = \mathcal{S}_h(\theta^*)$ and healthy spectral ratio ρ_0.
3: **Phase 2 – calibration:** sweep α with mode JOINT; record ρ, relative scatter $\mathcal{S}_h/\mathcal{S}_{single}$, collapse flag; locate α_c by the threshold crossing.
4: **Phase 3 – repair:** at a selected dose, train each arm on $L_p(H) + \alpha L_a(\widetilde{H})$, using its specified forward map $\widetilde{H}$ (or detached H for STF-0); backpropagate this loss and monitor the gradient diagnostic and relative scatter.
5: **Report:** collapse rate, recovery rate, ρ, $\mathcal{S}_h/\mathcal{S}_{single}$.

On NASA, the corrected protocol gives SOH RMSE 14.43 for joint and 12.74 for detachment, against the SOH-only reference 12.66. The paired difference is -1.69 percentage points (95% t interval $[-2.54, -0.83]$, nine of ten seeds improve). Observed-RUL MAE changes from 20.05 to 19.43 cycles, but the paired interval $[-3.37, 2.13]$ includes zero (exact two-sided Wilcoxon $p = 0.770$). The calibrated gate stays closed in all ten seeds: its two arms coincide with each other (14.47 SOH, 19.20 RUL), but not with joint because their encoder warm-up is detached. NASA has only five test cells; initialization intervals cannot establish population generalization.

In the historical ten-seed MATR experiment, the radial calibrated arm improves SOH over joint training (1.77 ± 0.47 versus 5.24 ± 0.54), while detachment reaches 1.68 ± 0.41. The cap32 rank arm remains near joint (5.26), so the radial result does not validate minimum-rank protection; Corollary 6 locates the saturation limit at full-width deletion. Censored RUL requires separate observed-error and lower-bound-shortfall metrics; a lower legacy MAE need not indicate better lifetime prediction. NASA's closed gate and its discrepancy from detachment further limit the diagnostic. Because these historical checkpoints were evaluation-selected, the corrected validation protocol provides the primary battery evidence.

A four-seed MATR ablation removes each weighting factor in turn. Omitting the rate factor gives fast-charge SOH RMSE 0.95; omitting the temperature factor gives extreme-temperature SOH RMSE 2.35. These results support the intended association between factors and operating regimes. A sweep over $\omega \in \{0.5, 1, 2\}$ evaluates sensitivity around the chosen value $\omega = 1$.

Table 24 reports the composition experiment. On HAR, accuracy changes from 0.943 with filtering alone to 0.933 with PCGrad and 0.940 with CAGrad. Relative scatter is at most 0.15 without filtering, compared with 0.97–1.14 after filtering. These measurements support compatibility for the tested filter-then-combiner order.

The synthetic testbed uses 17 log-spaced doses, six widths, and random projections of rank $\lfloor m/2 \rfloor$. Its near-flat relative-crossing slope ($\gamma = 0.07$) contrasts with the absolute-threshold estimate (1.53), as analyzed above. In RadioML, a five-seed sweep over $\alpha \in [3 \times 10^{-3}, 1]$, refined near the transition,

Algorithm 3 Released calibrated forward filters: radial STFCAL and capped STFCALRANK

1: **Input:** data $\mathcal{D}$, encoder width m, operating weight α, probe depth R, warm-up length W
2: **Warm-up:** train θ with the auxiliary branch detached ($\nabla_\theta L_a \equiv 0$) for W epochs; record the healthy anchor $\mathcal{S}^\star$ of (10)
3: **Probe:** obtain per-example gradients on an unfiltered frozen batch, radial projectors $\Pi_{c,i}$, and principal basis U. With $\varepsilon = 10^{-8}$, compute $\hat\rho = B^{-1}\sum_i \|\Pi_{c,i} g_a^{(i)}\|/(\|g_p^{(i)}\| + \varepsilon)$ and $\hat\rho_{\mathrm{res}}[r] = B^{-1}\sum_i \|(I - U_r U_r^\top) g_a^{(i)}\|/(\|g_p^{(i)}\| + \varepsilon)$ for $r = 0, \dots, R$.
4: **Calibrate:** $g \leftarrow \mathbf{1}[\alpha\hat\rho \ge 1]$; $\mathcal{R} \leftarrow \{r \le R : \alpha\hat\rho_{\mathrm{res}}[r] < 1\}$; choose $r \leftarrow \max(1, \min \mathcal{R})$ if nonempty, else $r \leftarrow R$. Record whether a feasible rank exists.
5: **Latch:** freeze g, r, U. If $g = 0$, pass H to the auxiliary head. Otherwise, STFCALRANK passes $H - C_B H U_r U_r^\top$; STFCAL passes $h_i - \langle h_i - \bar h, u_i\rangle u_i$, $u_i = (h_i - \bar h)/\max(\|h_i - \bar h\|, \varepsilon)$. Differentiate the resulting auxiliary loss through this map.
6: **Report:** gate, r, $\hat\rho$, $\mathcal{S}_h/\mathcal{S}^\star$

Table 13: The sixteen controlled toxifier cells behind Figure 7a. Every training auxiliary is a fixed projected-variance toxifier; the probe uses an isotropic variance surrogate. $\alpha_{\mathrm{probe}} = 1/\bar\rho$ is the inverse mean warm-up probe, while $\alpha_{1/2}$ interpolates the mean dose–scatter curve at relative scatter 0.5. Their ratio is a calibration-transfer diagnostic. Each probe value is a ten-seed mean of the archived warm-up records.

Domain	Head	Width m	Probe $\bar\rho$	α_{probe}	$\alpha_{1/2}$	Ratio
HAR	toxifier	32	30.2	3.32×10^{-2}	1.02×10^{-3}	32.5
HAR	toxifier	64	110.2	9.07×10^{-3}	1.41×10^{-3}	6.4
HAR	toxifier	128	552.0	1.81×10^{-3}	1.63×10^{-3}	1.1
HAR	toxifier	256	1504.9	6.65×10^{-4}	4.52×10^{-3}	0.15
Battery	toxifier	32	1.10	9.07×10^{-1}	9.51×10^{-1}	0.95
Battery	toxifier	64	0.71	1.42	8.95×10^{-1}	1.58
Battery	toxifier	128	0.85	1.18	9.03×10^{-1}	1.31
Battery	toxifier	256	0.28	3.56	1.18	3.01
RadioML	toxifier	32	0.14	7.13	1.73×10^{-1}	41.1
RadioML	toxifier	64	0.08	11.9	3.95×10^{-1}	30.3
RadioML	toxifier	128	0.05	19.8	2.41×10^{-1}	82.2
RadioML	toxifier	256	0.04	24.9	2.16×10^{-1}	115.6
NYUv2	toxifier	32	2.26	4.43×10^{-1}	1.14×10^{-1}	3.9
NYUv2	toxifier	64	2.01	4.98×10^{-1}	6.89×10^{-2}	7.2
NYUv2	toxifier	128	1.64	6.11×10^{-1}	3.56×10^{-2}	17.1
NYUv2	toxifier	256	1.29	7.78×10^{-1}	2.39×10^{-2}	32.6

gives crossings $\{0.17, 0.39, 0.24, 0.22\}$ at widths $\{32, 64, 128, 256\}$. The eightfold width range changes the crossing by at most a factor of 2.3, much less than the factor of eight implied by $\gamma = 1$. At $m = 64$, per-seed crossings span 0.29–0.54. Together with the near-zero fit R^2, this variation limits inference about the width slope.

The repository supplies the scripts and recorded results listed in its reproduction manifest; data-dependent reruns additionally require the stated datasets and preprocessing. New audits are identified separately from historical experiments. The drivers `cross_domain_mtl.py`, `battery_bmtl_v3.py`, and `nyu_bmtl.py` implement the STF family for the unified, battery, and dense-perception experiments. Algorithm 3 specifies the calibrated forward maps; each driver records application-specific details. PCGrad, CAGrad, and FAMO share these experiment pipelines, including the composition study. The remaining methods (MGDA, Aligned-MTL, Nash-MTL,

Table 14: Four scientific domains and the financial discovery application. Sample counts describe the listed experimental datasets; individual protocols specify their partitions. Controlled scientific experiments use variance-shrinkage auxiliaries. The financial application uses real heads, and the battery and NYUv2 application studies also evaluate practical reconstruction heads.

Domain	Signal	Primary task	Samples	Metric
HAR (UCI-HAR)	motion (561-d)	6-class activity	7,352 / 2,947	accuracy
RadioML (2016.10A)	I/Q (256-d)	11-class modulation	110,000 / 110,000	accuracy
Battery (NASA+MATR)	charge/discharge	SOH + RUL	18,696 / 3,210	RMSE / MAE
Finance (A-share)	1,680-d factors	5-d return rank	800 stocks $\times$ 2,059 d	rank-IC
NYUv2	RGB-D (144$\times$192)	13-class seg + depth + normals	795 / 654 scenes	pxacc / mIoU / RMSE / angle

Table 15: Financial application: mean OOS rank-IC $\pm$ population s.d. over 100 fold-seed cells (ten temporal folds, ten initialization seeds). Cells sharing a fold are dependent. After averaging seeds within folds and correcting three paired comparisons, no method differs significantly from joint training. These records use real auxiliary heads without the fixed variance toxifier.

Mode	OOS rank-IC	vs. joint
JOINT	$+0.060 \pm 0.055$	—
STF-0 (`.detach()`)	$+0.036 \pm 0.056$	-39%
STF-HARD	$+0.063 \pm 0.052$	$+5\%$
STF-SOFT	$+0.056 \pm 0.053$	-7%

IMTL-G, IMTL-L, GradNorm, and uncertainty weighting) use `moo_combiners.py`, which ports kernels from the authors' reference implementations. Comparisons use the recorded optimizer and schedule with published method defaults; batch-size exceptions are stated below.

JSON job manifests specify each experimental configuration by domain, width, mode, and seed, together with its command, output path, and log path. A deterministic runner skips existing outputs. The manifests cover collapse scans, width sweeps, filter composition, real auxiliary heads, battery weight sweeps, calibrated validation, NYUv2 experiments, and the three-domain multi-task baseline comparison. Experiments use fixed integer seeds, with counts reported for each study. Where available, JSON records retain individual seed outcomes; aggregate-only historical records are identified in the artifact audit.

The original unified runs use an NVIDIA A100 80 GB, Adam ($\mathrm{lr} = 10^{-3}$), bf16 autocast, and TF32 matrix multiplication. Calibration runs in fp32 outside autocast on detached features. Batch size is 256 for the unified protocol, real-head comparison, regularizer sweep, and MATR application; 128 for NASA conflict baselines versus 256 for its plain-summation runs; 32 for NYUv2; and 512 for financial walk-forward training. The NASA batch-size difference qualifies those historical comparisons. A unified-protocol configuration takes a few GPU-minutes; the battery application uses 150 epochs per configuration. Corrected battery runs use Adam at 2×10^{-3}, the parameter-constraint pilot uses SGD, and exact-rank controls use AdamW on CPU. Individual scripts and JSON configurations record these optimizer differences. Calibration records contain each seed's $\hat{\rho}$, residual statistics $\hat{\rho}_{\mathrm{res}}[0..R]$, gate, and selected rank. The script `calib_cert_b.py` repeats probes on a frozen healthy anchor at different batch sizes to produce Table 11.

Table 19 measures per-epoch training cost on UCI-HAR at four widths. We report the median of per-repetition median epoch times over ten repetitions, exclude epoch 0, and synchronize the device within each timed epoch. Runs begin after three consecutive idle-device readings, and method order rotates between repetitions to reduce timing drift. Contention materially affects the result: a non-idle run measured the reference at $0.180\,\mathrm{s}$ rather than $0.099\,\mathrm{s}$ and FAMO overhead at 0.0% rather than approximately 31%.

On the idle device, hard and soft filtering cost at most 21.8% on HAR. At $m = 256$, their overheads are 9.5% and 12.2%, respectively. For the tested radial implementation, projection costs $O(Bd)$ compared with the encoder's $O(Bd^2)$; this accounting does not include arbitrary full-rank eigendecompositions. Detachment and calibrated radial filtering reduce measured time by up to 65.0% and 47.8%. PCGrad, CAGrad, and FAMO cost 21.1–41.6%, 38.5–75.2%, and 19.6–31.1%, respectively. Across five domains and four widths, spectral-variant overhead is 6.3–26.0%, versus 5.8–98.2% for

Table 16: Historical MATR censoring ablation: SOH RMSE (%) and legacy RUL MAE (cycles), with leave-cell-out evaluation, $d = 128$, bf16 training, and fp32 evaluation. Legacy rows use four seeds; the censoring-aware row uses ten. Legacy supervision treats censored bounds as exact targets; the EOL-only variant additionally excludes $96/185$ cells, leaving 14 test cells. Censoring-aware supervision applies a one-sided hinge to the 59.2% censored samples and retains all cells. Evaluation-selected checkpoints and legacy RUL scoring limit interpretation.

	SOH RMSE (%)		RUL MAE (cyc)		
Mode	JOINT	STF-0	JOINT	STF-0	cells used
legacy (RUL=0 labels)	5.56 ± 0.17	2.05 ± 0.38	94.1 ± 5.7	97.0 ± 15.3	185
legacy + `--eol-only`	1.53 ± 0.13	0.53 ± 0.03	77.5 ± 2.3	76.6 ± 4.1	89
censoring-aware	5.24 ± 0.54	1.68 ± 0.41	209.0 ± 59.8	574.7 ± 80.7	185

The legacy rows reproduce the `nocens` supervision, every other setting held fixed.

Table 17: Historical battery-head comparison at $\alpha = 10$ (five seeds): SOH RMSE (%), legacy all-sample RUL MAE (cycles), and "scatter" $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$ (sample correction $d - 1$, no centering across examples), not (10). PCGrad, FAMO, and CAGrad yield higher SOH error than detachment in both datasets. Scatter alone does not establish operational collapse; Table 4 reports separate controlled-toxifier experiments. These evaluation-selected results are exploratory.

	MATR			NASA		
Combiner	SOH RMSE (%)	RUL MAE	scatter	SOH RMSE (%)	RUL MAE	scatter
single-task reference	0.51	694.6	–	9.17	18.5	–
JOINT (sum, plain)	6.43 ± 0.19	208.8	0.126	13.33 ± 1.07	18.7	0.214
PCGrad	4.02 ± 0.98	172.9	0.094	11.89 ± 0.73	18.6	0.164
FAMO	10.10 ± 2.17	295.3	0.435	16.02 ± 0.50	32.4	0.474
CAGrad	6.33 ± 0.45	198.0	0.166	13.04 ± 0.75	17.7	0.198
STF-0 (detach)	1.83 ± 0.51	623.4	0.198	8.81 ± 0.72	17.1	0.225

the comparison methods. Hard and soft filtering are cheaper than PCGrad and CAGrad in every measured configuration; FAMO is an exception to the broader ordering on CelebA and NYUv2 at $m = 256$. These are empirical timings for the specified implementations.

G NYUv2: Dense Perception Beyond Time Series

NYUv2 (Silberman et al., 2012; Ladický et al., 2014) contains 795 training and 654 test scenes with semantic segmentation (13 classes), depth, and surface normals. Images are resized to 144×192; segmentation pixels labeled -1 and zero-norm normal targets are masked. The convolutional encoder uses four stride-2 blocks with batch normalization and ReLU ($3 \to 32 \to 64 \to 128 \to 256$ channels), followed by a $d = 128$ representation and linear task heads. Losses are cross-entropy, L_1 depth error, and cosine normal error. We report final-epoch pixel accuracy, mIoU, depth RMSE/MAE, and mean angular error over ten seeds, without test-based checkpoint selection. The *proj* auxiliary is a fixed random variance-shrinkage projection to $d/4$, evaluated at $\alpha = 1$. The *ae* auxiliary is a trainable three-layer input-reconstruction head, evaluated at $\alpha = 250$, or 3.4 times its probe-based threshold. A separate unified-protocol experiment uses an MLP and a 9×12 coarse depth grid (108 outputs), with the optimizer, probe, and collapse criterion of Section 4.1.

After four detached warm-up epochs, the unified MLP probe gives $\hat{\rho} = 2.26 \pm 0.16$, 2.01 ± 0.10, 1.64 ± 0.09, and 1.29 ± 0.05 at widths $32, 64, 128, 256$. Per-seed ranges are 2.06–2.65, 1.90–2.20, 1.51–1.77, and 1.22–1.38. The inverse-mean calibration values are 0.44, 0.50, 0.61, and 0.78, yielding a probe-based width slope of 0.27 ($R^2 = 0.98$, 95% CI $[+0.15, +0.40]$).

This is not the scatter-crossing exponent. Half-height crossings on the same widths yield $\gamma = -0.77$; varying the crossing level from 0.3 to 0.7 gives $[-0.94, -0.73]$ with $R^2 \geq 0.99$, whereas HAR remains positive ($+0.34$ to $+0.46$). At $m = 64$, the crossing and calibration values are 0.069 and 0.50, a factor of 7 apart. Extending the grid four decades lower restores relative scatter to 0.99–

Table 18: Historical MATR auxiliary-weight sweep (seed 0): SOH RMSE (%) and legacy RUL MAE (cycles), with censoring-aware supervision and bf16 training/fp32 evaluation. Joint training uses reconstruction/distributional weight λ_{aux}; detachment makes encoder learning independent of this weight. SOH error generally increases with dose, although the $d = 32$ curve is not strictly monotone. Single-task rows are width-matched references; legacy RUL scores and evaluation-selected checkpoints are not independent lifetime-error estimates.

		SOH RMSE (%)↓			RUL MAE (cyc)↓		
Mode	λ_{aux}	d=32	d=128	d=256	d=32	d=128	d=256
JOINT	10^{-2}	2.72	1.36	1.52	478.0	386.3	377.5
	10^{-1}	2.69	2.23	2.87	230.4	399.7	411.0
	1	5.96	4.69	5.53	285.6	198.0	176.4
	10	6.52	6.58	6.01	178.5	197.0	182.0
STF-0 (dose-free)	—	2.36	1.18	1.51	607.2	627.4	540.2
single-task reference	—	0.60	0.33	0.57	951.8	761.7	325.8

Table 19: UCI-HAR per-epoch overhead relative to plain summation, batch 256. Entries are medians over ten repetitions of median epoch time, excluding epoch 0 and synchronizing each epoch. Method order rotates after three idle-device readings. A paired-ratio estimator differs by at most five percentage points. The first row gives reference times; negative overhead indicates speedup for the measured implementation.

Method	m=32	m=64	m=128	m=256
plain summation (s/epoch)	0.0813	0.0986	0.1283	0.1454
overhead relative to plain summation				
STF-0 (detach)	−35.8%	−48.1%	−59.5%	−65.0%
STF-HARD	+21.8%	+18.1%	+14.3%	+9.5%
STF-SOFT	+21.8%	+19.8%	+8.9%	+12.2%
STFCAL	−0.7%	−20.1%	−37.7%	−47.8%
PCGrad	+41.6%	+31.4%	+26.6%	+21.1%
CAGrad	+75.2%	+60.2%	+46.0%	+38.5%
FAMO	+31.0%	+31.1%	+23.9%	+19.6%

1.01 at $\alpha = 10^{-6}$, compared with 0.64–0.83 at the previous lowest dose 0.01, without materially changing the crossing. The opposite width trends therefore cannot be explained solely by the original grid's lower endpoint.

We report the probe because it determines the implemented gate, while retaining the crossing estimate as a separate diagnostic. The probe slope varies over a range of 0.17 across calibration times on NYUv2, versus 2.55 on HAR (Appendix H.3); this relative stability does not make it a capacity exponent. For the convolutional encoder, the *proj* probe gives $\hat{\rho} = 3.27 \pm 0.54$ (range 2.45–3.92), predicting approximately 0.31, while the *ae* probe gives 0.0136 ± 0.0015, predicting approximately 73.

Table 20 shows joint collapse in $10/10$ seeds for every tested $\alpha \geq 0.5$. Calibrated filtering achieves $0/10$ only for $\alpha \geq 1$, when all gates open and relative scatter is approximately 1.00; the recorded candidate rank is 3. At $\alpha = 0.5$, the gate opens in the five seeds with $\hat{\rho} > 2$, and all five avoid collapse (candidate rank 1). This is an empirical threshold comparison, subject to Corollary 7's scope restrictions. At lower doses, closed-gate outcomes are close to joint training.

The task metric can conceal shrinkage: joint depth RMSE falls overall from 1.335 to 1.092 (approximately 18%) while relative scatter falls from 0.62 to 0.04. Thus the high-dose joint model has lower depth error than the filtered models despite meeting the representation-collapse criterion. At every tested $\alpha \geq 1$, paired Wilcoxon tests on relative scatter give $p \approx 2 \times 10^{-3}$, the smallest two-sided value attainable with ten nonzero, same-sign paired differences.

Table 20: NYUv2 under the unified protocol (MLP, $m = 64$, ten seeds): depth RMSE in meters on a 9×12 grid and scatter relative to the same-seed single-task anchor. Calibration gives mean $\hat{\rho} = 2.01$ and inverse-mean threshold 0.50. Recorded candidate ranks are 1 at $\alpha \leq 0.5$ and 3 at $\alpha \geq 1$; the radial STFCAL arm does not use this rank. Full detachment yields RMSE 1.353 and relative scatter 1.000 at every dose.

α	0.03	0.1	0.3	0.5	1	3	10	30	100
JOINT RMSE	1.335	1.336	1.316	1.307	1.305	1.280	1.221	1.100	1.092
JOINT rel. scatter	0.62	0.45	0.30	0.26	0.22	0.16	0.10	0.04	0.04
JOINT collapsed	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10	10/10
STFCAL RMSE	1.339	1.327	1.314	1.328	1.354	1.353	1.353	1.354	1.354
STFCAL rel. scatter	0.63	0.45	0.31	0.63	1.01	1.01	1.01	1.00	1.01
STFCAL collapsed	0/10	0/10	4/10	5/10	0/10	0/10	0/10	0/10	0/10
calibrated gate open	0/10	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10

Table 21: NYUv2 three-task comparison, shared convolutional encoder ($d = 128$), split, and ten seeds; final-epoch evaluation. Blocks give single-task and auxiliary-free references, projection auxiliary (*proj*, $\alpha = 1$), and reconstruction (*ae*, $\alpha = 250$). Depth error is in meters and normal error in radians. Recorded "scatter" is within-example $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$, with correction $d - 1$; it measures a different quantity from across-example scatter in (10). Absolute task scores depend on this small-backbone protocol.

Configuration	seg pxacc	seg mIoU	depth RMSE	normal angle	scatter
single-seg	0.359	0.083	2.780	1.574	2.22
single-depth	0.076	0.032	1.159	1.573	5.08
single-normal	0.074	0.031	3.346	0.593	12.43
joint3 (no aux)	0.356	0.100	1.130	0.586	7.30
+ proj aux (α=1)	0.365	0.090	1.085	0.580	1.64
STF-0	0.356	0.100	1.131	0.585	7.30
STFCAL	0.357	0.100	1.134	0.586	7.37
STFCALRANK	0.357	0.095	1.115	0.583	2.65
+ ae aux (α=250)	0.350	0.092	1.168	0.580	3.99
STF-0	0.357	0.099	1.133	0.586	7.19
STFCAL	0.338	0.091	1.171	0.594	3.84

Table 21 compares dense tasks under a shared small-backbone protocol. Its recorded statistic is mean within-example feature variance, which differs from the across-example quantity in (10). With the projection auxiliary, the amplitude ratio is 0.22: segmentation mIoU decreases while depth RMSE improves, and radial filtering returns amplitude near the auxiliary-free reference. With reconstruction at $\alpha = 250$, filtered amplitude is 3.84, compared with 3.99 for joint training and 7.19 for detachment. These task and amplitude comparisons motivate an across-example covariance audit before assigning a collapse mechanism.

H Supplementary Experimental Results

Ten-seed calibrated experiments support Section 5. At high doses, joint versus STFCAL relative scatter is 0.016 versus 0.990 for HAR ($\alpha = 100$), 0.147 versus 1.001 for battery ($\alpha = 100$), and 0.014 versus 0.982 for RadioML ($\alpha = 3000$). In each case, collapse rates are $10/10$ versus $0/10$. At low doses with closed gates, outcomes are close: HAR at $\alpha = 0.001$ gives 0.525 versus 0.528, and battery at $\alpha = 0.01$ gives 0.963 versus 0.965.

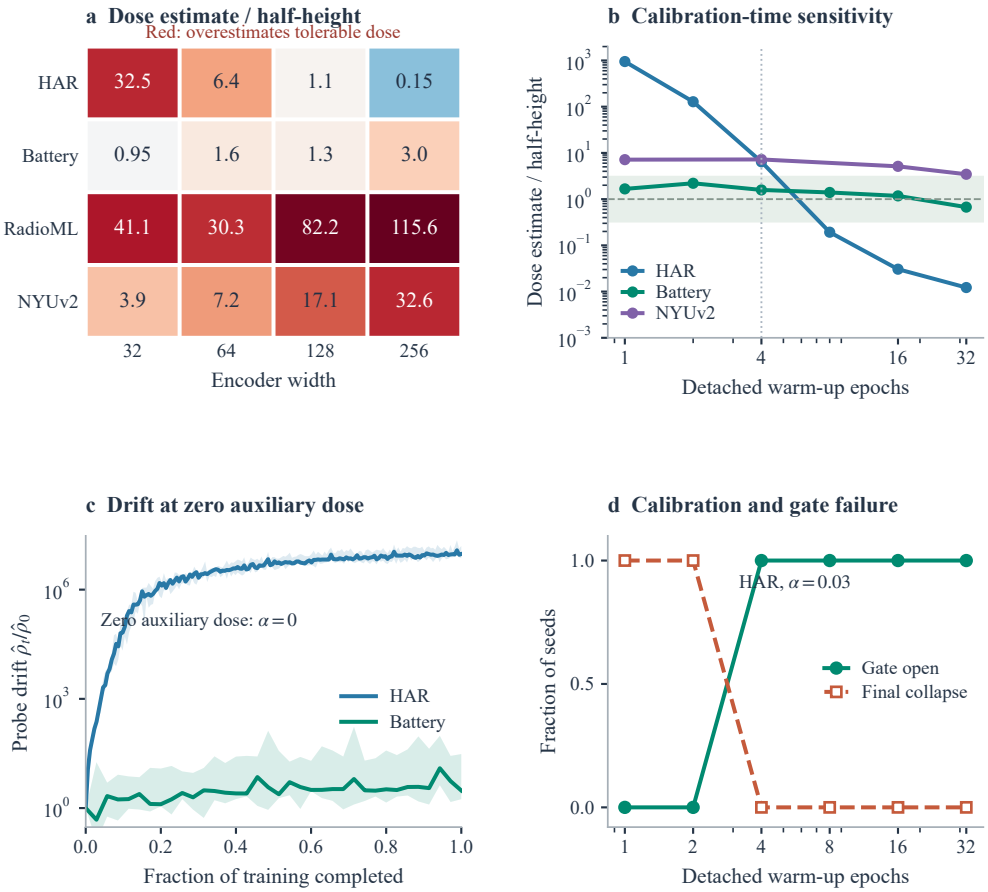


Figure 7: A frozen radial probe does not transfer uniformly. (a) $\alpha_{\text{probe}}/\alpha_{1/2}$ for 16 controlled domain–width cells. Red indicates an optimistic dose estimate; blue, a conservative one. (b) Calibration-time sensitivity at $m = 64$; shading marks a factor-3 reference band. Both panels use ten-seed mean probes. (c) Zero-dose probe drift: median and full range over five seeds, normalized to the initial value. (d) HAR at $\alpha = 0.03$: early calibration leaves all ten gates closed and all runs collapse; $W \geq 4$ opens every gate and prevents collapse. Values and crossing definitions are in Table 13 and Appendices H.2–H.3.

With Transformer encoders at $\alpha = 100$, HAR scatter changes from 0.030 to 1.041 and battery scatter from 0.036 to 1.000, again reducing collapse from $10/10$ to $0/10$. Battery SOH RMSE improves from 0.447 to 0.267. On CelebA at $\alpha = 100$ (five seeds), scatter increases from 0.003 to 0.934 and collapse falls from $5/5$ to $0/5$.

The real-head experiments use $m = 64$ and ten seeds (Table 6). Neither reconstruction head collapses at its tested dose; filtering has little effect or modestly reduces primary accuracy. The training-subject classifier presents a different trade-off: joint training achieves identity accuracy 0.646 against chance $1/21$, while filtering reduces it to 0.055 and improves primary accuracy. This is a training-head measurement, not a held-out privacy attack. For high-dose SNR regression, joint training collapses in every seed; hard and soft filtering do not restore accuracy, and detachment performs best among the tested interventions. Probe magnitudes differ substantially across these heads, but the false negatives discussed in Section 4.4 preclude a universal decision rule. Figure 9 shows the battery repair outcome, Table 23 the corresponding collapse and recovery rates, and Table 22 the intervention families.

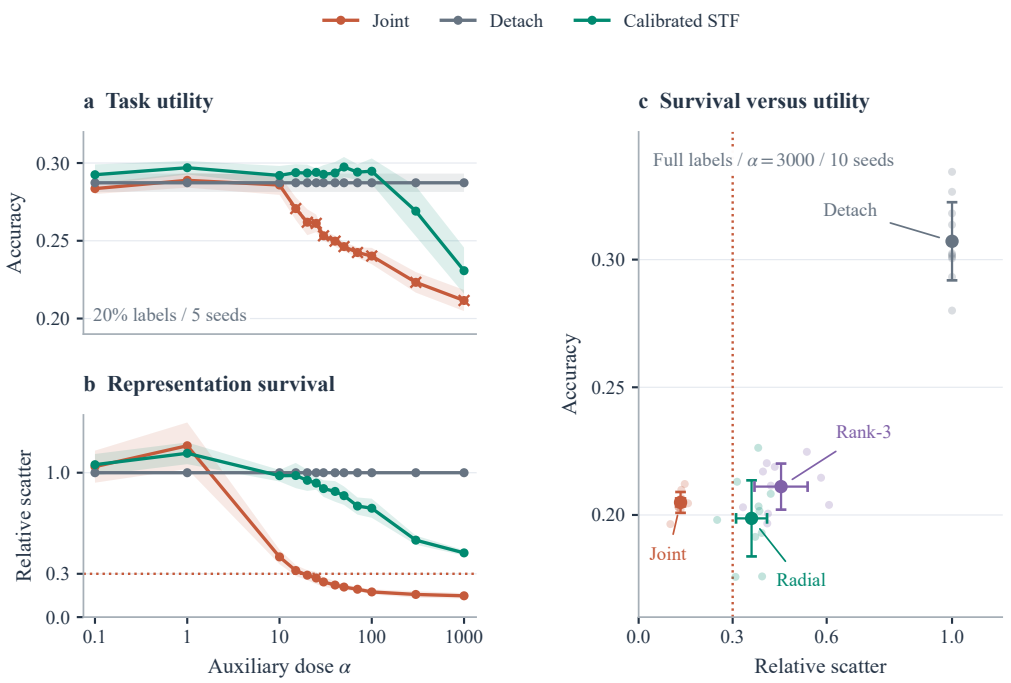


Figure 8: Filtering can retain an auxiliary gain, but scatter protection alone is insufficient. RadioML SNR regression, width 64. (a,b) With 20% primary labels, STF exceeds detachment at intermediate doses; the gain disappears at extreme doses despite scatter above 0.3. Five unique seeds per dose; bands show seed SD and crosses mark at least one collapsed joint run. (c) In the separate fully labeled dose-3000 protocol, rank-3 filtering avoids collapse in all ten seeds but underperforms detachment. Small points are seeds; large points and bars show means and SD. The dotted line marks the 0.3 criterion.

Table 22: Intervention families and their evaluation requirements. Performance comparisons under the stated implementations and protocols appear in Tables 4 and 25.

Family	Intervention	Relevant limitation
Task weighting (Chen et al., 2018; Liu et al., 2023)	scalar auxiliary strength	positive attenuation can help at a fixed dose
Gradient geometry (Yu et al., 2020; Liu et al., 2021a; Navon et al., 2022)	task-gradient combination	alignment does not determine variance change
Variance regularization (Bardes et al., 2022)	representation penalty	penalty weight and primary-task trade-off
Full detachment	remove auxiliary encoder gradient	also removes useful auxiliary learning
Selective filtering	remove selected directions	basis identification and subsequent drift
Parameter variance constraint	constrain the actual parameter update	fixed guard batch; finite-step feasibility must be checked

The results identify several distinct regimes: task error can improve as scatter shrinks, real auxiliaries can remain benign over the tested grid, and calibration can miss contraction. Frozen-basis error and sampling variation provide additional diagnostic checks. Numerical instability and loss of plasticity (Dohare et al., 2024) remain outside the fixed-operator model.

The battery encoder maps a per-cycle window $x \in \mathbb{R}^{T \times D}$ of normalized voltage, current, temperature, and resistance to $h \in \mathbb{R}^d$. It combines a TSMIXER feature/temporal-mixing block (Ekambaram

Table 23: Controlled high-dose collapse and task-recovery rates for three domains. Recovery is the fraction of seeds whose primary metric improves over the mean joint-training metric. HAR can retain high accuracy despite substantial shrinkage, so scatter recovery does not imply better accuracy. Battery uses 20 seeds per mode and RadioML ten; the historical HAR file provides aggregate rates. The joint recovery rate is set to zero by convention. All plotted rates are reproduced directly from the released records.

Domain	Metric	JOINT coll. / rec.	STF-0 coll. / rec.	STF-HARD coll. / rec.	STF-SOFT coll. / rec.
HAR	accuracy (flat)	1.0 / 0.0	0.0 / 0.1	0.0 / 0.2	0.0 / 0.1
RadioML	accuracy	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0
Battery	log-SOH RMSE	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0

et al., 2023) with a gated state-space block (Gu et al., 2022; Gu & Dao, 2024). SOH (L_2), RUL (L_1), reconstruction, and distributional heads share this representation. The real auxiliary heads are evaluated empirically and are not assumed to satisfy Definition 2.

RUL targets require careful handling of right censoring. If a cell has not reached end of life when recording stops, its remaining lifetime is known only to exceed a lower bound. Treating that bound as an exact target can label a healthy cell with zero remaining life. In MATR, 59.2% of samples are censored and $96/185$ cells do not reach EOL within their records. Initialization zeros in summary arrays cause a further $48/185$ cells to receive an EOL index of zero under a naive threshold search. Restricting the search to each trajectory's physically valid suffix recovers these cells. We retain censored cells and use $|\hat{r} - r|$ for observed lifetimes and $\max(0, r_{\mathrm{lb}} - \hat{r})$ for censored lower bounds. This one-sided loss is motivated by censored regression (Tobin, 1958), but is not a Tobit likelihood. Excluding cells without observed EOL would leave only 14 test cells in the historical split (Section 5).

To emphasize fast-charge, extreme-temperature, and rapid-degradation windows, BatteryMTL uses three multiplicative weighting factors:

$$\begin{aligned} w_i &= w_i^{(\mathrm{rate})} \cdot w_i^{(\mathrm{temp})} \cdot w_i^{(\mathrm{degrad})}, \\ w_i^{(\mathrm{rate})} &= 1 + \omega\, \mathbf{1}[\text{C-rate}_i > r_q], \\ w_i^{(\mathrm{temp})} &= 1 + \omega\, \mathbf{1}[T_i < t_{lo} \vee T_i > t_{hi}], \\ w_i^{(\mathrm{degrad})} &= 1 + \omega\, \frac{|\Delta\mathrm{SOH}_i|}{\max_j |\Delta\mathrm{SOH}_j|}, \end{aligned} \tag{25}$$

Equation (25) amplifies windows in which several factors are elevated, while also up-weighting windows with a single elevated factor. Thresholds are estimated from the training fold: r_q is the 70th C-rate percentile, and t_{lo}, t_{hi} are the 10th and 90th temperature percentiles. These relative thresholds distinguish regimes within MATR, where 99.8% of cycles exceed 1.5C and temperatures lie within 31–36°C. In contrast, fixed cuts at 1.5C and $0/40^\circ$C would mark nearly all cycles for rate and none for temperature. Section 5 reports regime-specific ablations; SOH-only and RUL-only models provide task-specific references, not guaranteed performance ceilings.

H.1 Composition and Regime Weighting

The following measurements expand the composition and weighting results in Sections 5 and 5.

At the tested high doses, applying PCGrad or CAGrad to the unfiltered battery branch leaves collapse rate at 1.0 and relative scatter at $0.029/0.030$ (Table 24). Hard filtering alone reduces collapse to 0.0, restores relative scatter to 1.00, and improves log-SOH RMSE from 0.611 to 0.294. Adding PCGrad or CAGrad afterward gives RMSE $0.283/0.291$ and scatter up to 1.18. HAR and RadioML show the same qualitative preservation of scatter under composition. These results establish compatibility for the tested order; they neither show that sequencing is necessary nor establish why the unfiltered combiners fail.

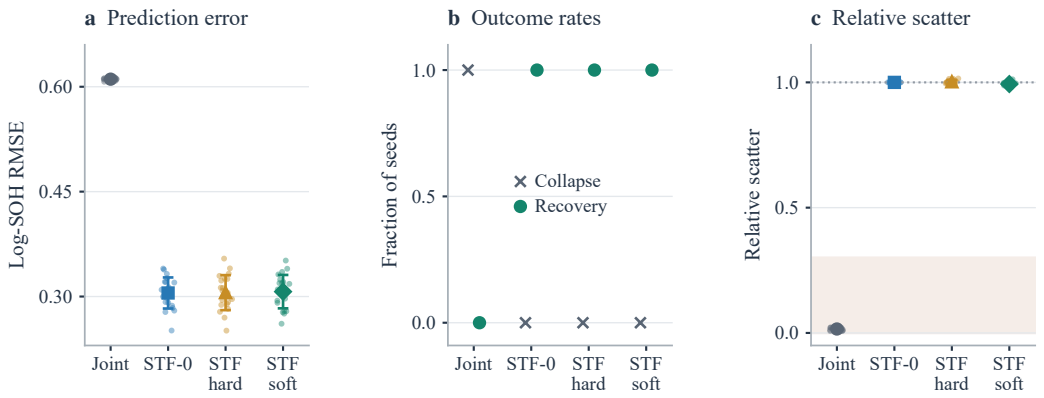


Figure 9: Controlled battery contraction can be prevented by filtering. NASA PCoE unified protocol, $m = 128$, $\alpha = 10^4$, 20 seeds per mode. (a) Mean log-SOH RMSE falls from 0.611 to about 0.305. (b) Collapse and task-recovery rates. (c) Scatter relative to the same-seed single-task reference; the dotted line marks 1.0 and shading the region below 0.3. Small points are seeds; large markers and whiskers show means and population SD. Log-SOH error and percentage-point SOH error in the separate application have different units.

Table 24: Filter-then-combiner experiments at high auxiliary doses (five seeds, three domains). Unfiltered branches retain collapse rates of 1.0 and relative scatter at most 0.15. The hard filter, alone or followed by PCGrad/CAGrad, avoids the collapse criterion in every seed. Primary metrics change modestly after composition; the reverse order was not tested. Bold identifies the filter-only reference within each block.

		unfiltered (JOINT)			filtered (STF-HARD)		
	combiner	sum	PCGrad	CAGrad	sum	PCGrad	CAGrad
Battery	log-SOH RMSE↓	0.611	0.592	0.603	**0.294**	0.283	0.291
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.03	0.03	**1.00**	1.05	1.18
HAR	accuracy↑	0.941	0.667	0.952	0.943	0.933	0.940
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.15	0.00	0.03	**0.97**	1.14	1.03
RadioML	accuracy↑	0.293	0.091	0.109	**0.301**	0.296	0.295
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.00	0.00	**0.97**	0.99	1.04

BatteryMTL's multiplicative regime weights use training-data percentiles for C-rate and temperature, together with degradation rate. The aggregate SOH results change little, while factor ablations associate improvements in fast-charge and extreme-temperature windows with the rate and temperature terms, respectively (errors 0.95 and 2.35 when the corresponding terms are removed). Table 10 reports the historical aggregate results.

H.2 Probe Dynamics During Training

The frozen calibration in Section 5 is motivated by the behavior of a continuously recomputed probe. We repeat unfiltered joint training with `--rho-trace`, using five seeds per configuration over

Table 25: Anti-collapse regularizers at the doses and seeds of Table 4 (ten seeds per configuration). Penalties of weight λ are added to joint training. Entries report collapse rate, primary metric in parentheses, and relative scatter (1.0 is the single-task reference). Joint training, detachment, and hard filtering are evaluated in the same runs. Avoiding the scatter threshold need not improve the primary task.

	HAR		Battery		RadioML	
	coll. / acc.↑	rel. scat.	coll. / RMSE↓	rel. scat.	coll. / acc.↑	rel. scat.
JOINT	1.00 (0.945)	0.15	1.00 (0.610)	0.022	1.00 (0.256)	0.008
STF-0	0.00 (0.945)	1.00	0.00 (0.297)	1.00	0.00 (0.341)	1.00
STF-HARD	0.00 (0.946)	0.99	0.00 (0.299)	1.00	0.00 (0.339)	0.97
VICReg (variance + covariance)						
λ=1	1.00 (0.950)	0.22	1.00 (0.610)	0.023	1.00 (0.249)	0.007
λ=3	0.20 (0.946)	0.31	1.00 (0.610)	0.023	1.00 (0.253)	0.007
λ=10	0.00 (0.944)	0.36	1.00 (0.609)	0.024	1.00 (0.256)	0.008
λ=30	0.00 (0.926)	0.36	1.00 (0.604)	0.033	1.00 (0.260)	0.008
λ=100	0.00 (0.902)	0.36	1.00 (0.451)	0.14	1.00 (0.301)	0.011
λ=300	0.00 (0.873)	0.36	0.00 (0.339)	0.59	1.00 (0.362)	0.035
λ=10^3	0.00 (0.843)	0.36	0.00 (0.390)	1.95	1.00 (0.320)	0.15
λ=3×10^3	0.00 (0.831)	0.36	0.00 (0.423)	3.70	0.10 (0.207)	0.45
λ=10^4	0.00 (0.821)	0.36	0.00 (0.423)	4.54	0.00 (0.215)	1.29
Barlow Twins (redundancy reduction)						
λ=1	1.00 (0.935)	0.11	1.00 (0.607)	0.11	1.00 (0.367)	0.016
λ=3	1.00 (0.765)	0.022	1.00 (0.607)	0.13	1.00 (0.330)	0.024
λ=10	1.00 (0.435)	0.006	1.00 (0.606)	0.15	1.00 (0.121)	0.074
λ=30	1.00 (0.352)	0.006	1.00 (0.606)	0.16	1.00 (0.105)	0.096
λ=100	1.00 (0.338)	0.006	1.00 (0.605)	0.16	1.00 (0.126)	0.072
λ=300	1.00 (0.347)	0.006	1.00 (0.606)	0.16	1.00 (0.141)	0.043
λ=10^3	1.00 (0.343)	0.006	1.00 (0.606)	0.16	1.00 (0.140)	0.042
λ=3×10^3	1.00 (0.365)	0.006	1.00 (0.607)	0.16	1.00 (0.148)	0.053
λ=10^4	1.00 (0.339)	0.006	1.00 (0.607)	0.16	1.00 (0.153)	0.044

1730 HAR steps and 350 battery steps. Monitoring does not alter the measured trajectories: all 28 shared BSCALE configurations match exactly in task metric, collapse rate, relative scatter, and $\hat{\rho}$ (maximum difference zero).

Three observations matter. First, HAR probe values rise sharply during training: from 0.017 to 1.9×10^5 at $m = 32$, 0.010 to 1.0×10^5 at $m = 64$, and 0.006 to 3.5×10^4 at $m = 256$. Second, similar terminal values occur when the toxifier is inactive ($\alpha = 0$) and at the lowest positive dose (10^{-4}), so a large probe value alone does not establish auxiliary-induced contraction. At $m = 32$, increasing the dose instead lowers the terminal probe, to 5.3×10^2 at 10^{-2} and 1.1×10^2 at 3×10^{-2}. Third, in ten HAR configurations whose final scatter lies between 0.50 and 1.04 of the reference, the gate $\alpha\hat{\rho} \geq 1$ opens in every seed between steps 110 and 590. A gate using this running statistic therefore activates on runs that do not meet the collapse criterion at the recorded endpoint.

Battery behaves differently: probe values remain $O(1)$, ranging from 0.29–0.49 initially to 0.3–2.9 at the final step. Among 35 seeds whose final scatter lies above the half-height level, the gate opens only once, at the last step. These domain-dependent trajectories motivate a specified calibration time; they do not establish that a fixed probe is universally preferable to every adaptive rule. The script `repro/scripts/rho_trace_analyze.py` reproduces both analyses.

H.3 Sensitivity to Calibration Time

The default calibration measures $\hat{\rho}$ after four detached warm-up epochs and fixes the gate and rank. We vary only this measurement time: $W \in \{1, 2, 4, 8, 16, 32\}$ at five widths 32–512 for HAR and battery, and $W \in \{1, 4, 16, 32\}$ at four widths for NYUv2, with ten seeds per cell. The $W = 4$ configuration exactly reproduces all 100 shared HAR and battery probe values. NYUv2 gives $2.258/2.008/1.637/1.285$, matching the rounded values in Appendix G.

Probe values depend on the training state. Across the warm-up sweep, HAR at $m = 32$ increases from 0.63 to 6.9×10^4, and at $m = 512$ from 51.9 to 1.6×10^4. Battery remains $O(1)$: 0.96 to 1.92 at $m = 32$ and 0.058 to 0.67 at $m = 512$. NYUv2 at $m = 32$ changes from 1.94 to 4.19, a factor of 2.2. Accordingly, width slopes fitted to $1/\hat{\rho}$ vary from -2.00 to $+0.55$ for HAR, $+0.37$ to $+0.97$ for battery, and $+0.11$ to $+0.27$ for NYUv2. These are calibration-dependent probe slopes, not interchangeable estimates of capacity scaling. Differences in primary-gradient magnitude, which appears in the denominator, can contribute to this sensitivity.

Measurement time also changes gate activation. For HAR at $m = 64$, the inverse-mean probe value is 1.34 at $W = 1$, 0.181 at $W = 2$, 9.07×10^{-3} at $W = 4$, and 1.7×10^{-5} at $W = 32$, compared with the measured transition interval $[0.001, 0.01]$. In the sixty-epoch, ten-seed experiment at $\alpha = 0.03$, gates remain closed for $W = 1$ and $W = 2$: both settings collapse in $10/10$ seeds, with accuracy 0.947 and relative scatter $0.19/0.20$, close to joint training $(0.949, 0.19)$. For $W \geq 4$, all gates open, no seed collapses, and scatter returns to 0.99–1.00. At $\alpha = 100$, all six warm-up settings activate the gate and avoid collapse.

A short warm-up therefore fails at the marginal dose in this experiment. Detached warm-up also changes the training trajectory, so closed-gate runs need not equal joint training from initialization. Checking whether $\alpha\hat{\rho} \geq 1$ verifies activation at the chosen operating dose; it does not independently validate protection. The script `repro/scripts/warmup_sensitivity.py` reproduces these results.

H.4 Scale-Invariant Audit of the Scatter Criterion

The operational criterion in (10) is a coordinate-dependent, scale-sensitive proxy. To measure what a contraction preserves, we run the released HAR protocol at three doses and audit the frozen representation with covariance geometry and linear probes on raw, per-dimension-standardized, whitened, 10%-label, and new-label-space features. Thirty cells use $m = 64$, 60 epochs, Adam (10^{-3}, batch 256), and seeds 0–9; the driver is `repro/scripts/representation_audit.py`, and the records are in `results/REPAUDIT/`.

Table 26: Scale-invariant audit of the controlled HAR toxifier. Relative scatter uses the same-seed healthy reference. Covariance trace is absolute; participation rank (PR), effective (entropy) rank, and standardized probes are scale-invariant. At $\alpha = 100$ the scatter criterion records $10/10$ collapses, while the standardized probe and the new-label-space probe are unchanged and PR rises.

α	rel. scatter	cov. trace	PR	eff. rank	standardized probe	primary acc.
0 (healthy)	1.000	578.7	2.00	2.57	0.947	0.941
0.1	0.130	8.51	4.97	5.52	0.958	0.956
100	0.016	0.125	4.90	5.02	0.955	0.952

The trace falls by $4{,}600\times$ and every seed meets the scatter criterion, yet per-dimension standardisation preserves activity-probe accuracy. A static-versus-dynamic activity probe stays at 0.999. A one-hidden-layer MLP probe (64 units, 120 epochs) similarly reaches 0.945 at $\alpha = 0$ and 0.954 at $\alpha = 100$ over five seeds. These measurements show a large change in feature amplitude with retained access to the measured task information.

H.5 Small-Dose Operator Audit

The main-text comparison holds each operator at the domain's supercritical dose. The complementary question is whether the default scalarizations contract the representation when the auxiliary weight is far below the calibrated threshold. Table 27 reports

five operators at three doses on HAR ($m = 64$, 60 epochs, three seeds); the driver is `repro/scripts/lowalpha_operator_audit.py`, and the records are in `results/LOWALPHA/`.

Table 27: Accuracy (scatter-collapse count over three seeds) at small auxiliary doses, HAR, $m = 64$. Plain summation is healthy at $\alpha = 0.001$, while the default scalarizations contract in every seed. This is operator conditioning rather than toxifier dose.

α	joint	MGDA	IMTL-G	Aligned	Nash	FAMO
0.001	0.943 (0/3)	0.551 (3/3)	0.242 (3/3)	0.927 (3/3)	0.425 (3/3)	0.176 (3/3)
0.01	0.950 (2/3)	0.409 (3/3)	0.301 (3/3)	0.884 (3/3)	0.461 (3/3)	0.176 (3/3)
0.1	0.953 (3/3)	0.182 (3/3)	0.183 (3/3)	0.832 (3/3)	0.565 (3/3)	0.181 (3/3)

At $\alpha = 0.001$, the tested operators contract features even though the auxiliary gradient is tiny and plain summation remains healthy. The low-dose behavior therefore depends on the default scalarization as well as the auxiliary dose. MGDA still collapses after lowering the learning rate to 10^{-4} (Table 28), which rules out the tested step-size reduction as a remedy. The fixed-dose comparisons establish the outcomes of these implementations; identifying their general performance requires separate optimization studies.

Table 28 compares the default learning rate with a tenfold smaller one at $\alpha = 0.001$. The loss-weighting families (UW, GradNorm, IMTL-L) become healthy at this low dose, as does plain summation; the gradient-geometry operators (PCGrad, CAGrad, FAMO, IMTL-G, Nash, Aligned, MGDA) collapse in every seed at both learning rates, and several of their accuracies worsen at the lower rate. The low-dose contraction is therefore operator-specific, not a single step-size artifact. At the high doses of Table 4, all ten operators collapse.

Table 28: Small-dose operator audit at $\alpha = 0.001$, HAR, $m = 64$, three seeds. Accuracy (scatter-collapse count) at the default 10^{-3} and a tenfold smaller 10^{-4} learning rate. The weighting families become healthy at low dose; the gradient-geometry operators collapse at both rates.

Method	$lr = 10^{-3}$	$lr = 10^{-4}$
joint (sum)	0.943 (0/3)	0.934 (0/3)
PCGrad	0.660 (3/3)	0.656 (3/3)
CAGrad	0.954 (3/3)	0.953 (3/3)
FAMO	0.176 (3/3)	0.176 (3/3)
UW	0.941 (0/3)	0.935 (0/3)
GradNorm	0.954 (0/3)	0.934 (0/3)
IMTL-L	0.944 (0/3)	0.935 (0/3)
IMTL-G	0.242 (3/3)	0.187 (3/3)
Nash-MTL	0.425 (3/3)	0.338 (3/3)
Aligned-MTL	0.927 (3/3)	0.790 (3/3)
MGDA	0.551 (3/3)	0.186 (3/3)

H.6 Readout Transfer Under Contraction

The next experiment measures the compatibility of a readout fitted to healthy features. For five HAR seeds, we train healthy ($\alpha = 0$) and contracted ($\alpha = 100$) encoders and split training examples into readout-fitting and calibration subsets. Table 29 evaluates four maps on held-out test data: the healthy reference; the unchanged healthy head and statistics on contracted features; the same head with input statistics refitted without new labels; and a new head fitted to the contracted features.

Proposition 1 isolates the scale mechanism. A unique bias-only winner gives a constant predicted class at sufficiently small positive scale; ties need a specified convention. Fixed input centering contributes to the effective offset even with a bias-free head. Figure 10 summarizes finite c_i^{bias} by their within-seed median and then the median across seeds. This statistic describes bias exit, distinct from the first prediction flip when shrinking from $c = 1$. The standard-deviation floor is inactive over the plotted range; exact invariance for every $c > 0$ assumes unfloored statistics.

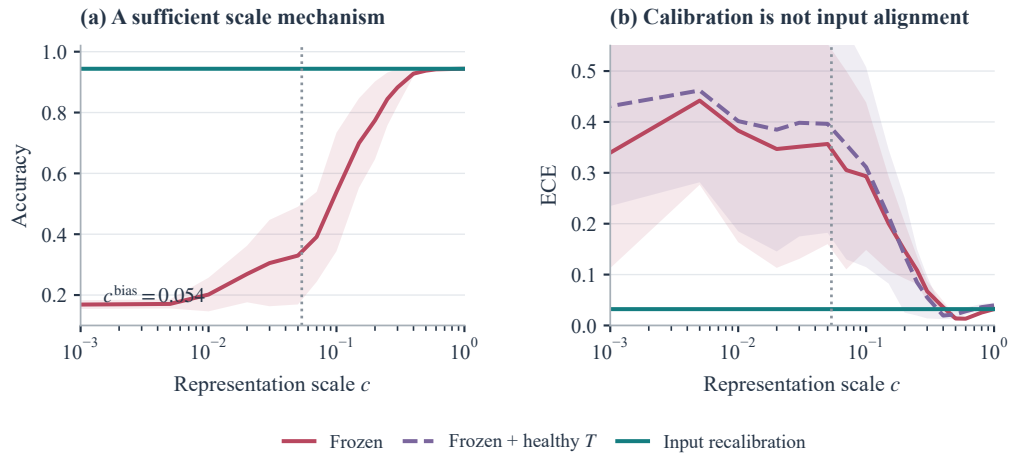


Figure 10: Pure scale change can invalidate a fixed affine readout. Healthy HAR features are multiplied by c before readout. (a) Frozen accuracy degrades; refitting input statistics cancels the imposed scale change. (b) ECE before and after applying the healthy temperature map. Five seeds, $m = 64$; bands show seed SD. The dashed line summarizes finite bias-exit thresholds: median of seed medians $c^{\text{bias}} = 0.054$, seed range 0.020–0.207. Bias exit and the first flip when shrinking from $c = 1$ are distinct boundaries.

Table 29: Historical transfer between separately trained healthy and contracted HAR encoders (five seeds, means). "Frozen" applies the healthy head and input statistics unchanged; the reported ECE is the raw softmax ECE, and applying the healthy temperature map raises frozen ECE to 0.512. Input recalibration reuses the weights but refits the feature statistics; retraining fits a new head. The frozen readout degrades with large NLL/ECE, while cheap input recalibration recovers accuracy but not the NLL of a retrained head. The last two rows apply the same healthy readout to the STF-0 and STF-cal encoders.

Readout	accuracy	NLL	ECE
healthy reference	0.945	0.187	0.031
frozen (healthy statistics)	0.241	2.516	0.430
input recalibration	0.952	0.396	0.043
retrained on contracted features	0.955	0.190	0.034
STF-0 encoder (same readout)	0.946	0.188	0.031
STF-cal encoder (same readout)	0.943	0.201	0.033

The historical readout-transfer study trains the healthy and contracted encoders separately from the same initialization; it measures compatibility across fitted models rather than continuation of one checkpoint. Its calibration subset is withheld from readout fitting but participates in supervised encoder training. Historical nominal-90% conformal coverage (0.802 healthy, 0 frozen, 0.806 recalibrated, 0.834 retrained) is consequently descriptive, not a split-conformal guarantee; its quantile also lacks the finite-sample order-statistic correction. The paired continuation study below excludes calibration examples from encoder and head fitting and uses the corrected quantile. The official HAR subject split still changes the test population, so we report empirical coverage without assuming calibration–test exchangeability.

The unchanged readout models a fixed deployment interface behind a changing encoder. Its NLL and ECE rise sharply; input-statistic recalibration restores accuracy and most of ECE, while a new head also restores NLL. Applying the same healthy readout to STF-0 and STF-cal encoders gives 0.946 and 0.943 accuracy (`stf_readout_audit.py`). The gap also appears at $m = 32$ (frozen 0.236, recalibrated 0.887, refitted 0.909), $m = 128$ (0.166, 0.955, 0.954), and with a Transformer (0.162, 0.950, 0.952). These tested widths and encoders exhibit the same separation between decodabil-

Table 30: Historical real-head transfer between separately trained encoders. Frozen reuses the healthy head and statistics; recalibration refits statistics; retrained fits a new head; detached reuses the healthy readout on the STF-0 encoder. Relative S is the same-seed scatter ratio; five seeds for SNR and subject-ID, three for I/Q.

Head	α	rel. S	healthy	frozen	recal.	retrained	detached
RadioML SNR	3000	0.137	0.315	0.099	0.094	0.229	0.315
RadioML I/Q	3000	1.053	0.312	0.195	0.195	0.351	0.312
HAR subject-ID	10	0.932	0.944	0.384	0.466	0.901	0.944

ity and compatibility. Driver: `repro/scripts/readout_transfer_audit.py`; records: `results/READOUT/`.

H.7 Paired Checkpoint Continuation

The paired experiment updates the encoder behind an already fitted consumer. We use the official HAR subject split and a prepared RadioML split with the same three-layer, width-64 MLP; the RadioML source-split audit appears below. A class-stratified 20% training reservation is made before any encoder update for calibration. The remaining 80% fits the encoder, primary head, and replacement probes. Reference readouts use per-feature standardization and the historical regularized LBFGS fit. Temperature fitting follows the fixed reference predictor; raw conformal scores use the corrected $\lceil (n+1)0.9 \rceil$ order statistic. Because HAR separates subjects between training and test, we report empirical coverage without assuming calibration–test exchangeability.

Each seed first trains a healthy encoder for 60 HAR or 25 RadioML epochs. A real SNR head is trained on detached features during this warm-up. We clone all model weights and Adam moments, retain the fitted consumer readout, and continue each branch for 30 HAR or 15 RadioML epochs. The learning rate is 10^{-3}, batch size 256, and every operation uses CPU fp32 except the double-precision linear readout fit and evaluation. Future minibatch permutations are generated once per seed and reused for joint training, detachment, STF-cal, and primary-only continuation. Initial parameter, optimizer-state, and minibatch-order hashes verify the pairing; all detached encoder and primary-head endpoints match primary-only continuation exactly. Evaluation never selects a checkpoint. The isotropic variance objective matches the historical readout audit; a fixed projected-variance objective tests the benchmark operator separately. The first five HAR variance seeds are followed by a predeclared replication on seeds 5–9, with the recipe unchanged and all outcomes retained.

Table 31 gives every sampled dose for both variance objectives. At isotropic $\alpha = 3$, frozen accuracy is 0.453, current-head accuracy 0.948, and recalibrated accuracy 0.947. Continuation thus causes a large compatibility loss while task decodability remains high. The fitted scalar residual shows additional feature drift, which matters at $\alpha = 100$: recalibration reaches 0.829 versus 0.935 after refitting. The imposed affine sweep and the trained continuations provide complementary tests of the scale component.

The SNR continuation provides a protocol boundary. At $\alpha = 3, 10, 30$, frozen accuracy is 0.293, 0.269, and 0.243, against 0.304 detached. At 30, recalibration reaches 0.253, refitting 0.285, and relative scatter 0.870. The severe collapse seen in historical from-initialization training is absent at these nominal doses after healthy warm-up. Calibration split, auxiliary state, warm-up, and total horizon also differ; the head-state factorial below tests one of these factors.

At the first minibatch of specified continuation epochs, we evaluate V and its gradient v on a fixed 256-example fitting batch. Loss gradients use the actual update minibatch. We record flow $r - \alpha c$, the applied increment s_θ, $v^\top s_\theta$, and the exact post-step change in V. A primary-only Adam counterfactual starts from identical pre-step weights and moments; its difference from the joint step measures the auxiliary's conditional effect at that optimizer state. Across SNR joint branches, flow and actual-step signs disagree on $33/75$ sampled updates. Table 32 includes negative-c checkpoints with a contracting auxiliary counterfactual, demonstrating why flow attribution and applied-step attribution need separate records.

Table 31: Paired checkpoint continuation with held-out calibration. Means ± population seed standard deviations. Each branch starts from the same healthy model and Adam state; the future minibatch order is shared. Trace is relative to the initial healthy checkpoint. HAR uses 60 healthy plus 30 continuation epochs; RadioML uses 25 plus 15. The real auxiliary head is trained on detached features during warm-up.

Case / dose	arm	frozen	input recal.	retrained	own	rel. trace
HAR variance, 10 seeds						
0	primary only	0.943 ± 0.007	0.943 ± 0.006	0.945 ± 0.005	0.944 ± 0.005	1.0897
0.1	joint	0.789 ± 0.161	0.951 ± 0.005	0.953 ± 0.002	0.951 ± 0.007	0.0213
0.1	STF-cal	0.943 ± 0.007	0.943 ± 0.006	0.945 ± 0.005	0.944 ± 0.005	1.0897
3	joint	0.453 ± 0.135	0.947 ± 0.005	0.952 ± 0.004	0.948 ± 0.006	0.0037
3	STF-cal	0.943 ± 0.007	0.943 ± 0.005	0.945 ± 0.005	0.941 ± 0.009	1.0868
100	joint	0.221 ± 0.067	0.829 ± 0.062	0.935 ± 0.016	0.796 ± 0.060	0.0003
100	STF-cal	0.941 ± 0.009	0.942 ± 0.005	0.944 ± 0.005	0.941 ± 0.006	1.0918
HAR projected, 5 seeds						
0	primary only	0.941 ± 0.007	0.940 ± 0.004	0.944 ± 0.004	0.943 ± 0.004	1.0636
0.1	joint	0.844 ± 0.128	0.932 ± 0.022	0.950 ± 0.005	0.945 ± 0.007	0.3460
0.1	STF-cal	0.941 ± 0.007	0.940 ± 0.004	0.944 ± 0.004	0.943 ± 0.004	1.0636
3	joint	0.652 ± 0.125	0.782 ± 0.077	0.946 ± 0.003	0.936 ± 0.011	0.3489
3	STF-cal	0.939 ± 0.011	0.940 ± 0.004	0.945 ± 0.004	0.943 ± 0.004	1.0737
100	joint	0.214 ± 0.056	0.398 ± 0.033	0.851 ± 0.083	0.724 ± 0.110	0.0640
100	STF-cal	0.939 ± 0.011	0.940 ± 0.003	0.944 ± 0.005	0.943 ± 0.004	1.0717
RadioML SNR, 5 seeds						
0	primary only	0.304 ± 0.011	0.306 ± 0.015	0.320 ± 0.010	0.319 ± 0.009	1.1045
3	joint	0.293 ± 0.011	0.300 ± 0.014	0.314 ± 0.012	0.312 ± 0.011	0.8280
3	STF-cal	0.296 ± 0.012	0.304 ± 0.015	0.318 ± 0.012	0.317 ± 0.012	0.9778
10	joint	0.269 ± 0.017	0.281 ± 0.016	0.302 ± 0.011	0.298 ± 0.013	0.7274
10	STF-cal	0.292 ± 0.015	0.301 ± 0.015	0.317 ± 0.011	0.315 ± 0.011	0.8520
30	joint	0.243 ± 0.020	0.253 ± 0.016	0.285 ± 0.012	0.277 ± 0.014	0.6394
30	STF-cal	0.286 ± 0.014	0.297 ± 0.013	0.312 ± 0.011	0.309 ± 0.010	0.7111

Table 32: Actual-step audit on a fixed 256-example fitting batch, using the first minibatch of predeclared continuation epochs. Disagree counts opposite signs for the gradient-flow derivative and actual Adam variance change. Aux. contracts compares the joint step with a primary-only Adam step from the same optimizer state. The final column counts negative flow auxiliary flux with a contracting auxiliary counterfactual. Checkpoints within a trajectory are dependent.

Case	α	updates	disagree	aux. contracts	$c < 0$, aux. contracts
HAR variance	0.1	80	28	80	0
HAR variance	3	80	22	79	2
HAR variance	100	80	23	76	3
HAR projected	0.1	40	14	31	0
HAR projected	3	40	11	25	1
HAR projected	100	40	8	29	2
RadioML SNR	3	25	10	12	1
RadioML SNR	10	25	13	10	0
RadioML SNR	30	25	10	10	1

Each new run uses one CPU thread, with at most four concurrent workers and explicit memory and time limits. The remote directories contain numerical code, inputs, checkpoints, and logs; analysis, writing, figures, and compilation run locally. Driver `paired_readout_continuation.py` writes the complete per-seed records to `repro/results/CPUCONT/`. Script `summarize_cpu_evidence.py` checks the paired controls and regenerates the numerical macros, tables, and figure. Seed intervals describe variation conditional on these datasets.

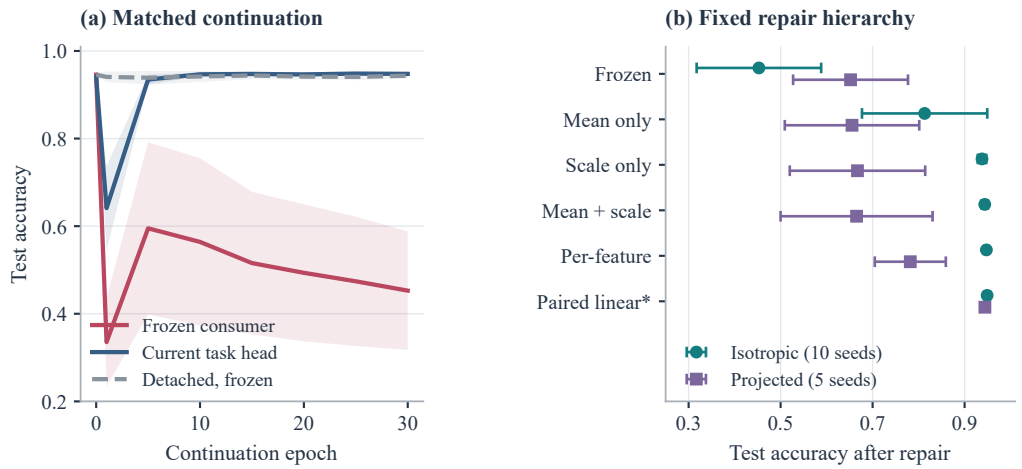


Figure 11: Input repair reveals the scale component of a learned compatibility failure. (a) Frozen-consumer and current-head trajectories at $\alpha = 3$, sharing healthy weights, Adam state, and future minibatches. (b) Fixed repairs fitted to training features without new labels. Global scale correction largely repairs isotropic contraction; projected contraction needs richer alignment. *Paired linear alignment requires old/new features of the same anchors. Shading and bars show population seed SD: 10 isotropic and 5 projected seeds.

H.8 A Fixed Unlabeled Repair Hierarchy

To distinguish scale from richer drift, we apply a fixed repair hierarchy to every saved endpoint from the 10 isotropic HAR, 5 projected HAR, and 5 SNR continuation seeds. Deterministic replay reconstructs each healthy readout and reproduces its accuracy, NLL, ECE, and all archived frozen and recalibrated endpoints within 10^{-7}. The reconstructed weights and statistics are saved; the encoders remain fixed throughout the audit.

Let $R = H_0 - \mathbf{1}\mu_0^\top$ and $X = H_1 - \mathbf{1}\mu_1^\top$ denote the old and new fitting features after centering. The global scale estimate is $c_{\mathrm{RMS}} = \|X\|_F/\|R\|_F$. Mean-only correction feeds $h - \mu_1 + \mu_0$ to the original preprocessing; scale-only feeds h/c_{RMS}; their combination feeds $(h - \mu_1)/c_{\mathrm{RMS}} + \mu_0$. Per-coordinate recalibration replaces each input mean and standard deviation. These estimators use marginal fitting statistics, with no new labels or test-statistic fitting.

The richer repairs are fixed post-hoc controls. Paired diagonal regression predicts each old coordinate from its new counterpart; scaled orthogonal Procrustes fits a rotation and global scale. Covariance alignment maps centered features by $C_1^{-1/2}C_0^{1/2}$, following CORAL (Sun et al., 2016), with eigenvalues floored at 10^{-6} of the largest. Paired ridge maps new standardized anchors to old standardized anchors with $\lambda = 10^{-3}$. These constants are fixed independently of test outcomes. Covariance and marginal-statistic repairs need only marginal features; paired regressions and Procrustes need old/new features for the same anchors. Jointly trained BC-Aligner (Hu et al., 2022b) has a different fitting procedure.

At isotropic dose 3, scale-only correction reaches 0.938, versus 0.813 for mean correction, 0.944 for their combination, and 0.947 for per-coordinate statistics. A strong scale-sensitive readout component survives substantial feature drift. At projected dose 3, scale-only reaches 0.667, per-coordinate statistics 0.782, and paired ridge 0.944, close to the refitted probe. At isotropic dose 100, the corresponding scores are 0.603, 0.829, and 0.915. The effective repair therefore depends on the drift. Covariance matching also leaves covariance-preserving rotations unidentified. Numerical controls recover known shifts, scales, positive diagonal changes, and scaled rotations, and include a covariance-identical sign flip.

Table 33: Fixed repair hierarchy on archived paired checkpoints. Means across 10 HAR isotropic, 5 HAR projected, and 5 RadioML SNR seeds. All maps use fitting features only, without labels or test-statistic fitting. PD, Proc., and ridge require paired old/new anchors; covariance alignment uses only marginal second moments. Ridge is fixed at 10^{-3} in standardized coordinates; covariance eigenvalue flooring is 10^{-6} of the largest eigenvalue. Full seed dispersions, NLL, ECE, prediction agreement, and fitting/held-out residuals accompany the records.

Case	α	frozen	mean	scale	mean +scale	diag.	PD	Proc.	cov.	ridge
HAR isotropic	0	0.943	0.941	0.943	0.941	0.943	0.943	0.941	0.945	0.946
HAR isotropic	0.1	0.789	0.941	0.946	0.950	0.951	0.952	0.949	0.930	0.950
HAR isotropic	3	0.453	0.813	0.938	0.944	0.947	0.949	0.948	0.912	0.949
HAR isotropic	100	0.221	0.280	0.603	0.782	0.829	0.841	0.855	0.774	0.915
HAR projected	0	0.941	0.938	0.941	0.938	0.940	0.940	0.940	0.944	0.944
HAR projected	0.1	0.844	0.896	0.850	0.898	0.932	0.924	0.931	0.553	0.950
HAR projected	3	0.652	0.655	0.667	0.665	0.782	0.711	0.862	0.493	0.944
HAR projected	100	0.214	0.362	0.333	0.354	0.398	0.354	0.431	0.383	0.769
RadioML SNR	0	0.304	0.306	0.303	0.306	0.306	0.302	0.310	0.312	0.310
RadioML SNR	3	0.293	0.296	0.294	0.299	0.300	0.291	0.302	0.302	0.303
RadioML SNR	10	0.269	0.275	0.271	0.283	0.281	0.267	0.280	0.285	0.287
RadioML SNR	30	0.243	0.246	0.247	0.254	0.253	0.233	0.248	0.257	0.266

All nine arms, all doses, and the primary-only, detached, and STF-cal endpoints are retained. Records include accuracy, NLL, ECE, agreement with healthy predictions, fitting/calibration/test alignment residuals, and map hashes. Fitting residuals and evaluation residuals are reported separately. Drivers: `checkpoint_repair_audit.py` and `readout_alignment.py`; records: `repro/results/CPUCONT/mechanism/d*.json`.

H.9 Real Reconstruction Under Matched Continuation

We extend the unchanged paired-continuation driver to a real reconstruction decoder, $64 \to 256 \to d_{\text{input}}$ with a ReLU hidden layer and mean squared error. All five seeds and the fixed grids are retained: HAR doses $1, 10, 100$, with 60 healthy and 30 continuation epochs; RadioML I/Q doses $30, 300, 3000$, with 25 healthy and 15 continuation epochs. During warm-up the decoder learns on detached features. Every continuation branch starts from identical encoder, primary head, decoder, and Adam states and reuses the same future minibatch order. All detached encoder and primary-head endpoints again match primary-only continuation bitwise.

The trained decoder is evaluated on fitting, calibration, and test data at the fixed endpoint. The reported ratio averages seed-wise test-MSE ratios to the dose-matched detached decoder. STF-cal evaluation uses its forward filter and frozen gate on ordered batches of 256; raw-input decoder error is also archived. Reconstruction progress, primary-task benefit, and frozen-consumer preservation are scored separately.

At HAR dose 100, reconstruction improves while frozen accuracy is 0.826 and current-head accuracy 0.941; the primary-only current head reaches 0.943. The paired frozen difference is -0.115, with descriptive seed-level 95% interval $[-0.216, -0.014]$. Relative scatter is 0.449 against primary-only continuation, with no seed below 0.3. Per-coordinate repair reaches 0.863 versus 0.941 after refitting. These results establish reconstruction progress and repairable consumer loss; primary-task benefit is tested in the separate low-label study. RadioML I/Q is a near-null control: at 3000, frozen accuracy is 0.301 versus 0.304 detached and trace is 1.061 of healthy. All grid cells and paired seed intervals are retained.

The prepared RadioML cache is checked read-only against the original dictionary using the data loader's per-modulation 50/50 split, NumPy `RandomState(0)`, and original dictionary iteration order. Every input, modulation label, and SNR target tensor must match exactly; the record stores source/cache and index hashes. This is a seeded benchmark partition, not a claim of a unique official RadioML split. Calibration is then reserved within its training portion before encoder updates.

Table 34: Paired continuation with real reconstruction auxiliaries (five seeds per domain). All branches share the healthy encoder, primary head, Adam state, and future batches. MSE ratio compares held-out reconstruction error with the dose-matched detached decoder; values below one indicate improved reconstruction. Trace is relative to the healthy encoder. All entries are seed means.

Case	α	arm	frozen	recal.	refit	own	rel. trace	MSE ratio
HAR recon.	0	primary	0.941	0.940	0.944	0.943	1.064	–
HAR recon.	1	joint	0.941	0.940	0.945	0.946	1.073	0.886
HAR recon.	1	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	1	STF-cal	0.942	0.942	0.944	0.945	1.027	3.779
HAR recon.	10	joint	0.930	0.932	0.943	0.943	0.658	0.784
HAR recon.	10	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	10	STF-cal	0.939	0.941	0.944	0.945	0.756	3.799
HAR recon.	100	joint	0.826	0.863	0.941	0.941	0.208	0.765
HAR recon.	100	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	100	STF-cal	0.914	0.932	0.941	0.937	0.288	3.778
RadioML I/Q	0	primary	0.304	0.306	0.320	0.319	1.105	–
RadioML I/Q	30	joint	0.303	0.305	0.319	0.319	1.106	0.995
RadioML I/Q	30	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	30	STF-cal	0.303	0.305	0.319	0.319	1.106	0.995
RadioML I/Q	300	joint	0.303	0.305	0.319	0.318	1.098	0.987
RadioML I/Q	300	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	300	STF-cal	0.303	0.305	0.319	0.318	1.098	0.987
RadioML I/Q	3000	joint	0.301	0.305	0.319	0.318	1.061	1.007
RadioML I/Q	3000	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	3000	STF-cal	0.301	0.305	0.319	0.318	1.061	1.007

Drivers: `reconstruction_task_audit.py` and `radioml_split_audit.py`; full records: `repro/results/CPUCONT/{mechanism,task_validation}/`.

H.10 Waveform CNN Continuation

The paired real-head studies above use small MLPs. We therefore predeclare a waveform CNN on the identical prepared RadioML I/Q inputs, with five seeds and reconstruction doses $30, 300, 3000$. The flattened vector is reshaped to 2×128. The encoder uses a padded kernel-7 convolution ($2 \to 8$ channels), ReLU and max-pooling by two; a padded kernel-5 convolution ($8 \to 16$), ReLU and pooling by two; and a linear map from the resulting 16×32 features to width 64. There is no stochastic layer or normalization. The linear primary head and $64 \to 256 \to 256$ reconstruction decoder are unchanged.

We reuse the unchanged paired training engine: 25 healthy epochs with a detached auxiliary, 15 continuation epochs, batch 256, Adam at 10^{-3}, independent calibration reservation, complete model/optimizer cloning, matched future orders, and actual-step counterfactual audits. This study tests joint training, detachment, and primary-only continuation, for seven branches per seed; it does not test STF-cal on the CNN. The source records contain both the unchanged engine hash and the wrapper/architecture hash. Healthy heads are replayed deterministically and checked against their archived accuracy, NLL, and ECE before the consumer audit.

The healthy CNN readout reaches 0.424, compared with the MLP's 0.297 on the same split. At dose 3000, frozen accuracy is 0.415 versus 0.418 detached; recalibration gives 0.425, refitting 0.442, and the current head 0.437. Relative trace is 1.032 and the reconstruction MSE ratio to detachment is 0.917. The full dose grid and a pure-scale sweep for each healthy CNN readout are recorded. This stronger encoder shows little real-auxiliary consumer damage under the tested recipe, setting a useful boundary on transfer from the MLP failures. Driver: `cnn_reconstruction_continuation.py`; records: `repro/results/CPUCONT/frontier/c*.json`.

Table 35: Waveform CNN reconstruction continuation on the same prepared RadioML inputs and split as the MLP study. Five seeds; 25 healthy plus 15 continuation epochs; the original heads, optimizer, cloned-state and batch controls are retained. Entries are means, with population seed SD for frozen and own accuracy; all dispersions are archived. MSE ratio compares reconstruction error with the matched detached decoder. All metrics here use the full original test corpus, separately from the monitor/remainder audit.

α	arm	frozen	recal.	refit	own	rel. trace	MSE ratio
0	primary	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	–
30	joint	0.417 ± 0.018	0.427	0.443	0.437 ± 0.015	1.060	0.918
30	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000
300	joint	0.418 ± 0.018	0.428	0.443	0.437 ± 0.015	1.053	0.851
300	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000
3000	joint	0.415 ± 0.023	0.425	0.442	0.437 ± 0.015	1.032	0.917
3000	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000

Table 36: Conditional SNR head-state factorial at dose 30 (five seeds). Shared encoder/primary parameters and their Adam states, the frozen consumer, and future minibatches are identical at branching. Auxiliary weights and auxiliary Adam state form the four cells. Warm/retained replays the original model/Adam endpoint bitwise. Fresh probes share a fixed initialization seed across cells. Means with population seed SD for frozen accuracy; full paired intervals and early actual-step audits are archived.

Weights	aux. Adam	frozen	recal.	refit	own	rel. trace	SNR MSE
warm	retained	0.243 ± 0.020	0.253	0.284	0.277	0.639	0.460
warm	reset	0.241 ± 0.020	0.254	0.283	0.277	0.643	0.460
initial	retained	0.244 ± 0.025	0.245	0.281	0.275	0.726	0.459
initial	reset	0.226 ± 0.033	0.227	0.283	0.277	0.888	0.460

H.11 Conditional Auxiliary Head-State Factorial

To test whether auxiliary-head state explains the difference between historical SNR training and warm continuation, we branch from five archived healthy references. Encoder and primary weights, their Adam state, the frozen consumer, and future minibatches stay identical. At dose 30, warm versus initial pre-warm auxiliary weights are crossed with retained versus reset auxiliary Adam state, followed by 15 continuation epochs. The initial/retained cell isolates the interaction. The design precedes outcome computation; seed 0 is the complete implementation check and remains in the five-seed analysis. The dose is fixed throughout.

Warm/retained reproduces the original model and Adam endpoint bitwise in all five seeds. Resetting auxiliary moments changes frozen accuracy from 0.243 to 0.241. Initial/reset reaches 0.226 with trace 0.888, versus 0.639 for warm/retained. More retained variance therefore need not give higher consumer accuracy. No factorial cell reproduces the historical severe scatter collapse: auxiliary state alone leaves the protocol difference unexplained at this reference and horizon.

Dense early audits and selected later audits record 620 dependent step observations, not 620 independent trials. First-step primary-only counterfactual increments match across all four cells; changing only auxiliary moments also leaves the first encoder increment unchanged, as expected from the parameterwise optimizer. Shared-state and batch hashes, CPU checkpoint validation, bitwise endpoint replays, and seed-paired intervals are archived in `head_state_verification.json` and `head_state_summary.json`. The 20 endpoint states include five exact replays and 15 new conditional trajectories; they are outside the fixed consumer-certificate bank below. Numerical driver: `auxiliary_state_factorial.py`.

H.12 Readout-Nullspace Protection with a Useful Real Auxiliary

The old consumer applies $z_0(x) = W_0 D_0^{-1}(h_0(x) - \mu_0) + b_0$, with fixed positive diagonal $D_0 = \mathrm{diag}(s_0)$. Let $A = W_0 D_0^{-1}$ and $P_\perp = I - A^\dagger A$, the orthogonal projector onto $\ker A$. We use

Table 37: All predeclared primary-accuracy contrasts in percentage points, ten independent confirmation seeds. Paired two-sided 95% descriptive t intervals are conditional on the fixed dataset/split/recipe. GD controls the guarded architecture itself; GJ–GD isolates its auxiliary benefit. Noninferiority was declared at -0.5 point, not selected from the outcomes.

Contrast	mean (pp)	paired 95% interval	required lower limit
J–D	0.97	[0.63, 1.31]	0
GJ–GD	0.89	[0.60, 1.18]	0
GJ–D	1.25	[0.92, 1.58]	0
GJ–J	0.28	[0.08, 0.47]	−0.50

established projection and backward-compatibility ideas (Farajtabar et al., 2020; Shen et al., 2020) as a constructive control for measuring retained benefit and compatibility.

Proposition 10 (Compatibility and remaining feature freedom)**.** *For any updated encoder h_θ and retained deterministic reference encoder h_0, define*

$$\widetilde{h}_\theta(x) = h_0(x) + P_\perp\big(h_\theta(x) - h_0(x)\big). \tag{26}$$

The frozen affine consumer has exactly the reference logits on every input in real arithmetic. It leaves $m - \operatorname{rank}(A)$ feature directions free. It is also the unique Euclidean-nearest feature vector to $h_\theta(x)$ among all vectors with the reference logits. Conversely, every feature displacement preserving these logits belongs to $\ker A$; if A has full column rank, no nonzero compatible displacement remains.

Proof. Since $AP_\perp = 0$, the logit difference is $AP_\perp(h_\theta - h_0) = 0$. Rank–nullity gives the free dimension and the converse. The construction is the orthogonal projection onto the affine set $h_0(x) + \ker A$, proving the minimum-distance statement. The fixed normalizer's mean and the head's bias cancel in the difference. □

The row-vector implementation is $\widetilde{H} = H_0 + (H_\theta - H_0)P_\perp$. A float64 SVD of the normalized head A fixes the projector before continuation. Because the reference branch remains present, the logit identity holds for later encoder parameters and arbitrary inputs in real arithmetic. It also preserves fixed post-processing and its resulting NLL and ECE. Corpus-level float64 checks measure the numerical residual. This functional identity is stronger than the checkpoint-local flow statement and covers a different object from the sampled disagreement bound.

The committed plan separates development from confirmation. Development seeds 200–202 test HAR reconstruction with 10% of fitting primary labels at doses $1, 10, 100$, and RadioML SNR with 20% at $0.1, 1, 10$. A training reservation made before any model or readout fit supplies validation only. Candidates share healthy states, Adam moments, and future minibatches. Healthy readouts use labeled fitting examples; all fitting inputs and auxiliary targets remain available. HAR reconstructs the 561-dimensional input, while SNR metadata are standardized on fitting data. The rule maximizes the minimum mean validation gain among J–D, GJ–GD, and GJ–D. The frozen choice record retains all six candidates, including null and adverse results; development computes no test metrics.

The selected recipe is RadioML SNR dose 1, width 64, Adam 10^{-3}, batch 256, 25 healthy plus 25 continuation epochs. Ten new seeds 300–309 receive this unchanged recipe and all four branches: ordinary detachment/joint training (D/J), and guarded detachment/joint training (GD/GJ). Classification logits and the projector use float64 in every arm; encoder/auxiliary parameters and Adam states remain fp32. The guard and losses are fixed before continuation. Training-anchor prediction drift is logged at fixed epochs with a 0.05 disagreement threshold; it is a diagnostic trace rather than an adaptive controller or warning-lead-time claim. Endpoints are fixed; neither checkpoint nor seed selection uses test outcomes.

Confirmation requires five predeclared conditions: positive paired lower 95% limits for J–D, GJ–GD, and GJ–D; GJ–J above -0.5 percentage point; and logit error below 10^{-8} with zero prediction changes. All five pass. Intervals are paired seed-level t intervals conditional on this dataset, split, and recipe. Current-head accuracy is 0.288 for GJ, 0.285 for J, 0.279 for GD, and 0.275 for D. The guarded comparison GJ–GD isolates the auxiliary's contribution, with SNR MSE 0.766 of matched guarded detachment.

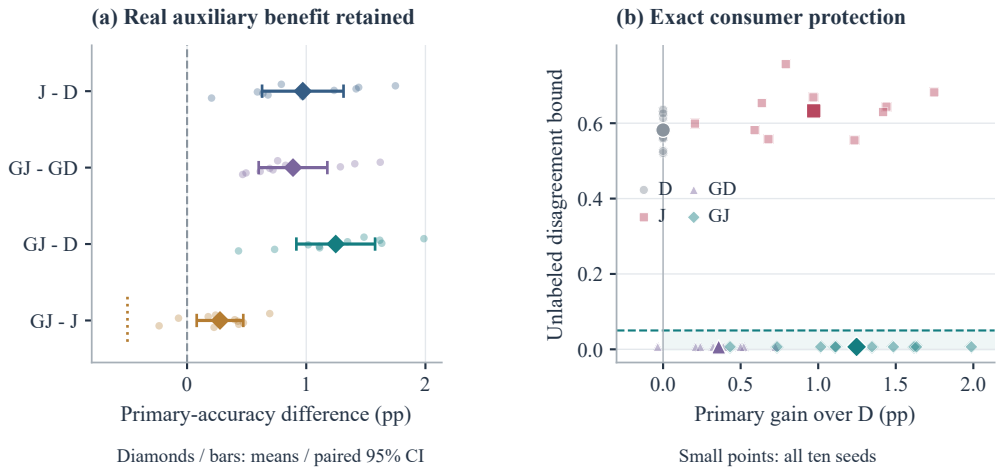


Figure 12: Independent confirmation retains auxiliary benefit and old outputs. (a) All predeclared primary-accuracy contrasts across ten new seeds, with paired 95% intervals. Zero marks the benefit threshold; -0.5 applies to the final noninferiority contrast. (b) All forty consumers: primary gain over D versus the unlabeled disagreement bound. The dashed line is the five-point relative-error budget; shading marks positive gain with a certified consumer. Ordinary consumers remain uncertified despite nearly unchanged aggregate frozen accuracy. Bounds cover the fixed corpus remainder.

The separate confirmation bank contains forty fixed consumers, total tail probability 0.05, an independent $1{,}024$-input monitor per seed, and a 0.05 relative classification-error budget. GD and GJ each certify in $10/10$ seeds with a bound near 0.0065; D and J each certify in $0/10$. Ordinary consumers change many predictions while aggregate frozen accuracy remains nearly unchanged. J's frozen ECE rises from 0.048 to 0.155; guarded predictions, NLL, and ECE match the reference, with maximum logit error 4.6×10^{-14}. This bank is distinct from the earlier $3{,}015$-candidate family.

All ten normalized heads have rank 11, leaving 53 free directions. Stacking normalized matrices protects several affine heads on the same reference features, at the cost of a smaller nullspace; full column rank leaves no compatible displacement. Nonlinear consumers need their own constraint. The reference encoder remains at inference. On one Xeon Platinum 8570 CPU thread, batch $1{,}024$, the mean ratio of per-seed median latencies is $2.37\times$, using nine repetitions after three warm-ups. Reference fitting features are cached during training. Thus the confirmed low-label result includes the cost of retaining and evaluating the reference.

A direct dual-routing baseline also retains both encoders, serving the old consumer from h_0 and the current head from J. It inherits J's primary metric and the reference legacy metrics. The guarded control instead emits one shared fixed-width vector for both heads and adds a projection. Its distinctive constraint is a common representation; it has no measured efficiency advantage over dual routing. The nearest-vector property concerns Euclidean feature distance.

The full study verifies 104 development/confirmation model–Adam states, shared initial states, primary-label masks, future-order hashes, and all consumer bounds. An independent process reconstructs all forty saved consumers and replays their accuracy/NLL/ECE and bounds exactly. Numerical drivers are `benefit_compatibility_cpu.py` and `verify_benefit_consumers_cpu.py`; local analysis is `summarize_benefit_compatibility.py`. The committed plan, frozen validation choice, full records, verification and all negative development cases accompany the artifact.

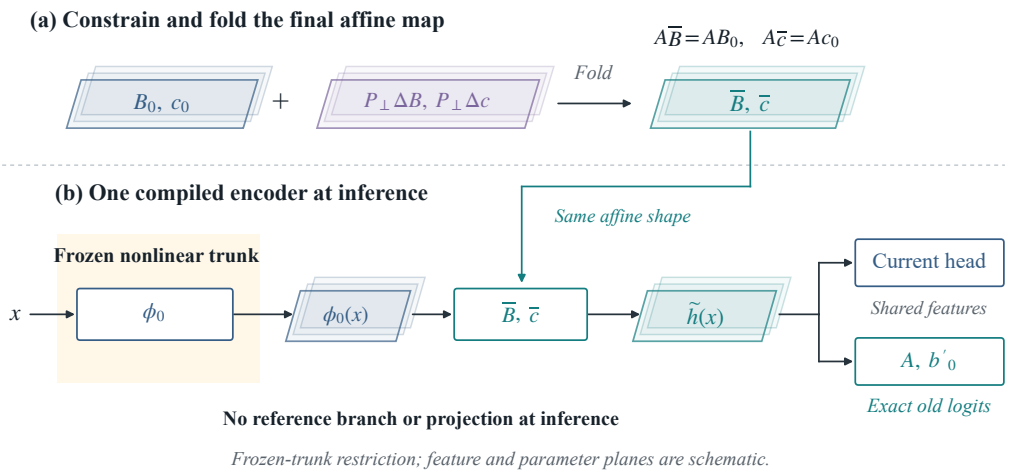


Figure 13: A frozen trunk allows the compatibility constraint to fold into one encoder. With $\Delta B = B - B_0$ and $\Delta c = c - c_0$, constrained weights and biases retain their native shapes (Equation 27); $b'_0 = b_0 - A\mu_0$ is the consumer's effective bias. Inference evaluates one encoder while preserving the reference logits. Freezing the nonlinear trunk restricts feature learning. Planes are schematic.

H.13 Folding Compatibility into a Single Encoder

The reference branch makes compatibility expensive. Freezing the nonlinear trunk lets us fold the constraint into its final affine map, retaining ordinary inference shapes but restricting feature learning. A new plan, committed after the earlier confirmation and before execution, tests this tradeoff as a separate study.

Corollary 8 (Foldable affine compatibility)**.** *Let $h_0(x) = B_0\phi_0(x) + c_0$, with a retained deterministic trunk ϕ_0. For arbitrary trainable B, c, set*

$$\overline{B} = B_0 + P_\perp(B - B_0), \qquad \overline{c} = c_0 + P_\perp(c - c_0). \tag{27}$$

Then $\overline{h}(x) = \overline{B}\phi_0(x) + \overline{c}$ has the reference affine-consumer logits for every input in real arithmetic. Deploying $\overline{B}, \overline{c}$ needs one trunk and one final affine map of the original shapes, with no reference branch or projection. If the trunk has d outputs, this parameterization leaves $(m - \operatorname{rank} A)(d + 1)$ affine degrees of freedom. Changing the trunk is outside this guarantee.

Proof. Both $A(\overline{B} - B_0)$ and $A(\overline{c} - c_0)$ vanish. Folding is a substitution of one weight and bias, not a second inference operation. Each of the $d + 1$ weight/bias columns can vary in $\ker A$. □

The construction uses constrained affine adaptation, related to nullspace updates and mergeable frozen-backbone methods (Farajtabar et al., 2020; Pham et al., 2026; Hu et al., 2022a). With a linear current head it remains a linear predictor of fixed trunk features. Folding preserves the constrained function exactly; whether that function retains auxiliary benefit is the empirical question.

RadioML uses SNR regression, dose 1 and 20% of fitting primary labels, inherited from the prior validation choice. HAR uses input reconstruction, dose 10 and 10% labels, the positive max-min candidate in the previously archived three-seed HAR validation grid. No new grid is searched. Each task uses ten new seeds 400–409, the same prepared test split, a stratified 20% training reservation before any model/head fit, width 64, Adam 10^{-3} and batch 256. HAR has 60 healthy plus 30 continuation epochs; RadioML has $25 + 25$. The last encoder affine map is natively float64 in every arm, as are classification logits; trunk and auxiliary parameters are fp32. The healthy consumer is fitted to that same mixed-precision interface. This explicit precision change prevents pooling with the earlier encoder-fp32 experiment.

Six branches clone each healthy model and full Adam state, with identical fitting-label masks and future minibatches: full-network D/J; frozen-trunk, unconstrained-last-layer LD/LJ; and frozen-trunk,

Table 38: Folded single-encoder continuation, ten new seeds per task. D/J update the full encoder; LD/LJ freeze the nonlinear trunk; CD/CJ additionally constrain the last affine map to preserve the normalized healthy consumer. All use the same native float64 final affine and float32 trunk. Means with population seed SD; the 120-consumer bank uses a five-point relative-error budget.

Task	arm	own accuracy	frozen accuracy	frozen NLL	ECE	certified
SNR	D	0.267 ± 0.008	0.243 ± 0.032	2.262	0.169	0/10
	J	0.278 ± 0.009	0.248 ± 0.026	2.198	0.166	0/10
	LD	0.242 ± 0.015	0.221 ± 0.025	2.261	0.139	0/10
	LJ	0.242 ± 0.015	0.219 ± 0.029	2.255	0.132	0/10
	CD	0.242 ± 0.015	0.244 ± 0.015	2.061	0.039	10/10
	CJ	0.242 ± 0.015	0.244 ± 0.015	2.061	0.039	10/10
HAR recon.	D	0.918 ± 0.008	0.913 ± 0.012	0.482	0.062	6/10
	J	0.919 ± 0.005	0.898 ± 0.019	0.540	0.069	2/10
	LD	0.918 ± 0.006	0.913 ± 0.011	0.440	0.059	6/10
	LJ	0.918 ± 0.006	0.906 ± 0.015	0.450	0.062	5/10
	CD	0.918 ± 0.006	0.918 ± 0.005	0.355	0.052	10/10
	CJ	0.918 ± 0.007	0.918 ± 0.005	0.355	0.052	10/10

Table 39: All declared folded-control accuracy contrasts in percentage points, mean with paired descriptive 95% interval. Conditional on each fixed dataset/split/recipe. The first four rows have required lower limits $0, 0, 0, -0.5$ respectively; all other rows diagnose freezing and constraint costs regardless of sign.

Contrast	RadioML SNR	HAR reconstruction
J–D	$+1.11$ [0.90, 1.32]	$+0.04$ [-0.30, 0.37]
CJ–CD	$+0.01$ [-0.02, 0.04]	$+0.01$ [-0.08, 0.09]
CJ–D	-2.42 [-3.20, -1.64]	-0.03 [-0.43, 0.36]
CJ–J	-3.53 [-4.23, -2.83]	-0.07 [-0.47, 0.33]
LD–D	-2.45 [-3.21, -1.69]	-0.02 [-0.43, 0.39]
LJ–J	-3.53 [-4.21, -2.85]	-0.08 [-0.47, 0.30]
CJ–LJ	-0.003 [-0.039, 0.033]	$+0.010$ [-0.078, 0.099]
CD–LD	$+0.018$ [-0.025, 0.061]	-0.020 [-0.093, 0.052]
LJ–LD	$+0.028$ [0.001, 0.055]	-0.024 [-0.068, 0.021]

constrained-last-layer CD/CJ. Heads continue learning in every branch. The constraint is fixed before continuation; the fixed training-anchor diagnostic is not an adaptive policy. Validation/test inputs never train the encoder or healthy consumer. Frozen trunk parameters *and their Adam states* are checked unchanged. Every branch uses the declared endpoint and every seed is retained.

The five criteria require positive lower paired 95% limits for J–D, CJ–CD, and CJ–D; CJ–J above -0.5 percentage point; and preserved old logits and predictions. Neither task passes all five. On SNR, unrestricted auxiliary gain is 1.11 points, $[0.90, 1.32]$, while compiled gain is 0.01, $[-0.02, 0.04]$. CJ–J is -3.53, $[-4.23, -2.83]$, below the noninferiority margin. The unconstrained frozen-trunk comparison LJ–J is -3.529 points, $[-4.213, -2.846]$; adding the constraint, CJ–LJ, changes accuracy by -0.003, $[-0.039, 0.033]$. Freezing accounts for the dominant measured cost in this recipe. LJ–LD retains a small 0.028-point gain, $[0.0006, 0.0554]$, far below unrestricted training.

HAR gives J–D of 0.04 points, $[-0.30, 0.37]$, and CJ–CD of 0.01, $[-0.08, 0.09]$. Both leave auxiliary benefit unresolved. CJ–J meets the half-point margin. Table 39 gives every planned contrast, including adverse and near-null outcomes. All intervals describe paired variation over ten seeds on each fixed dataset, split, and recipe.

All 40 CD/CJ consumers retain old predictions, NLL, and ECE, with maximum logit error 3.6×10^{-14}. Folded and training-time features are bitwise identical on the scored corpus. A separate fixed family of 120 consumers uses total $\delta = 0.05$, $1{,}024$ uniformly sampled monitor inputs per seed, and a five-point relative-error budget. All 40 compiled consumers certify on the corpus remainder; monitor information stays outside training.

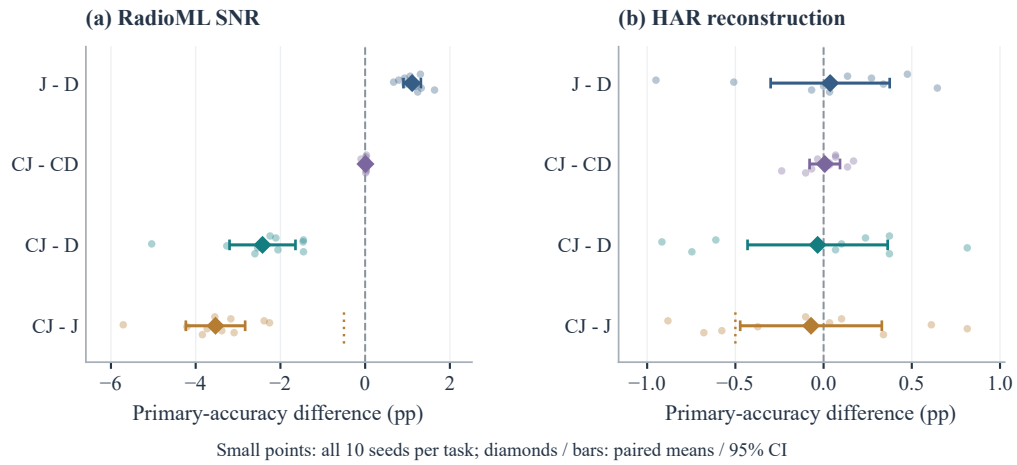


Figure 14: Folding retains compatibility but loses the SNR auxiliary gain. All four declared primary-accuracy contrasts, ten new seeds per task. Diamonds and bars show means and paired descriptive 95% intervals. D/J update the full encoder; CD/CJ freeze the trunk and constrain its last affine map. The -0.5 marker applies only to CJ–J. SNR retains an unrestricted gain; neither task passes the full five-criterion target. Panel scales differ.

Deployed encoders retain exactly three affine evaluations and the baseline parameter shapes, with no reference or projector parameters. At batch 1, mean ratios of per-seed median CPU latency are $1.00\times$ for SNR and $1.00\times$ for HAR. At batch 1,024, they are $1.01\times$ and $1.00\times$. Measurements cover the encoder and current head, using one Xeon Platinum 8570 thread, five warm-ups and 21 alternating timed repetitions per path; the baseline uses the same mixed precision. The corresponding two-encoder paths average $2.22\times$ and $2.16\times$ at batch 1,024. These deployment measurements establish the cost reduction, not useful adaptation.

Local checks cover all 180 healthy, endpoint model–Adam, and compiled states. A separate tensor-based process reconstructs all 120 consumers and exactly replays accuracy, NLL, ECE, and sampling bounds. Structural checks cover the normalized-head constraint, Adam steps, folding, reference and projector removal, nonzero gradients, and failure under trunk drift. The artifact includes the committed plan, drivers, complete outcomes, and verification. This control removes reference computation but loses the confirmed SNR benefit, motivating a trainable nonlinear channel.

H.14 A Compatible Nonlinear Channel: Development Gate and Limits

A narrow side network allows nonlinear learning while the base remains fixed. We combine side-tuning (Zhang et al., 2020) with established compatibility projections (Farajtabar et al., 2020; Shen et al., 2020):

$$\widetilde{h}(x) = f_0(x) + P_\perp g_\psi(x), \qquad A\widetilde{h}(x) = Af_0(x). \tag{28}$$

The healthy base f_0 stays fixed, and a two-ReLU side MLP g_ψ learns its nonlinear prefix. Both heads receive one shared 64-vector. Projection preserves affine-consumer logits; the following experiment measures whether the added flexibility also retains useful learning.

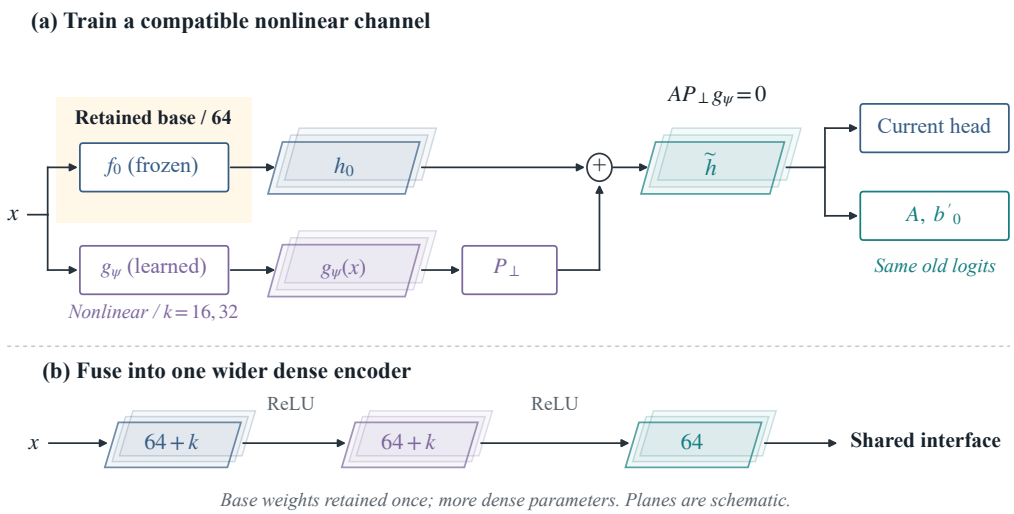


Figure 15: A compatible nonlinear channel adds trainable capacity. (a) The base stays fixed while the side learns, with $AP_\perp g_\psi = 0$. (b) Equal-depth paths fuse into a dense width-$64 + k$ MLP: the middle matrix is block diagonal and the final side block is projected. The base is embedded once, all dense zero blocks count toward storage and computation, and the output stays 64-dimensional. Planes are schematic; parameter ratios and the validation screen are reported separately.

For base weights (U_1, U_2, U_3) and side weights (V_1, V_2, V_3), the fused matrices are

$$\overline{U}_1 = \begin{bmatrix} U_1 \\ V_1 \end{bmatrix}, \quad \overline{U}_2 = \begin{bmatrix} U_2 & 0 \\ 0 & V_2 \end{bmatrix}, \quad \overline{U}_3 = \begin{bmatrix} U_3 & P_\perp V_3 \end{bmatrix}. \tag{29}$$

Stacking the first two biases and using final bias $b_3 + P_\perp c_3$ gives the same function with three affine layers and two ReLUs. The fused model embeds the base once and absorbs the separate projection. Its wider dense matrices still count all zero blocks: for side widths $16/32$, encoder parameter ratios are $1.301/1.623$ on SNR and $1.279/1.569$ on HAR, at native float64. Full-data runtime evaluation was contingent on passing the development screen. The side's final weight and bias start at zero; prefix weights copy the first k healthy units and their $k \times k$ second-layer block.

The new plan was committed before numerical execution, after observing the earlier studies. Development uses seeds 500–502, the original width-64 base, Adam 10^{-3}, batch 256, and the same stratified 20% training reservation before model/head fitting. SNR uses 20% of fitting primary labels, dose 1, and $25 + 25$ healthy/continuation epochs. HAR uses a new 1% fitting-label reconstruction regime, doses $1, 10$, and $60 + 30$ epochs. Both test side widths $16, 32$. All feature-encoder arithmetic is float64, with fp32 head parameters and float64 classification logits. These changes prevent pooling with earlier confirmation results.

Each candidate compares full-network D/J, unconstrained side UD/UJ, and constrained side TD/TJ, with detached/joint auxiliary gradients. Healthy base/head parameters and their Adam states are retained; new side Adam states start empty. The common parameter ordering, paired side initialization, fitting-label masks and future minibatches are checked. The declared score is the minimum of three mean validation gains (J–D, TJ–TD, TJ–D) and the mean TJ–J contrast plus 0.5 point; ties prefer smaller width, then dose. A task enters ten-new-seed test confirmation only if its best score is strictly positive. No development test outcomes are computed.

The highest-scoring SNR candidate uses $k = 32$, dose 1. Its validation TJ–TD is 0.32 points, with descriptive paired 95% interval $[-0.17, 0.80]$, but TJ–D is -0.69, $[-1.93, 0.55]$, giving a negative score. The highest-scoring HAR candidate uses $k = 16$, dose 1: TJ–TD is 0.70, $[-1.76, 3.17]$, but J–D is exactly tied in aggregate correct counts, giving score zero. Neither qualifies for confirmation. The artifact retains all six candidates. These selected three-seed validation estimates describe the development screen.

Table 40: Complete nonlinear-channel development grid: three seeds per task, validation only. D/J are full-network branches; TD/TJ are constrained side branches. All entries are percentage-point means. The score is $\min\{\overline{\mathrm{J}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{TD}}, \overline{\mathrm{TJ}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{J}}+0.5\}$. Confirmation requires strictly positive score, evaluated from integer correct counts. No candidate qualifies. HAR uses a new 1% fitting-label regime, SNR 20%.

Task	k	α	J–D	TJ–TD	TJ–D	TJ–J	score	confirm
SNR	16	1	+0.34	+0.53	−1.73	−2.07	−1.73	no
SNR	32	1	+0.34	+0.32	−0.69	−1.02	−0.69	no
HAR	16	1	+0.00	+0.70	+0.34	+0.34	0.00	no
HAR	32	1	+0.00	+0.39	+0.25	+0.25	0.00	no
HAR	16	10	+1.50	+0.32	−0.05	−1.54	−1.04	no
HAR	32	10	+1.50	−0.29	−0.43	−1.93	−1.43	no

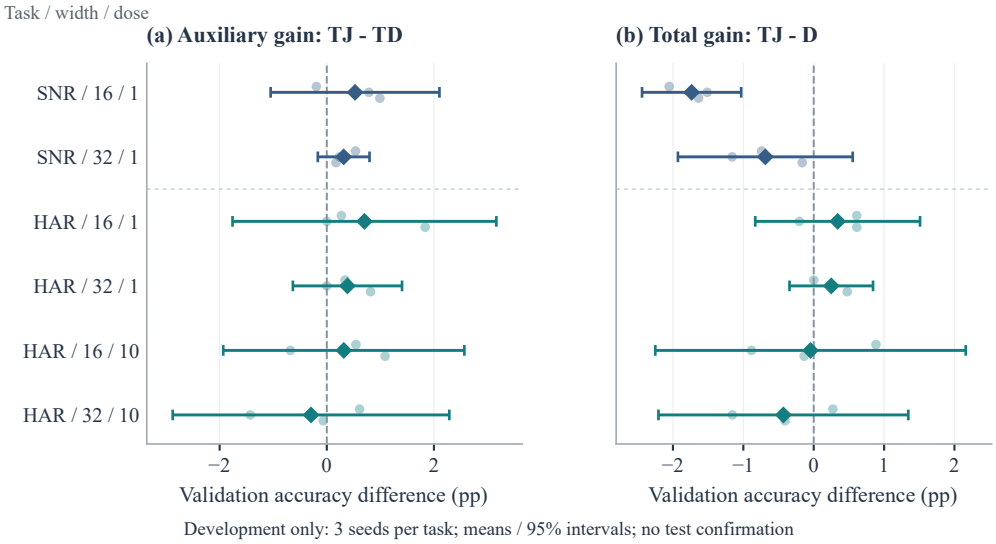


Figure 16: Neither task passes the nonlinear-channel development screen. Auxiliary and total gains for all six candidates, with three seed points and descriptive paired 95% intervals per row. Labels give task, side width, and dose. The screen additionally requires positive J–D and the TJ–J margin (Table 40). No candidate proceeds to test confirmation or full-data latency evaluation.

The strict-positive gate is evaluated from integer correct counts and exact rational means. Before confirmation, an arithmetic check corrected a floating mean near 3.7×10^{-17} for the tied HAR contrast to zero. Both serializations and the correction audit are retained; the grid and decision rule stay fixed. Since the screen fails, the planned independent-test, auxiliary-MSE, and CPU-budget conjunction is not evaluated, and the 200-consumer confirmation family is not instantiated.

All 165 saved state files are locally checked: six healthy states, 87 endpoint model–Adam states and 72 deployed side models. A separate tensor-only process reconstructs all 87 validation consumers; accuracy, NLL and ECE replay exactly, with auxiliary errors also checked. All 36 constrained-side consumers retain their old validation predictions/NLL/ECE; maximum logit error is 2.5×10^{-14}. Fused/unfused feature error is at most 1.1×10^{-14}. The nonlinear prefix changes in every constrained branch, and frozen base weights and Adam states remain unchanged. Structural tests include arbitrary nonlinear changes, native-precision fusion, full-rank/no-nullspace and unconstrained/base-drift negative controls.

The construction supports nonlinear updates with retained affine-consumer outputs, but the tested candidates do not close the utility and cost target. A matched side with its prefix frozen would be

needed to isolate the prefix's contribution to performance. Broader architectures, nonlinear downstream consumers, new populations, and adaptive warning policies remain open extensions.

H.15 A Fixed-Corpus Unlabeled Consumer Check

For a present checkpoint, we complement the variance audit with unlabeled disagreement against the healthy consumer. This uses the standard relative-loss inequality and exact finite-population sampling. It sits alongside unlabeled performance estimation (Garg et al., 2022) and update certificates (Usama & Qadir, 2026), with reference predictions and an explicit abstention option.

For a fixed corpus of N inputs, let f_0 be the healthy consumer, f a fixed candidate, and R_T the mean 0–1 error on subset T. For every assignment of labels,

$$R_T(f) - R_T(f_0) \le D_T(f, f_0), \qquad D_T = |T|^{-1} \sum_{i \in T} \mathbf{1}\{f(x_i) \neq f_0(x_i)\}. \tag{30}$$

The inequality follows pointwise: error can increase only where predictions change. It is sharp when the labels equal the healthy predictions. Conversely, changing predictions can improve accuracy, so large disagreement does not establish harm.

Uniformly sample n monitor inputs without replacement, independently of the fixed candidate family. If c predictions disagree on this monitor and the unknown full-corpus disagreement count is K, then c has the hypergeometric distribution with parameters (N, K, n). With family size M and total tail probability δ, define

$$K^+(c) = \max\{K \in \{c, \dots, N - n + c\} : F_{N,K,n}(c) \ge \delta/M\}, \qquad U = \frac{K^+(c) - c}{N - n}. \tag{31}$$

Here F is the exact hypergeometric CDF. Inverting its lower tail bounds the full-corpus disagreement count, hence $D_T \le U$ on the unmonitored remainder. For each candidate the failure probability is at most δ/M; a union bound gives simultaneous coverage for all M candidates, allowing selection from that fixed family. This random-partition statement does not require independent adjacent HAR windows. It covers the fixed corpus, rather than a new population or future optimizer trajectory. The bound concerns classification error; it does not certify NLL, ECE, or conformal coverage. Monitor-adaptive encoder training or unconstrained construction of new repair candidates would require a different analysis.

We predeclare $n = 1{,}024$, $\delta = 0.05$, and budgets 0.05 and 0.10 for relative loss increase. The bank contains all 300 endpoints from the 30 earlier paired runs plus 35 CNN endpoints, each evaluated with the same nine fixed repair arms: $M = 3015$. All maps use the full fitting feature set; a restricted anchor budget would need its own evaluation. A seeded uniform monitor partition is shared across candidates in a run, and old/new reference predictions on those monitor inputs are required. Healthy reference predictions can be cached for a fixed corpus or obtained from the retained healthy encoder; this access is additional to paired training anchors for richer repairs. The first repair in the predeclared hierarchy satisfying $U \le \varepsilon$ is selected, or the audit abstains. Monitor labels never enter this choice. Candidate remainder accuracy and loss increase are scored only after choices are fixed; archived healthy/frozen metrics are also checked independently for reproducibility.

Across 105 joint endpoints, the 5-point budget selects 18 candidates and abstains on 87; the 10-point budget selects 49 and abstains on 56. The latter includes 12 unchanged frozen consumers and 37 repaired consumers. Observed selected-consumer loss increases exceed budget in 0 and 0 cases, respectively. The healthy-reference scatter heuristic makes 16 and 11 false-safe declarations. This comparison uses scatter relative to healthy features on the same monitor, which is available without a continued primary-only counterfactual; it is distinct from retrospective comparison with a fully trained single-task endpoint. The fixed unsigned probe's safe declarations and primary-only controls are also retained in the numerical summaries.

At the 10-point budget, HAR supplies 49 selections; RadioML supplies 0 of 45, including its near-null continuation cases. Disagreement can leave an accuracy-preserving or improving candidate uncertified when many decisions change. The check bounds relative 0–1 error on the fixed corpus; absolute accuracy and warning lead time require other evidence. All candidates, decisions, bounds, remainder scores, map hashes, and source links are archived. Checks cover finite-population coverage, boundary cases, monotonicity, sharpness, selection, abstention, and temperature-only

Table 41: Fixed-family consumer audit. Each corpus supplies 1,024 uniformly sampled unlabeled monitor inputs; remaining labels score the decisions afterward. The 3,015 fixed endpoint/repair candidates share total tail probability 0.05. *Scatter false safe* uses a scatter ratio ≥ 0.3 relative to healthy features on the same monitor, despite frozen-consumer loss increase above budget. Selection uses the first certified candidate, including identity/frozen; uncertified cases abstain. Counts describe dependent checkpoints, not independent trials.

Case	budget	joint	scatter false safe	frozen certified	candidate selected	selected false safe
HAR isotropic	0.05	30	0	0	6	0
HAR isotropic	0.10	30	0	3	24	0
HAR projected	0.05	15	10	0	3	0
HAR projected	0.10	15	8	0	10	0
HAR recon.	0.05	15	3	1	9	0
HAR recon.	0.10	15	3	9	15	0
RadioML SNR	0.05	15	3	0	0	0
RadioML SNR	0.10	15	0	0	0	0
RadioML I/Q MLP	0.05	15	0	0	0	0
RadioML I/Q MLP	0.10	15	0	0	0	0
RadioML I/Q CNN	0.05	15	0	0	0	0
RadioML I/Q CNN	0.10	15	0	0	0	0

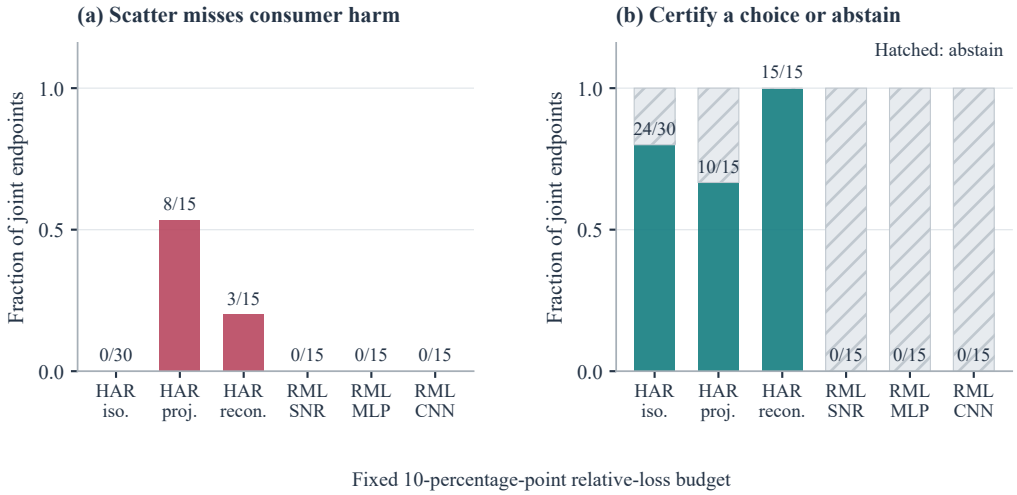


Figure 17: Unlabeled disagreement supports conservative repair selection. Fixed 10-percentage-point relative-error budget. Left: a monitor-scatter rule can declare a harmful frozen consumer safe. Right: the first certified repair is selected from the fixed hierarchy; otherwise the rule abstains. The final two groups use I/Q reconstruction. Bars show fractions of joint endpoints, with counts annotated. Within-seed branches are dependent. Simultaneous coverage applies to the fixed candidate family on the unmonitored corpus remainder.

NLL drift. Drivers: `consumer_risk.py` and `checkpoint_consumer_audit.py`; analysis: `summarize_consumer_frontier.py`.

H.16 Dissociation Between Contraction and Utility

The audits above separate three quantities that are often conflated: variance contraction, scale-invariant information, and readout utility. Table 42 collects the six cases that separate them. These are case studies; the battery, SNR, I/Q, and subject-ID rows use the protocols cited in their sections.

Table 42: Six cases separating contraction, decodability, and readout utility. Relative S is the same-seed scatter ratio. The scale check is the standardized probe where available; PR measures spectral concentration in the battery row. Utility names the primary or frozen-readout metric. SNR and subject-ID readout studies use five seeds, I/Q three; the battery row uses the corrected MATR application.

Case	Rel. S	Scale check	Utility	Diagnosis
HAR toxifier (100)	0.016	probe 0.955 (healthy 0.947)	frozen 0.241; input recal 0.952; retrained 0.955	contraction; information intact; readout broken
MATR battery	3.55	PR 2.49	SOH 5.57 vs 2.10 detached	no contraction; negative transfer
RadioML SNR (3000)	0.137	n/a	own 0.205; frozen 0.099; retrained 0.229; detached readout 0.315	relative collapse; readout near chance; sampled flow flux mostly negative
RadioML I/Q (3000)	1.05	n/a	own 0.344; frozen 0.195; retrained 0.351; detached readout 0.312	no contraction; frozen readout broken by drift
HAR subject-ID (10)	0.93	n/a	own 0.887; frozen 0.384; retrained 0.901	no contraction; frozen readout broken by drift
HAR reconstruction	1.00	n/a	0.942	benign head

The six cases show why no single scalar orders representation quality. In HAR, scatter contracts by a factor of 60 while task information remains linearly available and input recalibration repairs the readout. MATR has negative transfer without contraction. SNR combines relative collapse and readout failure, but mostly negative sampled flow flux leaves its Adam attribution unresolved. I/Q retains scatter 1.05 and refitted accuracy 0.351 versus 0.312 detached, while the frozen readout falls to 0.195. Subject-ID similarly gives joint accuracy 0.887 at scatter 0.93, yet frozen accuracy 0.384 versus 0.901 refitted. The last row is benign at its tested dose. Each case requires its own measurements of variation, decodability, and compatibility.

H.17 RadioML Dose–Response and Natural Scale

Table 43 varies the auxiliary dose for two real RadioML heads at $m = 64$, 25 epochs and three seeds, fully labeled: SNR regression (the head that collapses at the reported dose) and I/Q reconstruction (a benign head). The detached reference is 0.306 accuracy with $0/3$ collapses for SNR and 0.307 with $0/3$ for reconstruction; its relative scatter is 1.031 and 1.101, respectively.

Table 43: RadioML dose–response, fully labeled. Joint training, three seeds, $m = 64$. "Coll." is the scatter-collapse count; $\alpha\rho$ is the nominal dose times the measured per-example radial ratio. The SNR onset lies between $\alpha = 3$ and 10, and the head collapses in every seed from $\alpha = 30$ onward; the reconstruction head never collapses at any tested dose.

α	SNR acc.	SNR coll.	SNR rel. S	SNR $\alpha\rho$	I/Q acc.	I/Q coll.	I/Q rel. S
1	0.329	0/3	0.932	0.28	0.305	0/3	1.054
3	0.318	0/3	0.600	1.28	0.308	0/3	1.038
10	0.277	2/3	0.291	5.84	0.312	0/3	1.096
30	0.248	3/3	0.211	22.6	0.303	0/3	1.005
100	0.229	3/3	0.163	70.3	0.312	0/3	1.057
300	0.214	3/3	0.146	187	0.321	0/3	1.252
1000	0.207	3/3	0.144	557	0.338	0/3	1.267
10000	0.204	3/3	0.140	4138	0.320	0/3	0.641

SNR collapse begins between $\alpha = 3$ and 10, with all seeds below the criterion by 30; the historical 3000 comparison is deep within this observed regime. I/Q reconstruction stays near or above reference scatter through 1000 and reaches 0.641 at 10000, above 0.3. Thus nominal dose alone does not order the two heads. Their measured radial loadings differ sharply: at 30, SNR has $\alpha\rho = 22.6$, while I/Q loading is near zero. The signed-flux audit below checks what those unsigned measurements can explain.

Table 44 adds STF-cal at the three doses around and above the SNR onset. At $\alpha = 30$, calibrated filtering reaches 0.303 against joint 0.248 and detached 0.306, with no collapse; at $\alpha = 100$ and 300 the recovery shrinks to 0.278 and 0.236 even though the gate opens in every seed. The reconstruction gate stays shut (gate rate 0.00) at all three doses and STF-cal stays at 0.308. This is the dose-dependent selective-intervention behavior claimed in the main text: filtering helps near the onset but not at every dose.

Table 44: STF-cal at the SNR onset (RadioML, fully labeled, three seeds, $m = 64$). Gate rate and mean margin are for the calibrated arm; joint and STF-0 show the reference endpoints.

α	joint acc.	STF-0 acc.	STF-cal acc.	gate / margin
30	0.248	0.306	0.303	1.00 / 14.1
100	0.229	0.306	0.278	1.00 / 47.1
300	0.214	0.306	0.236	1.00 / 141.3

H.18 Readout Dose Scan and Signed Flux

Table 45 repeats the healthy-readout transfer across the SNR dose scan at $m = 64$, 25 epochs, and five seeds. The frozen readout is 0.176 at $\alpha = 3$ (relative scatter 0.558, above the 0.3 criterion), 0.121 at $\alpha = 10$ (relative scatter 0.274, at the criterion), and near chance from $\alpha = 30$ onward. Retraining recovers 0.322 at $\alpha = 3$ and 0.229 at $\alpha = 3000$, and exceeds the joint encoder's own accuracy at every dose, so the representation retains linearly recoverable information. The readout therefore degrades before the operational scatter criterion fires; a scatter threshold alone is not a deployment certificate. Detachment preserves 0.315 at every dose. Driver: `repro/scripts/real_head_readout_transfer.py`; records: `results/READOUT/rml_snr_readout_a*.json`.

Table 45: Historical RadioML SNR readout dose scan (five seeds, $m = 64$, 25 epochs). Own uses the joint primary head; frozen and recalibrated reuse the healthy weights; retrained fits new weights; detached uses the healthy readout on the STF-0 encoder. Relative S is the same-seed scatter ratio.

α	rel. S	healthy	frozen	recal.	retrained	detached	own
3	0.558	0.315	0.176	0.146	0.322	0.315	0.315
10	0.274	0.315	0.121	0.114	0.283	0.315	0.273
30	0.198	0.315	0.102	0.099	0.256	0.315	0.247
100	0.160	0.315	0.106	0.090	0.240	0.315	0.229
300	0.145	0.315	0.103	0.097	0.234	0.315	0.214
3000	0.137	0.315	0.099	0.094	0.229	0.315	0.205

The signed flux audit uses the same driver with `--flux-trace`. On a fixed diagnostic batch of 384 test examples we compute $V = \|C_B H\|_F^2/(2B)$, $v = \nabla_\theta V$, $r = -v^\top G_p$, and $c = v^\top G_a$, and record them at every epoch end; inside the first two epochs we also sample every 10 mini-batches. At $\alpha = 3000$ the seed-median c_t is -3.97 at epoch 1, -49.3 at epoch 5, -66.1 at epoch 10, and -107.5 at epoch 25; the aggregate signed cosine is below -0.90 at the endpoint. Of 75 epoch-end checkpoints, 74 have negative c_t; the exception is $+0.76$. Its possible early effect cannot be dismissed by comparing it with larger endpoint magnitudes or by sparse epoch-end sampling. I/Q reconstruction gives $|c_t| < 10^{-5}$ at every epoch end, consistent with no variance flux. The SNR head thus shows relative collapse without consistently positive sampled gradient-flow flux. These observations alone do not identify its Adam-step contributions; Appendix H.7 audits actual updates separately. Records: `results/READOUT/rml_snr_flux*.json` and `results/READOUT/rml_iq_flux.json`.

At $\alpha = 3000$, every calibrated SNR gate opens and the selected deletion reaches its cap of 3 (calibrated $\rho = 0.44$–0.49, margins 1.3–1.5×10^3). Relative scatter remains 0.25–0.36; frozen accuracy rises only from 0.094 (joint seed median) to 0.136, with recalibration 0.124 and refitting 0.212. The selected rank and reliably open gate therefore fail to restore compatibility at this dose. Records: `results/READOUT/rml_snr_readout_stfcal.json`.

I ANTI-COLLAPSE REGULARIZERS AT MATCHED DOSES

Table 25 evaluates two regularizer families at the doses and seeds of Table 4: VICReg's variance hinge plus off-diagonal covariance penalty, and Barlow Twins' redundancy-reduction penalty. Each is applied to the shared representation with weight λ. VICReg's invariance term is omitted because this experiment has no augmented-view pair. All rows use the same joint-training baseline; these are adaptations of the regularizers to this setting, not evaluations of their complete original learning pipelines.

Regularization can clear the operational collapse threshold without restoring the single-task reference or improving the task. In the coarse grid, VICReg first avoids collapse in all seeds at $\lambda = 10$ for HAR, 300 for battery, and 10000 for RadioML. Barlow Twins does not do so at any tested weight.

On HAR, VICReg scatter stays near 0.36 from $\lambda = 10$ to 10^4, while accuracy falls from 0.944 to 0.821. At the lowest clearing weight, accuracy is close to joint training (0.945) and detachment. On battery, scatter increases through the reference level, but the best tested RMSE is 0.339 at $\lambda = 300$ (relative scatter 0.59), compared with 0.297 for detachment. Larger weights worsen RMSE to 0.390 at 10^3 and 0.423 at 10^4; detachment is better in nine of ten seeds. On RadioML, $\lambda = 10000$ gives zero collapse and scatter 1.29, but accuracy drops to 0.215, below both collapsed joint training (0.256) and detachment (0.341). These cases separate the scatter criterion from predictive utility.

The Barlow Twins adaptation yields absolute scatter as low as 0.011, 0.012, and 0.013 for HAR, battery, and RadioML, respectively, or factors of 172, 6, and 24 below their references. It fails to clear the collapse threshold throughout the tested grid, with HAR accuracy reaching approximately 0.34–0.44 and RadioML approximately 0.11 in the low-accuracy configurations.

Additional ten-seed sweeps use $\lambda \in \{2, 5, 20, 50, 200, 500\}$ on HAR and $\lambda \in \{200, 500, 700, 1500, 2000, 5000\}$ on battery. The recorded battery transition is bracketed by 150 and 200, and the lowest measured battery RMSE remains 0.339 at $\lambda = 300$, above detachment. HAR scatter remains near 0.36 above its crossing. RadioML was not refined: the sampled collapse rate is 1.0 at 1000, 0.10 at 3000, and 0.0 at 10000. The experiments therefore locate an operational transition over a finite grid rather than a precise critical weight.